Climate Dynamics

에서 살아남기

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HISTORY AND THE NATURE OF THE PROBLEM

1. HISTORICAL DEVELOPMENT OF GENERAL CIRCULATION THEORY

1.1. Early Observations: Halley and the Trade Winds

무역풍의 발견과 Halley의 관측

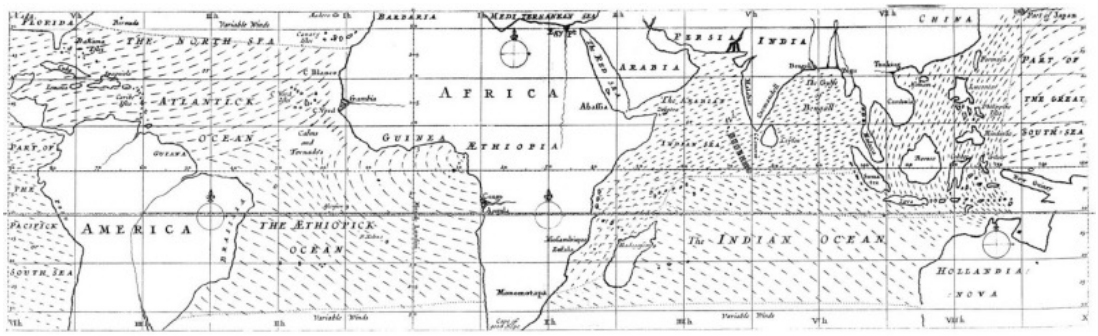
Christopher Columbus's westward voyages (1492–) revealed a remarkably persistent pattern: steady easterly winds in the low latitudes that reliably carried ships from Europe toward the Caribbean. These **trade winds** became the empirical foundation of atmospheric science, long before any dynamical theory existed.

Definition 1.1 | Trade Winds

The *trade winds* are the prevailing surface-level easterly winds found in the tropics, blowing from the subtropical high-pressure belts toward the equatorial low-pressure trough (the Intertropical Convergence Zone, ITCZ). 적도 저압대 쪽으로 부는 지속적인 편동풍; 대항해시대의 핵심 동력

Edmund Halley (1656–1742) offered the first physical explanation. In his 1686 paper, Halley argued that **differential solar heating** drives the trade winds: intense heating near the equator causes air to rise, drawing surface air equatorward from both hemispheres. He also produced the first known map of surface winds over the tropical oceans.

Halley의 설명은 위도에 따른 solar heating의 차이를 동력으로 제시했다는 점에서 혁명적이었지만, 왜 바람이 남북이 아닌 동서 방향을 갖는지는 설명하지 못했다 (지구 자전 효과를 고려하지 않음).



Halley's 1686 map of surface winds over the tropical oceans—the first known attempt to chart the global wind pattern from ship observations. Halley가 1686년에 발표한 열대 해양 표면풍 지도; 관측 기반 최초의 전구 바람 지도.

1.2. Hadley's Single-Cell Model

George Hadley (1685–1768) addressed the missing piece in his 1735 paper "Concerning the Cause of the General Trade-Winds." His key insight was to incorporate the *rotation of the Earth*.

Definition 1.2 | Thermally Direct Cell

A *thermally direct cell* is a meridional overturning circulation in which warm air rises in the region of greatest heating and cold air sinks in the region of greatest cooling. The sense of overturning is "consistent with" the thermal forcing. 열적으로 직접적인 순환: 따뜻한 곳에서 상승, 차가운 곳에서 하강

Hadley's reasoning proceeded as follows:

1. Strong solar heating at the equator causes air to rise.
2. At upper levels, this air flows poleward.
3. Conservation of angular momentum requires that this poleward-moving air acquire a westerly (west-to-east) component, since it originated at a larger distance from the rotation axis.

4. The return flow at the surface, moving equatorward, is deflected *westward*, producing the observed easterly trade winds.

Angular Momentum Argument

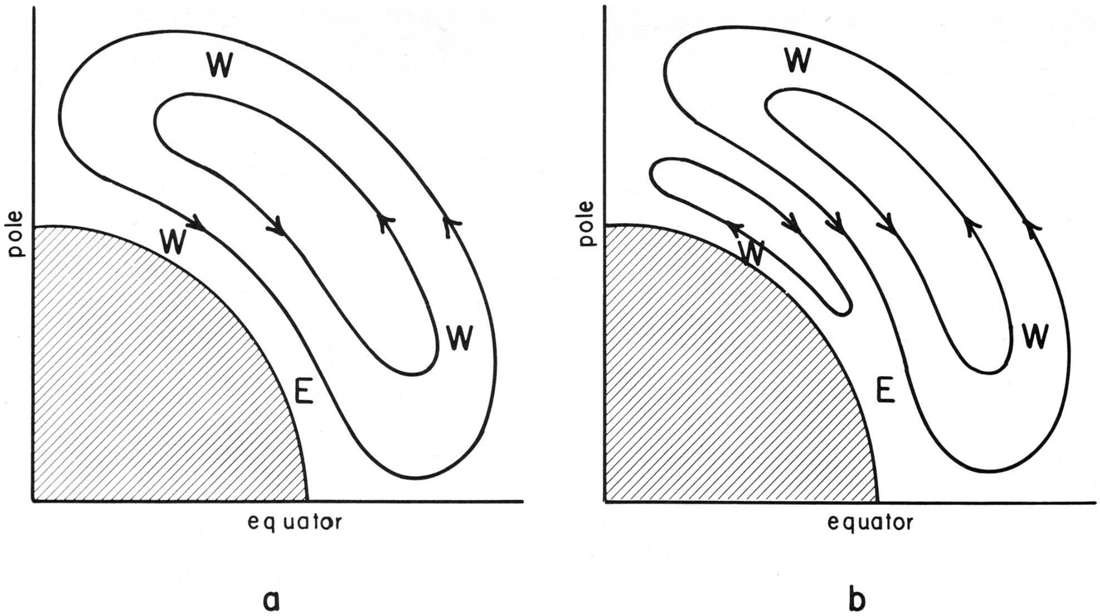
The absolute angular momentum per unit mass of a parcel on a rotating sphere is

$$M = (u + \Omega a \cos\phi)\, a \cos\phi, \tag{1.1}$$

where u is the zonal wind, Ω the planetary rotation rate, a the Earth's radius, and ϕ the latitude. If a parcel at rest at the equator ($\phi = 0$, $u = 0$) moves to latitude ϕ while conserving M, then

$$u(\phi) = \Omega a \frac{\sin^2\phi}{\cos\phi}. \tag{1.2}$$

Hadley는 적도에서 극까지 하나의 거대한 thermally direct cell을 상상했지만, 실제 Hadley circulation은 열대(위도 ∼30°)에 국한된다. 이유: 각운동량 보존에 의해 상층 서풍이 너무 강해져서 중위도에서는 baroclinic instability에 의해 순환이 깨진다 (Sec. 3.3 참조).
Ref: (Lorenz, 1983, *Bull. Amer. Meteor. Soc.*)



Schematic of Hadley's single thermally direct cell: warm air rises at the equator, flows poleward aloft, sinks at the pole, and returns equatorward at the surface. Hadley의 단일 직접순환 모식도: 적도 상승 → 상층 극향 → 극 하강 → 지표 적도향.

1.3. Thomson and Ferrel: The Three-Cell Model

The single-cell picture survived for over a century before being challenged.

James Thomson (1857) recognized that what we now call the **Coriolis force**—the apparent deflection experienced by air moving over a rotating Earth—would prevent a simple equator-to-pole overturning. He proposed a *multi-cell* structure.

William Ferrel (1856, 1859) independently developed a **three-cell model** of the meridional circulation:

- **Hadley cell** (0–30°): thermally direct
- **Ferrel cell** (30–60°): thermally *indirect*
- **Polar cell** (60–90°): thermally direct

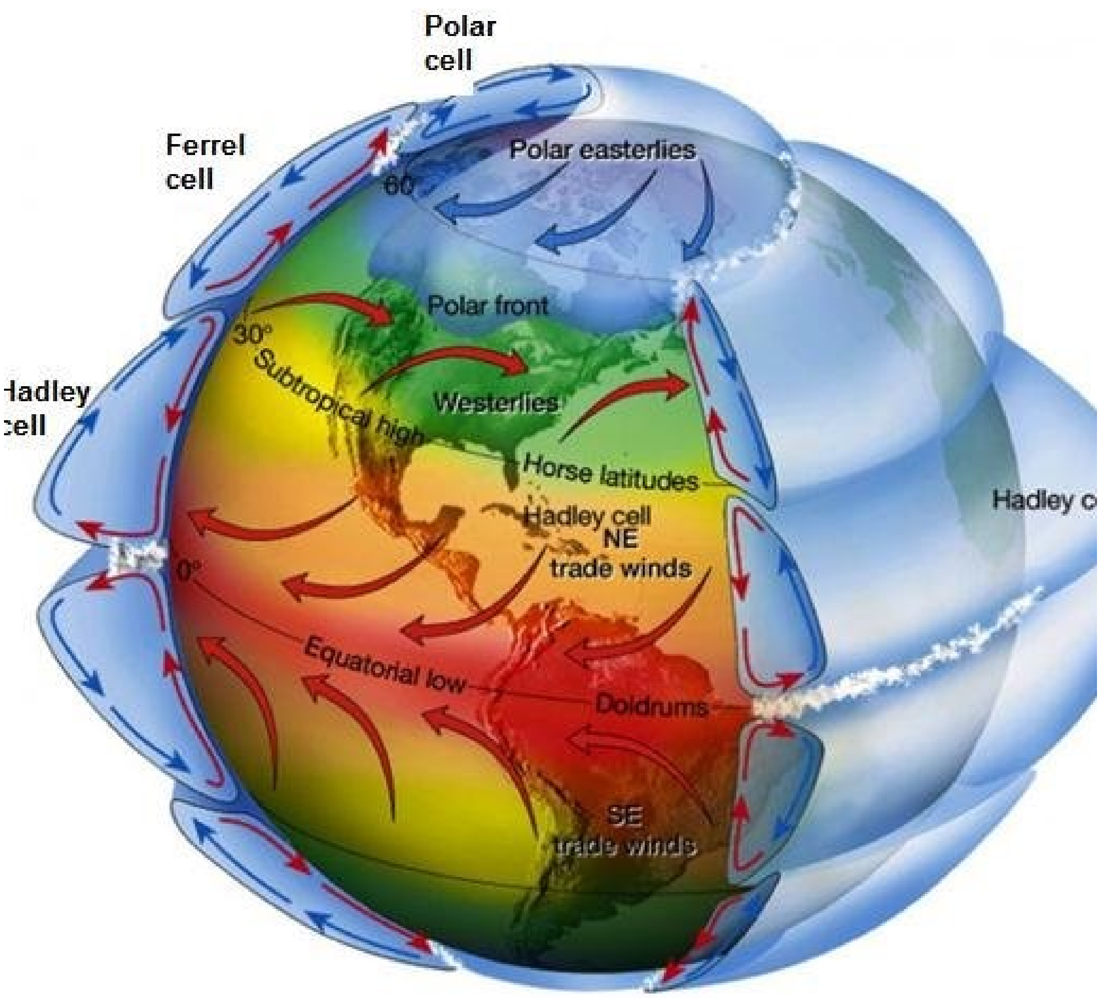
Hadley cell → 직접순환
Ferrel cell → 간접순환

Polar cell → 직접순환
3개 cell이 맞물려서 전구 순환 형성

Definition 1.3 | Thermally Indirect Cell

A *thermally indirect cell* is a meridional overturning circulation whose sense of overturning is *opposite* to that which thermal forcing alone would produce: relatively cool air rises on the poleward side, while relatively warm air sinks on the equatorward side. The Ferrel cell is the canonical example. 열적 강제와 반대 방향으로 순환하는 cell; 중위도 Ferrel cell이 대표적

Thomson은 multi-cell 구조를 제안했으나 thermally indirect cell의 개념을 명확히 구분하지 못했다. Ferrel은 Thomson과 달리 thermally indirect cell (Ferrel cell)을 극지방까지 확장하지 않았으며, 중위도에 국한된 간접순환으로 올바르게 인식했다.



The three-cell model of the meridional circulation: Hadley cell (thermally direct, 0–30°), Ferrel cell (thermally indirect, 30–60°), and Polar cell (thermally direct, 60–90°), shown on a 3-D globe view. 3-cell 모델의 3차원 모식도: Hadley, Ferrel, Polar cell의 배치와 지표풍 방향을 확인하라.

Remark 1.4
The Ferrel cell is not driven by direct thermal forcing. Rather, it is maintained by the convergence of eddy momentum fluxes associated with mid-latitude baroclinic waves. In this sense, it is a fundamentally *eddy-driven* circulation.

2. THE NATURE OF THE PROBLEM

2.1. Differential Heating

The starting point of any theory of the general circulation is the observation that the Earth's climate system is in approximate **radiative equilibrium** at the top of the atmosphere (TOA):

$$\overline{Q_{\mathrm{abs}}} = \overline{F_{\mathrm{OLR}}}, \tag{2.1}$$

where the overbar denotes a global and annual mean. The absorbed solar radiation Q_{abs} must, on average, equal the outgoing longwave (infrared) radiation F_{OLR}.

> 전지구 평균적으로 흡수되는 태양 에너지 = 방출되는 적외선 에너지. 하지만 위도별로 보면 이 균형이 깨진다 — 그것이 순환의 근본 원인.

The crucial feature is the **latitude dependence**:

- **Absorbed solar radiation** $Q_{\mathrm{abs}}(\phi)$ is *strongly* latitude-dependent, peaking near the equator. 입사 태양에너지: 위도에 강하게 의존 (적도 $\gg$ 극)
- **Outgoing longwave radiation** $F_{\mathrm{OLR}}(\phi)$ is only *weakly* latitude-dependent, because the meridional temperature gradient is much smaller than it would be in radiative equilibrium. 방출 적외선: 위도 의존성이 약함 (대기-해양 순환이 열을 수송하기 때문)

This produces a systematic **radiation surplus** at low latitudes and a **radiation deficit** at high latitudes:

$$R_{\mathrm{net}}(\phi) = Q_{\mathrm{abs}}(\phi) - F_{\mathrm{OLR}}(\phi) \begin{cases} > 0 & \text{(tropics: surplus)}, \\ < 0 & \text{(high latitudes: deficit)}. \end{cases} \tag{2.2}$$

Daily-Mean TOA Insolation

The daily-mean insolation at the top of the atmosphere depends on latitude ϕ and the solar declination angle δ_s:

$$\bar{S}(\phi) = \frac{S_0}{\pi}\left(h_0 \sin\phi \, \sin\delta_s + \cos\phi \, \cos\delta_s \, \sin h_0\right), \tag{2.3}$$

where S_0 is the solar constant and h_0 is the hour angle at sunset.

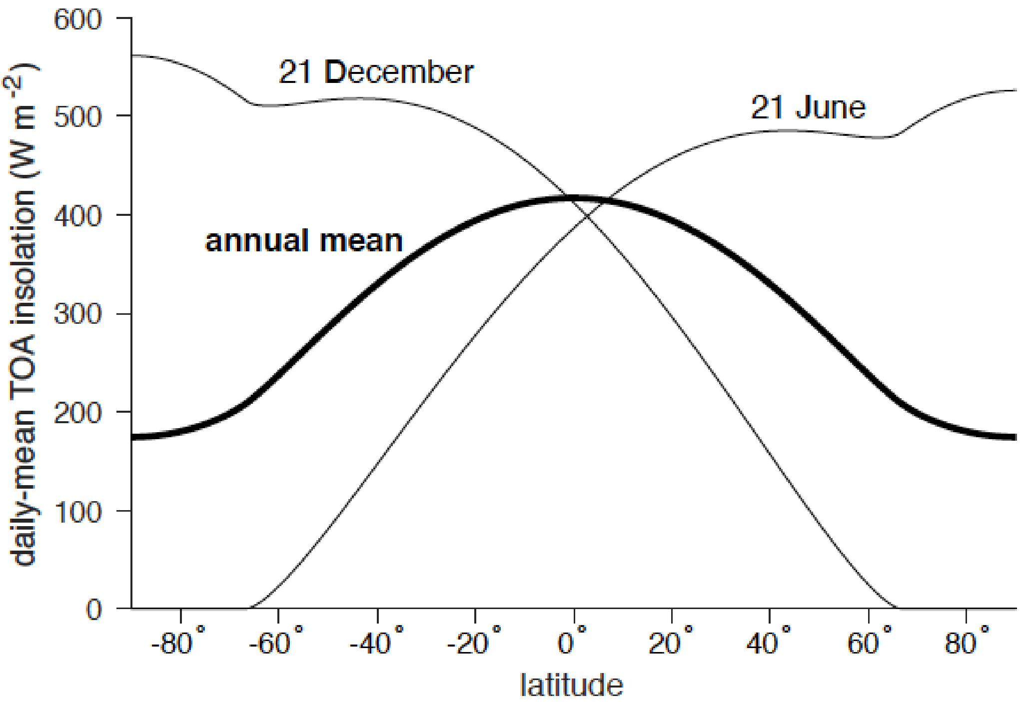
> June solstice: 일사량이 북극에서 최대 (24시간 일조 효과). December solstice: 남극에서 최대. Annual mean: 적도 부근에서 최대, 극에서 최소. 결론적으로 위도에 따른 net radiation의 불균형이 대기-해양 순환의 원동력.

2.2. The Atmosphere as a Heat Engine

The general circulation can be viewed as a **thermodynamic heat engine** that converts thermal energy into kinetic energy.

> **Definition 2.1 | Atmospheric Heat Engine**
>
> The atmosphere absorbs heat Q_A at a relatively *warm* temperature T_{warm} (primarily in the tropics, dominated by latent heat release) and rejects heat Q_L at a relatively *cool* temperature T_{cool} (primarily in the extratropics, through radiative cooling). The net mechanical work done per cycle is
>
> $$W = Q_A - Q_L. \tag{2.4}$$
>
> 열대에서 열 흡수 → 중고위도에서 열 방출 → 그 차이가 역학적 일(운동에너지)로 전환

On a thermodynamic T–s (temperature–entropy) diagram, the atmospheric cycle traces a clockwise loop. The area enclosed by this loop equals the net work output W.



Daily-mean top-of-atmosphere insolation as a function of latitude for June solstice, December solstice, and the annual mean. Note the polar maximum during the respective summer solstice due to 24-hour daylight. 하지(June)에는 북극에서 일사량 최대, 동지(Dec)에는 남극에서 최대; 연평균은 적도 최대.

The specific entropy s of dry air is related to potential temperature θ by

$$s = c_p \ln \theta + \text{const}, \tag{2.5}$$

where c_p is the specific heat at constant pressure.

Remark 2.2
The efficiency of the atmospheric heat engine is far below the Carnot limit. Typical estimates give an efficiency of only a few percent, reflecting the fact that most of the absorbed energy is re-radiated rather than converted to kinetic energy. 대기의 열효율은 Carnot 효율보다 훨씬 낮다 — 대부분의 에너지는 다시 복사로 방출됨

2.3. Available Potential Energy

Not all potential energy in the atmosphere is available for conversion to kinetic energy. This motivates the concept of **available potential energy** (APE).

Definition 2.3 | Available Potential Energy (APE)

The *available potential energy* (Lorenz, 1955) is the difference between the total potential energy of the atmosphere and the minimum potential energy that could be achieved by an adiabatic rearrangement of mass to a statically stable reference state:

$$\text{APE} = \text{PE}_{\text{actual}} - \text{PE}_{\text{ref}}. \tag{2.6}$$

Only this fraction is available for conversion to kinetic energy. 전체 위치에너지 중 운동에너지로 전환 가능한 부분만이 APE; 실제로는 매우 작은 비율 (∼0.5%)

The energy conversion chain in the atmosphere proceeds as follows:

$$\underbrace{\text{APE}}_{\text{differential heating}} \xrightarrow{\text{baroclinic processes}} \underbrace{\text{KE}}_{\text{winds}} \xrightarrow{\text{friction}} \underbrace{\text{Internal Energy}}_{\text{heat}}. \tag{2.7}$$

정상 상태(steady state)에서:

$$\text{APE} \to \text{KE 전환율} = \text{마찰에 의한 KE 소멸율}.$$

대기 운동에너지의 거의 전부는 수평 규모 $\gtrsim 100\,\text{km}$의 수평풍에 존재한다. 연직 운동의 KE 기여는 무시할 수 있을 정도로 작다.

Remark 2.4

The atmosphere maintains a quasi-steady kinetic energy budget: the rate of generation of KE from APE is balanced by the rate of frictional dissipation. Observationally, virtually all KE resides in the horizontal wind on scales larger than $\sim 100\,\text{km}$.

3. HEURISTIC MODELS

3.1. Radiative-Convective Equilibrium

Before considering the full three-dimensional circulation, it is useful to consider two idealised one-dimensional equilibria.

Definition 3.1 | Pure Radiative Equilibrium

A state in which the net emission of infrared radiation at every level equals the absorbed solar radiation at that level:

$$\frac{\partial F_{\text{net}}}{\partial z} = Q_{\text{solar}}(z) \quad \text{at each altitude } z. \tag{3.1}$$

각 고도에서 적외선 방출 = 흡수된 태양복사. 대류 없이 복사만으로 결정되는 온도 프로파일.

The pure radiative equilibrium temperature profile yields an unrealistically high surface temperature (∼330 K) and an extremely steep lapse rate in the lower troposphere — far exceeding the threshold for convective instability.

Definition 3.2 | Radiative–Convective Equilibrium (RCE)

A state in which the vertical temperature profile is determined jointly by *radiative transfer* and *convective adjustment*. Where the radiative equilibrium lapse rate exceeds a critical value (typically the moist or dry adiabatic lapse rate), convection acts to restore the lapse rate to the critical value.
복사 평형의 비현실적 온도 기울기를 대류가 조정 → radiative-convective equilibrium

순수 복사 평형(pure radiative equilibrium)은 지표면 온도가 비현실적으로 높고, 하부 대류권의 기온 감률이 대류 불안정 임계값을 크게 초과한다. 대류 조정(convective adjustment)이 필요하며, 이것이 RCE의 핵심 아이디어이다.
Ref: (Manabe and Strickler, 1964)

3.2. The Cycling of Mechanical Energy

A useful thought experiment illustrates how differential heating generates mechanical energy.

Two-Fluid Thought Experiment

Consider a container divided by a vertical partition into two halves, each filled with an *immiscible fluid* of different density ($\rho_1 > \rho_2$).

1. **Initial state.** Dense fluid on the left, light fluid on the right. The centre of mass is at some height z_{cm}.
2. **Partition removed.** The dense fluid slides under the light fluid. An overturning circulation develops.
3. **Final state.** Dense fluid on the bottom, light fluid on top. The centre of mass has dropped: $\Delta z_{\text{cm}} < 0$.

The potential energy released equals the kinetic energy generated (which is then dissipated by friction):

$$\Delta\text{PE} = -\int \rho\, g\, \Delta z_{\text{cm}}\, dV \;\longrightarrow\; \text{KE} \;\longrightarrow\; \text{Internal Energy (friction)}. \tag{3.2}$$

이것이 바로 Hadley circulation의 아날로그: 적도에서 아래쪽이 가열되고 (가벼운 유체 생성), 극에서 상부가 냉각됨 (무거운 유체 생성). → 밀도차에 의한 overturning → PE가 KE로 전환.

Maintenance Requirements

In the absence of continued forcing, the overturning would cease once the stratification reaches a stable equilibrium (dense below, light above). **Sustained circulation requires both horizontal**

and vertical gradients of diabatic heating:

- **Horizontal gradient**: equator-to-pole differential heating creates the meridional density (temperature) gradient.
- **Vertical gradient**: heating at low levels and cooling at upper levels continuously destabilises the stratification, maintaining the overturning.

수평 경도 + 연직 경도 둘 다 있어야 순환이 유지된다. 하나만으로는 정상 상태 순환 불가능.

3.3. The Influence of Planetary Rotation

Timescales and the Dominance of Rotation

The radiative relaxation timescale of the troposphere is

$$\tau_{\text{rad}} \sim 30 \text{ days}, \tag{3.3}$$

which is much longer than the rotation period (∼1 day). This means that the Coriolis force has ample time to influence the developing circulation, and the tidal (diurnal) response is secondary.

> 복사 이완 시간(∼30일) ≫ 자전 주기(∼1일) → Coriolis force가 대기 순환의 구조를 근본적으로 결정한다. (만약 자전이 없다면 단순한 적도–극 순환이 될 것)

Spin-Up Thought Experiment on an Aquaplanet

The influence of rotation can be understood through a conceptual spin-up experiment (Held, 2019, Ch. 6, *Meteorological Monographs*):

(a) **Non-rotating planet.** Differential heating drives a simple, hemispheric-scale, equator-to-pole thermally direct overturning (one Hadley cell per hemisphere spanning the entire latitude range). 자전 없음 → 단순한 적도-극 직접 순환 (Hadley의 원래 구상)

(b) **Introduce rotation → pressure redistribution.** As flow develops, the Coriolis force deflects poleward upper-level flow to the east and equatorward surface flow to the west. Pressure gradients are redistributed by the evolving wind field. 자전 도입 → Coriolis 편향 → 기압장 재배치

(c) **Geostrophic adjustment → zonal flow.** The flow adjusts toward *geostrophic balance*: the Coriolis force balances the pressure gradient force. The circulation becomes predominantly *zonal* (east–west), governed by the **thermal wind relation**:

$$f \frac{\partial u_g}{\partial z} = -\frac{g}{T_0}\frac{\partial T}{\partial y}, \tag{3.4}$$

where $f = 2\Omega \sin\phi$ is the Coriolis parameter, u_g the geostrophic zonal wind, and $\partial T/\partial y$ the meridional temperature gradient. 열풍(thermal wind): 남북 온도 경도가 있으면 상층으로 갈수록 서풍 증가

(d) **Baroclinic instability.** As the meridional temperature gradient steepens under continued differential heating, the jet stream intensifies until it reaches a critical shear. At that point the zonally symmetric state becomes unstable to wave-like perturbations.

Definition 3.3 | Baroclinic Instability

Baroclinic instability is the hydrodynamic instability that arises when the vertical shear of the zonal wind (equivalently, the meridional temperature gradient via thermal wind balance) exceeds a critical threshold. It is the primary mechanism by which the zonally symmetric Hadley circulation breaks down in the extratropics.

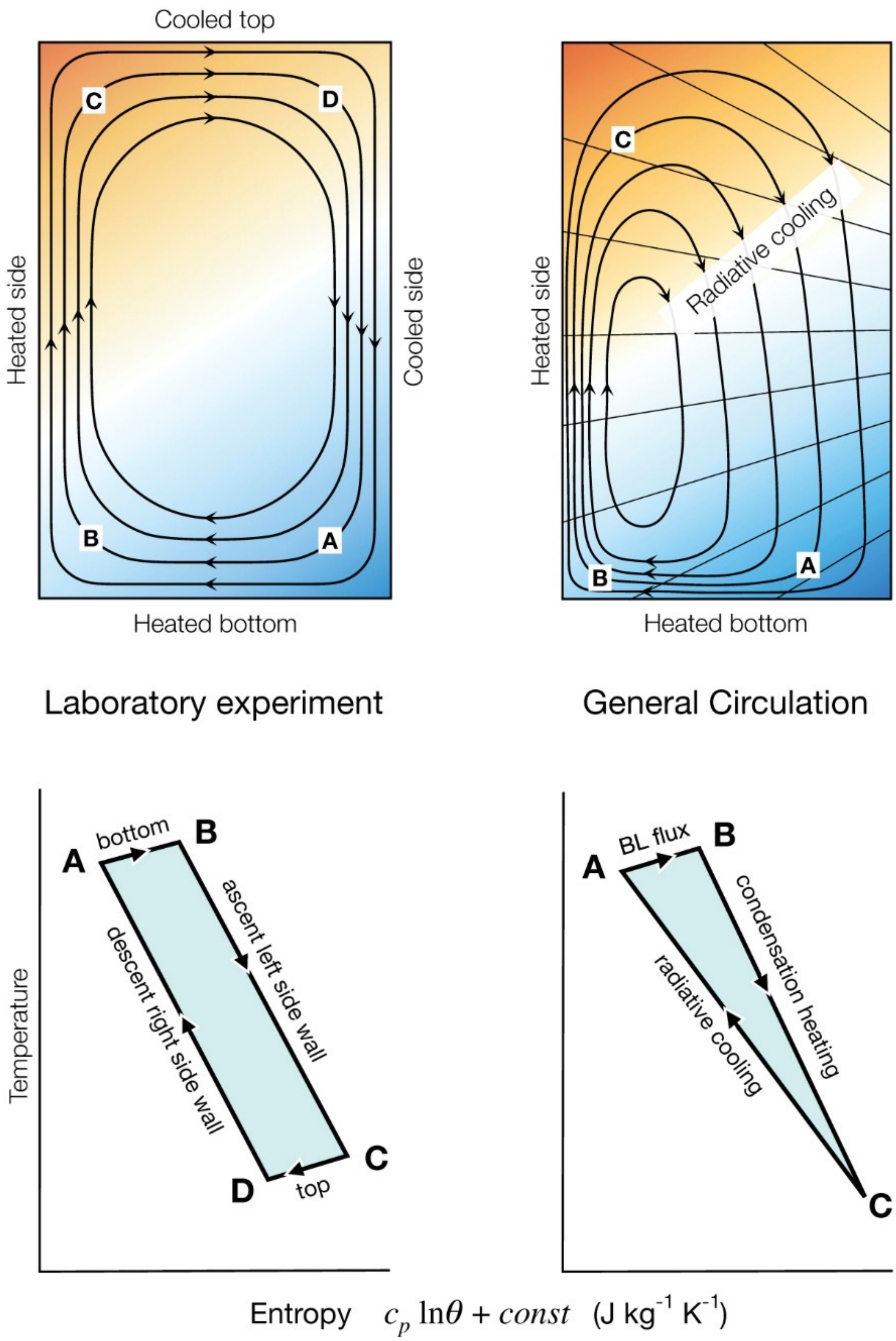
- Fastest-growing mode: synoptic-scale waves with wavelength ∼3500–4000 km.
- These waves (mid-latitude cyclones and anticyclones) accomplish most of the poleward heat and

momentum transport in the extratropics.
경압 불안정: Hadley cell의 한계를 넘어선 위도에서 대칭적 순환이 파동으로 깨짐. 가장 빨리 성장하는 파장 $\sim 3500\,\mathrm{km}$ → 중위도 저기압/고기압의 규모와 일치

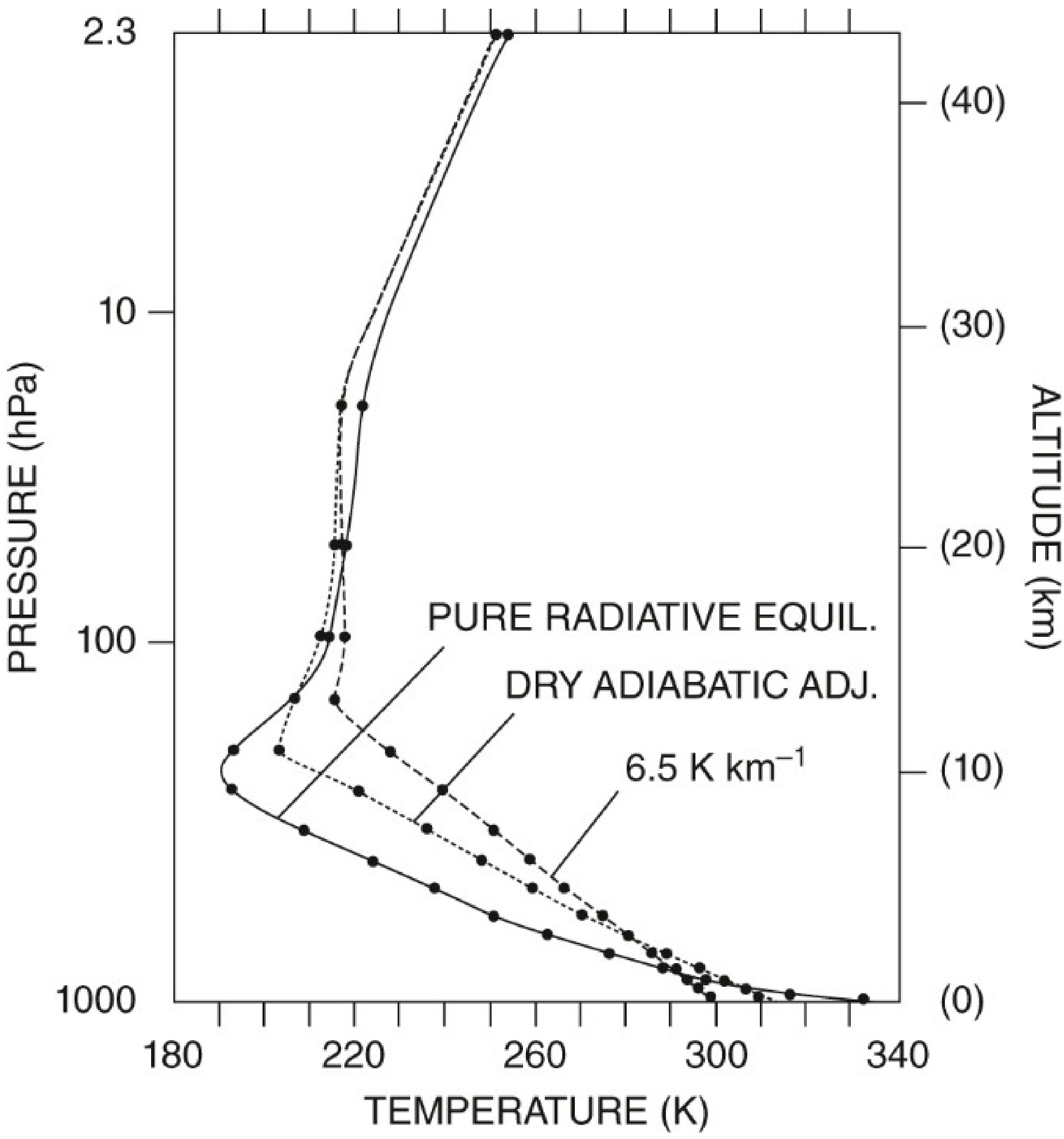
정리하면, 전구 대기 순환의 핵심 문제는 다음과 같다:

1. 위도별 복사 불균형이 적도–극 온도 경도를 만든다.
2. 대기는 열기관으로서 이 온도 경도를 이용해 운동에너지를 생성한다.
3. 지구 자전으로 인해 단순한 직접 순환은 열대에 국한되고 (Hadley cell), 중위도에서는 경압 불안정이 지배한다.
4. 최종 결과: three-cell structure + synoptic-scale wave activity.

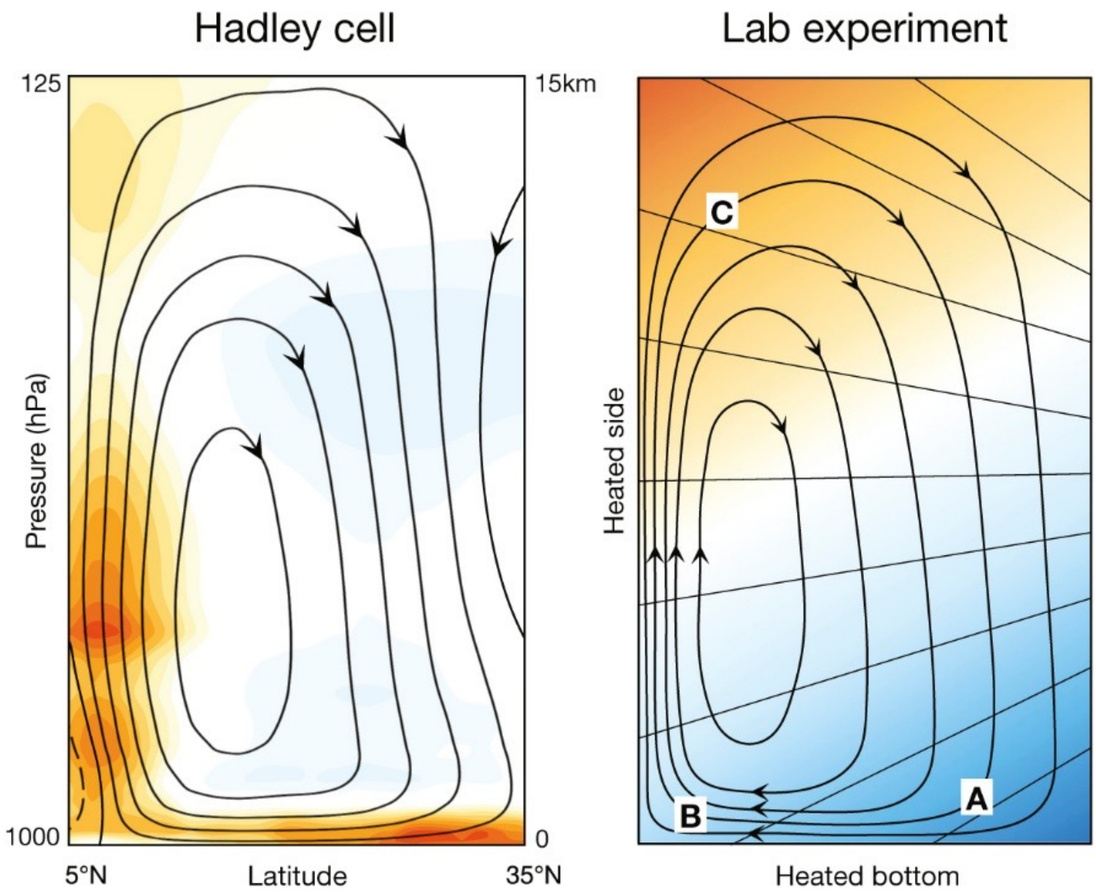
Ref: (Held, 2019, Ch. 6, *Meteorological Monographs*)



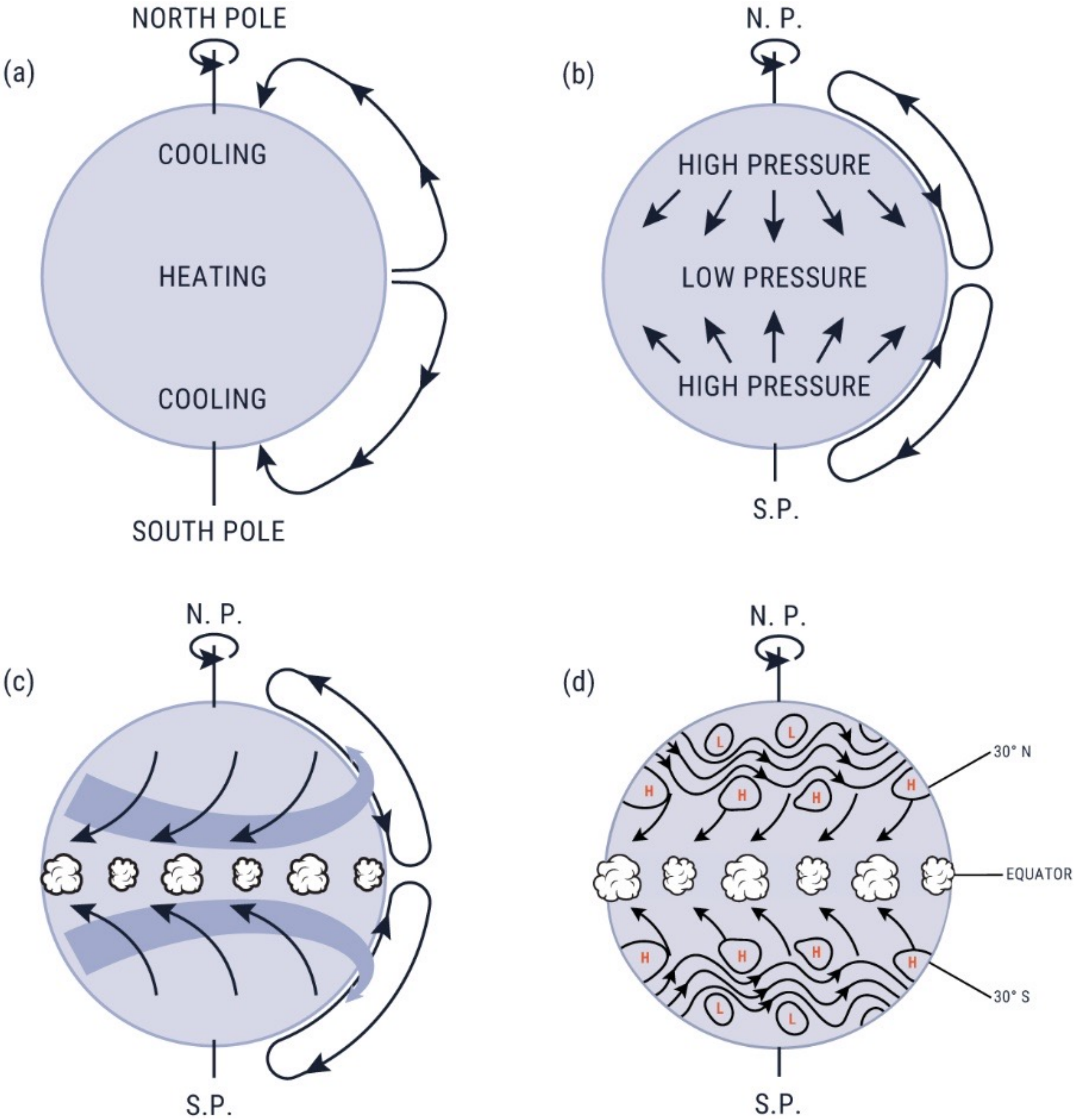
T–s diagram of the general circulation: the clockwise loop encloses an area equal to the net work W produced by the atmospheric heat engine, shown alongside the analogous laboratory experiment. 대기 열기관의 T–s 다이어그램; 시계방향 루프의 면적 = 순 역학적 일 W.



Vertical temperature profiles from Manabe & Strickler (1964): pure radiative equilibrium (dashed) versus radiative–convective equilibrium (solid). The radiative-only profile produces an unrealistically steep lapse rate near the surface; convective adjustment restores a more realistic profile. 순수 복사 평형(dashed)은 지표 근처 기온감률이 비현실적으로 급함; 대류 조정 후(solid) 현실적 프로파일로 변함.



Laboratory experiment: steady-state thermally driven overturning circulation in a non-rotating annulus heated from below on one side and cooled from above on the other. 비회전 실험 수조에서의 열적 직접순환; 한쪽 가열 + 반대쪽 냉각 → 정상상태 overturning 발생.



Four-panel spin-up experiment on an aquaplanet: (a) non-rotating thermally direct cell, (b) introduction of Coriolis deflection, (c) geostrophic adjustment producing strong zonal flow, (d) onset of baroclinic instability breaking the zonal symmetry. (a) 비회전 직접순환 → (b) Coriolis 편향 → (c) 지균조정 → (d) 경압 불안정 발생; 순환 구조가 단계적으로 복잡해짐.

DECOMPOSITION OF THE GENERAL CIRCULATION

4. DECOMPOSITION OF THE GENERAL CIRCULATION

4.1. Time Average (Reynolds Decomposition)

Definition 4.1 | Time Average

For any atmospheric variable $u = u(\mathbf{x}, t)$, the **time average** over a period τ is

$$\overline{u} \;=\; \frac{1}{\tau}\int_0^{\tau} u(t)\,dt \tag{4.1}$$

where τ can be hour, day, month, season, year, etc.

Definition 4.2 | Reynolds Decomposition

Any variable u is decomposed into its time-mean and its deviation:

$$u \;=\; \overline{u} \;+\; u' \tag{4.2}$$

- u': transient (perturbation) component
- $\overline{u}$: stationary (time-mean) component

정상 성분(시간 평균)과 변동 성분(편차)으로 분리

Properties of the Time Average

Proposition 4.3 | Properties of the time average operator

For any variables u, v and constants a, b:

1. **Linearity:** $\overline{au + bv} = a\,\overline{u} + b\,\overline{v}$
2. **Idempotence:** $\overline{\overline{u}} = \overline{u}$: 이미 평균을 낸 것에 다시 평균을 내도 같다
3. **Mean of perturbation:**
$$\overline{u'} = 0 \tag{4.3}$$
평균에서의 편차를 다시 평균하면 0
4. Commutes with differentiation: $\overline{\partial u/\partial x} = \partial\overline{u}/\partial x$

Proof Property 3 follows directly from the decomposition:

$$\overline{u'} = \overline{u - \overline{u}} = \overline{u} - \overline{\overline{u}} = \overline{u} - \overline{u} = 0. \qquad \blacksquare$$

Product Rule

Proposition 4.4 | Reynolds Average of a Product

For any variables u and v:

$$\overline{uv} \;=\; \overline{u}\,\overline{v} \;+\; \overline{u'v'} \tag{4.4}$$

Proof Substitute $u = \overline{u} + u'$ and $v = \overline{v} + v'$:

$$\begin{aligned}
\overline{uv} &= \overline{(\overline{u} + u')(\overline{v} + v')} \\
&= \overline{\overline{u}\,\overline{v} + \overline{u}\,v' + u'\,\overline{v} + u'v'} \\
&= \overline{u}\,\overline{v} + \overline{u}\,\underbrace{\overline{v'}}_{=0} + \underbrace{\overline{u'}}_{=0}\,\overline{v} + \overline{u'v'} \\
&= \overline{u}\,\overline{v} + \overline{u'v'}.
\end{aligned} \qquad \blacksquare$$

Remark 4.5

The term $\overline{u'v'}$ is the **covariance** of u and v.

공분산 = 상관 관계를 나타냄. 표준편차로 나누면 상관계수(correlation coefficient): $r_{uv} = \overline{u'v'}/(\sigma_u \sigma_v)$

4.2. Space Average (Zonal Decomposition)

Definition 4.6 | Zonal Average

The **zonal average** (average along a latitude circle) is

$$[u] \;=\; \frac{1}{2\pi}\int_0^{2\pi} u\,d\lambda \tag{4.5}$$

where λ is the longitude.

Definition 4.7 | Eddy (Wave) Component

The departure from the zonal mean is the **eddy** or **wave** component:

$$u^* \;=\; u - [u] \tag{4.6}$$

eddy = 동서 평균에서의 편차, 즉 파동 성분

Properties of the Zonal Average

Proposition 4.8 | Properties of the zonal average operator

Analogous to the time average:

1. $[au + bv] = a[u] + b[v]$: linearity
2. $[[u]] = [u]$: idempotence
3. **Eddy mean vanishes:**
$$[u^*] = 0 \tag{4.7}$$
4. Commutes with $\partial/\partial t$, $\partial/\partial\phi$, $\partial/\partial p$ (but not $\partial/\partial\lambda$)

Proposition 4.9 | Zonal Average of a Product

For any variables u and v:
$$[uv] \;=\; [u]\,[v] \;+\; [u^*v^*] \tag{4.8}$$

Proof Substitute $u = [u] + u^*$, $v = [v] + v^*$:

$$\begin{aligned}[uv] &= [([u] + u^*)([v] + v^*)] \\ &= [u]\,[v] + [u]\,\underbrace{[v^*]}_{=0} + \underbrace{[u^*]}_{=0}\,[v] + [u^*v^*] \\ &= [u]\,[v] + [u^*v^*].\end{aligned}$$

■

Remark 4.10

The structure is identical to the Reynolds decomposition. The zonal mean operator $[\cdot]$ satisfies the same algebraic properties as the time mean $\overline{(\cdot)}$, so all product rules carry over by substitution.

4.3. Mixed (Combined) Decomposition

Applying both the time average and the zonal average yields a **four-component decomposition**. First apply the time average, then the zonal average:

Definition 4.11 | Four-Component Decomposition

$$\boxed{u = \underbrace{[\overline{u}]}_{\text{(i)}} + \underbrace{\overline{u}^*}_{\text{(ii)}} + \underbrace{[u']}_{\text{(iii)}} + \underbrace{u'^*}_{\text{(iv)}}} \tag{4.9}$$

시간 평균 → 동서 평균을 순서대로 적용해서 4개의 성분으로 분해한다.

Each component has a distinct physical meaning:

Component	Notation	Alt. notation	Physical meaning
(i)	$[\overline{u}]$	$\overline{u}_Z$	Mean Meridional Circulation (MMC)
(ii)	$\overline{u}^*$	$\overline{u}_E$	Stationary wave (eddy)
(iii)	$[u']$	u'_Z	Transient zonally symmetric
(iv)	u'^*	u'_E	Transient wave (eddy)

- MMC: 시간, 경도 모두에서 평균적인 흐름 (예: Hadley cell)
- Stationary eddy: 시간 평균에서는 존재하지만 경도 방향으로 파동인 성분 (예: 지형파)
- Transient zonally symmetric: 시간에 따라 변하지만 경도 방향으로 균일한 성분
- Transient eddy: 시간에도 변하고 경도 방향으로도 파동인 성분 (예: 이동성 저기압)

Remark 4.12

Subscript notation is often used for compactness:

- $(\cdot)_Z \equiv [\cdot]$: zonal-mean part
- $(\cdot)_E \equiv (\cdot)^*$: eddy part

so that $u = \overline{u}_Z + \overline{u}_E + u'_Z + u'_E$.

4.4. Flux and Transport Decomposition

General Flux Decomposition

Consider the meridional transport of a scalar quantity X by the meridional wind v. Using the four-component decomposition, the time-and-zonal mean of the flux vX can be written as:

Proposition 4.13 | Flux Decomposition

$$\overline{(vX)}_Z = \underbrace{\overline{v}_Z\,\overline{X}_Z}_{\text{MMC}} + \underbrace{\overline{v'_Z X'_Z}}_{\substack{\text{transient}\\\text{meridional}}} + \underbrace{(\overline{v}_E\,\overline{X}_E)_Z}_{\substack{\text{stationary}\\\text{eddy}}} + \underbrace{\overline{(v'_E X'_E)}_Z}_{\substack{\text{transient}\\\text{eddy}}} \tag{4.10}$$

Proof Write $v = \overline{v} + v'$ and $X = \overline{X} + X'$:

$$vX = (\overline{v} + v')(\overline{X} + X') = \overline{v}\,\overline{X} + \overline{v}\,X' + v'\,\overline{X} + v'X'.$$

Taking the time average:

$$\overline{vX} = \overline{v}\,\overline{X} + \overline{v'X'}.$$

Now decompose $\overline{v} = \overline{v}_Z + \overline{v}_E$ and $\overline{X} = \overline{X}_Z + \overline{X}_E$, and similarly for the primed quantities. Taking the zonal average of each term:

$$\begin{aligned}\overline{(vX)}_Z &= [\overline{vX}] = [\overline{v}\,\overline{X}] + [\overline{v'X'}]\\ &= \overline{v}_Z\,\overline{X}_Z + (\overline{v}_E\,\overline{X}_E)_Z + \overline{v'_Z X'_Z} + \overline{(v'_E X'_E)}_Z.\end{aligned}$$ ■

총 수송 = MMC에 의한 수송 + 경도 방향 transient + 정상 에디 + 이동성 에디

Example 1: Angular Momentum Transport

Definition 4.14 | Angular Momentum per unit mass

The angular momentum per unit mass about Earth's axis is

$$M = (\Omega a\cos\phi + u)\,a\cos\phi \tag{4.11}$$

- Ω: Earth's angular velocity
- a: Earth's radius
- ϕ: latitude
- u: zonal wind

$\Omega a\cos\phi$: 지구 자전에 의한 기여, u: 대기 운동에 의한 기여

The northward transport of angular momentum, integrated over all longitudes and through a pressure layer from p_1 to p_2, is

$$M_\phi = \frac{2\pi a^2\cos^2\phi}{g}\int_{p_1}^{p_2}\Big[\underbrace{\Omega a\cos\phi\,\overline{v}_Z}_{\Omega\text{-term}} + \underbrace{\overline{(uv)}_Z}_{\text{eddy term}}\Big]\,dp \tag{4.12}$$

- Ω-term: 지구 자전 각운동량의 남북 수송 → $\overline{v}_Z$에 의해 결정
- eddy term: 대기 운동 각운동량의 남북 수송 → $\overline{(uv)}_Z$에 의해 결정

Continuity Constraint on Meridional Wind

The continuity equation in pressure coordinates, integrated zonally and over the full pressure depth, constrains the meridional transport.

Proposition 4.15 | Mass balance constraint

$$\int_0^{p_0} \overline{v}_Z \, dp \;=\; 0 \tag{4.13}$$

Proof The continuity equation in (y, p) coordinates gives

$$\frac{\partial \overline{v}_Z}{\partial y} + \frac{\partial \overline{\omega}_Z}{\partial p} = 0.$$

Integrating over all pressure levels from 0 to p_0 (surface):

$$\frac{\partial}{\partial y} \int_0^{p_0} \overline{v}_Z \, dp + \left[\overline{\omega}_Z\right]_0^{p_0} = 0.$$

Since $\overline{\omega}_Z = 0$ at both $p = 0$ (top) and $p = p_0$ (surface), the second term vanishes, hence

$$\int_0^{p_0} \overline{v}_Z \, dp = \text{const}.$$

Since $\overline{v}_Z \to 0$ at the poles (by symmetry), the constant must be zero. ∎

공기 질량은 축적되지 않는다 — no accumulation of mass. 따라서 연직 적분한 $\overline{v}_Z = 0$: 어떤 위도에서든 남쪽으로 가는 공기와 북쪽으로 가는 공기의 질량이 같다.

Momentum Transport Decomposition

Applying the four-component flux decomposition to uv:

$$\overline{(uv)}_Z \;=\; \underbrace{\overline{u}_Z\,\overline{v}_Z}_{\approx 0} + \underbrace{\overline{u'_Z v'_Z}}_{\substack{\text{transient}\\\text{meridional}}} + \underbrace{(\overline{u}_E\,\overline{v}_E)_Z}_{\substack{\text{stationary}\\\text{eddy}}} + \underbrace{\overline{(u'_E v'_E)}_Z}_{\substack{\text{transient}\\\text{eddy}}} \tag{4.14}$$

Remark 4.16

Key observation: The term $\overline{u}_Z\,\overline{v}_Z \approx 0$ because the mass-balance constraint (4.13) forces $\overline{v}_Z$ to change sign in the vertical; the product with $\overline{u}_Z$ largely cancels upon integration.

$\overline{u}_Z\overline{v}_Z$는 연직 적분하면 거의 0에 가깝다!

Remark 4.17

The remaining terms have distinct latitudinal dominance:

- **Low latitudes:** The first pair ($\overline{u}_Z\overline{v}_Z$ and $\overline{u'_Z v'_Z}$) dominates. This is associated with the **Hadley circulation** (MMC).
- **Mid–high latitudes:** The second pair ($(\overline{u}_E\overline{v}_E)_Z$ and $\overline{(u'_E v'_E)}_Z$) dominates. This is associated with **eddies** (baroclinic waves).

저위도: Hadley cell이 운동량 수송 주도 / 중 고위도: 에디(파동)가 주도

Example 2: Heat (Enthalpy) Transport

The same decomposition applies to the meridional heat transport $\overline{(vT)}_Z$:

$$\overline{(vT)}_Z \;=\; \underbrace{\overline{v}_Z\,\overline{T}_Z}_{\text{MMC}} + \underbrace{\overline{v'_Z T'_Z}}_{\substack{\text{transient}\\\text{meridional}}} + \underbrace{(\overline{v}_E\,\overline{T}_E)_Z}_{\substack{\text{stationary}\\\text{eddy}}} + \underbrace{\overline{(v'_E T'_E)}_Z}_{\substack{\text{transient}\\\text{eddy}}} \tag{4.15}$$

Remark 4.18 | Heat vs. Momentum Transport

There is a fundamental asymmetry between heat and momentum transport:
- **Heat transport** is **down-gradient**: heat flows from warm (equator) to cold (poles), reducing the equator-to-pole temperature gradient.
- **Momentum transport** is **up-gradient**: eddies transport westerly momentum *toward* the jet stream (mid-latitudes), intensifying the existing gradient.

열 수송은 경도(gradient)를 따라 (따뜻한 곳 → 차가운 곳), 운동량 수송은 경도를 거슬러 (제트 기류 쪽으로 집중)!
- 열: equator → pole, gradient를 약화
- 운동량: mid-lat jet를 강화, gradient를 증가

Variance Decomposition

Setting $X = v$ (i.e., examining the flux of v by itself) gives the **variance decomposition**:

Proposition 4.19 | Variance Decomposition

$$\overline{(v^2)}_Z = (\overline{v}_Z)^2 + \overline{(v'_Z)^2} + (\overline{v}_E^2)_Z + \overline{(v_E'^2)}_Z \tag{4.16}$$

분산(variance)도 같은 방식으로 분해된다. 각 항은 각 성분이 총 변동에 기여하는 정도를 나타낸다:
- $(\overline{v}_Z)^2$: MMC에 의한 분산
- $\overline{(v'_Z)^2}$: 경도 방향 transient에 의한 분산
- $(\overline{v}_E^2)_Z$: 정상 에디에 의한 분산
- $\overline{(v_E'^2)}_Z$: 이동성 에디에 의한 분산

Remark 4.20

Unlike the flux decomposition where cross-terms can be nonzero, the variance decomposition has the clean property that each term is non-negative. Therefore the total variance is an exact sum of contributions from each component. 분산 항은 전부 ≥ 0이므로 각 성분의 기여를 정량적으로 비교 가능

OBSERVED ATMOSPHERIC CIRCULATION

5. ANNUAL MEAN CIRCULATION

The observed atmospheric circulation provides the empirical foundation upon which all theoretical understanding of climate dynamics is built. In this chapter we survey the three-dimensional structure of wind, pressure, temperature, precipitation, and SST fields from reanalysis data and satellite observations, emphasising both the **zonally symmetric** (mean) and **eddy** components.

관측 자료는 주로 ERA5/NCEP reanalysis와 위성 관측에서 가져온 것이다.

5.1. Wind Fields

850 hPa – Lower Troposphere

Remark 5.1 | Key features at 850 hPa

- **Trade winds**: north-easterly in the NH, south-easterly in the SH, driven by the pressure gradient between subtropical highs and the equatorial trough.
- **Midlatitude westerlies**: especially strong and zonally symmetric in the SH ($\sim$15–21 m s^{-1}), reflecting the absence of large continents.
- **Subtropical high belt**: weak wind speeds near 30° N/S where the Hadley cell descends.

850 hPa 바람장에서 가장 눈에 띄는 것은 남반구 편서풍의 동서 대칭성이다. 대륙이 적어 지형 강제가 약하기 때문이다. 반면 북반구는 대륙과 해양의 비대칭 때문에 파동 구조가 뚜렷하다.

500 hPa – Mid-Troposphere

Remark 5.2 | Key features at 500 hPa

- Predominantly **westerly flow** at all latitudes poleward of $\sim$15°.
- SH: nearly zonally symmetric, with wind speeds of 20–30 m s^{-1}.
- NH: two distinct jet maxima —
  1. East Asia ($\sim$30°N, 130°E)
  2. Eastern North America ($\sim$40°N, 70°W)

  reflecting stationary wave forcing by the Himalaya–Tibet and Rocky Mountain topography.

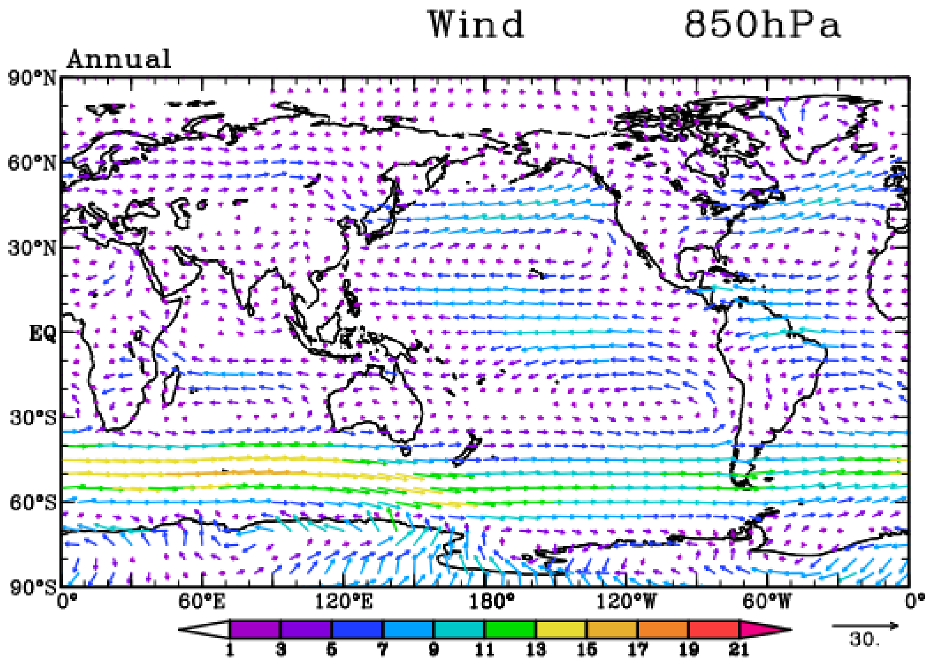
500 hPa는 대류권 중간으로, 지형에 의한 정상파(stationary wave)의 영향을 잘 보여준다.

200 hPa – Upper Troposphere

Remark 5.3 | Key features at 200 hPa

- **Subtropical jet streams** dominate the upper troposphere near 30° N/S.
- East Asian jet: strongest on Earth in the annual mean, exceeding **40 m s^{-1}**, anchored by the Tibetan Plateau.
- Atlantic jet: secondary maximum over the eastern seaboard of North America.
- SH: the subtropical jet is more continuous (nearly circumpolar) but the amplitude of individual maxima is weaker than in the NH.

200 hPa에서의 아열대 제트는 열대와 중위도 사이의 온도 경도에 의해 thermal wind 관계를 통해 설명된다 (Section 5.4). 동아시아 제트가 특별히 강한 이유는 티벳 고원에 의한 diabatic heating과 정상파 강제 때문이다.

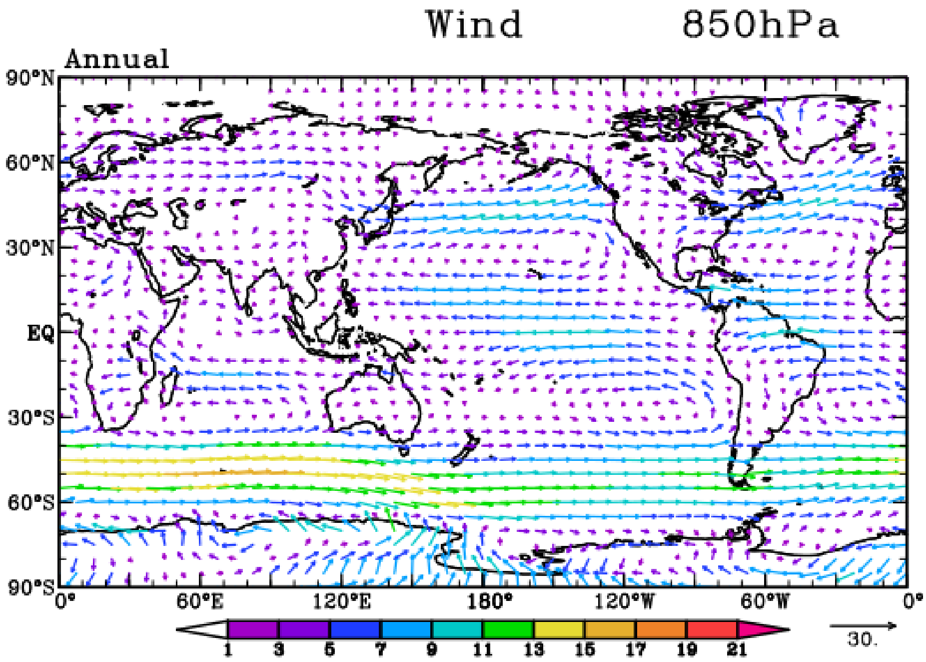


Annual-mean wind field at 850 hPa (vectors and isotachs, $m\,s^{-1}$).

5.2. Sea-Level Pressure

Definition 5.4 | Major Pressure Centres

The annual-mean SLP field is organised into a small number of recurring centres:



Annual-mean wind field at 500 hPa.

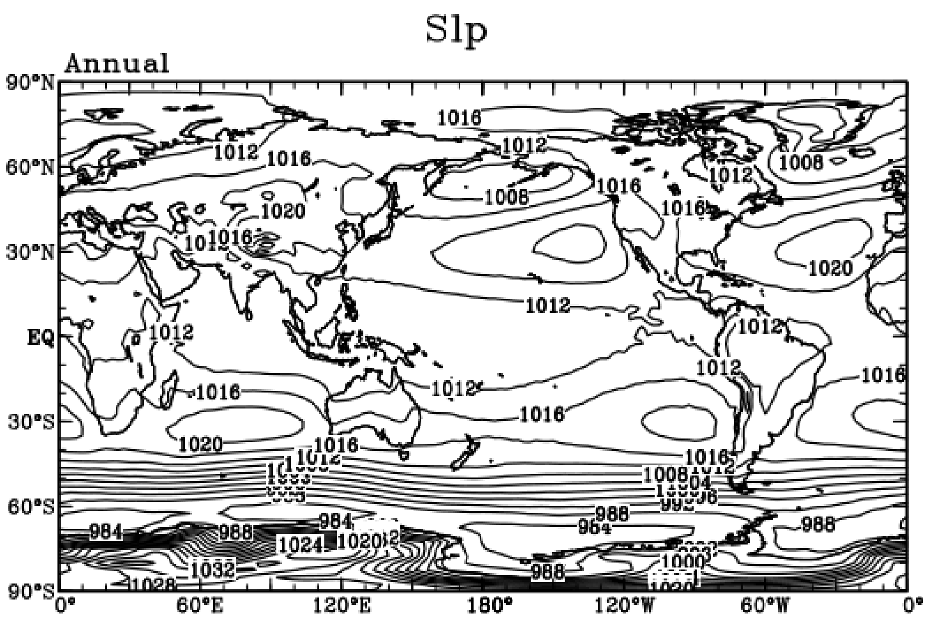
Subtropical Highs (∼1016–1020 hPa)
Located over each ocean basin near 30° N/S:

- North Pacific High
- North Atlantic (Azores/Bermuda) High
- South Pacific High
- South Atlantic High
- South Indian Ocean High

Subpolar Lows

- Aleutian Low (∼50°N, 170°W)
- Icelandic Low (∼62°N, 20°W)
- SH circumpolar trough (∼65°S, ∼984–988 hPa)

Equatorial Trough
∼1008–1012 hPa, slightly north of the equator in the annual mean.



Annual-mean wind field at 200 hPa, highlighting the subtropical jet streams.

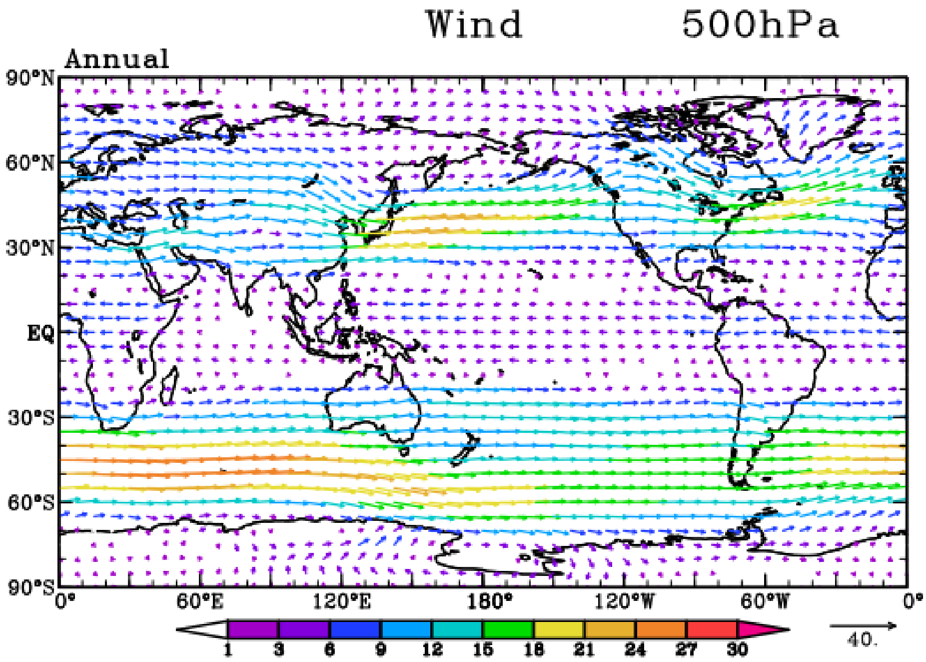
NH는 대륙–해양 대비가 크기 때문에 기압 분포가 비대칭적이다. SH는 대륙이 적어 기압 패턴이 거의 동서 대칭이며, 특히 65°S 부근의 순환극 저기압 골(circumpolar trough)은 매우 깊다.

5.3. Zonal Mean vs. Eddy Decomposition

Any atmospheric field $A(\lambda, \phi, p, t)$ can be decomposed as

$$A = [A] + A^*, \tag{5.1}$$

where $[A]$ is the zonal mean and $A^* \equiv A - [A]$ is the **eddy** (departure from the zonal mean).



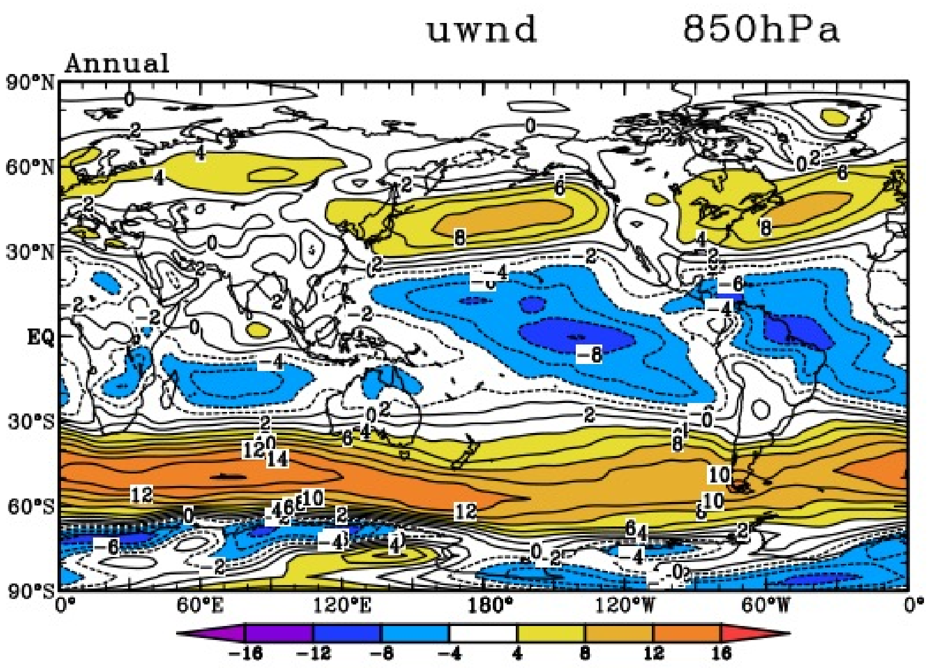
Annual-mean sea-level pressure (hPa). Contour interval: 4 hPa.

Zonal-Mean Zonal Wind

Remark 5.5 | Zonal-mean zonal wind structure

- Two **westerly jet maxima** at ~200 hPa near 30°N and 30°S, with peak speeds of ~30 m s^{-1} in the annual mean.
- Weak tropical easterlies throughout the troposphere.
- Surface easterlies (trades) equatorward of ~30° and surface westerlies poleward of ~30°.

Zonal-Mean Meridional Wind



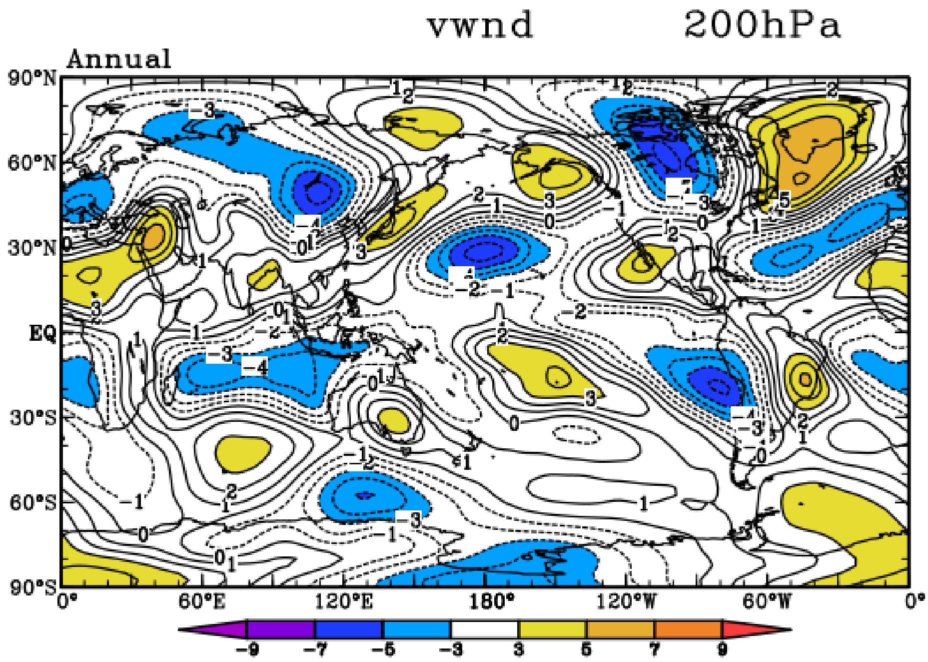
Annual-mean $[\bar{u}]$ ($\mathrm{m\,s^{-1}}$), latitude–pressure cross-section.

$[\bar{v}]$는 $[\bar{u}]$에 비해 한 자릿수 이상 작지만, 해들리 셀을 통한 질량 수송을 반영하는 중요한 양이다.

Geopotential Height: Full Field and Eddy Field

Remark 5.6 | Eddy geopotential height

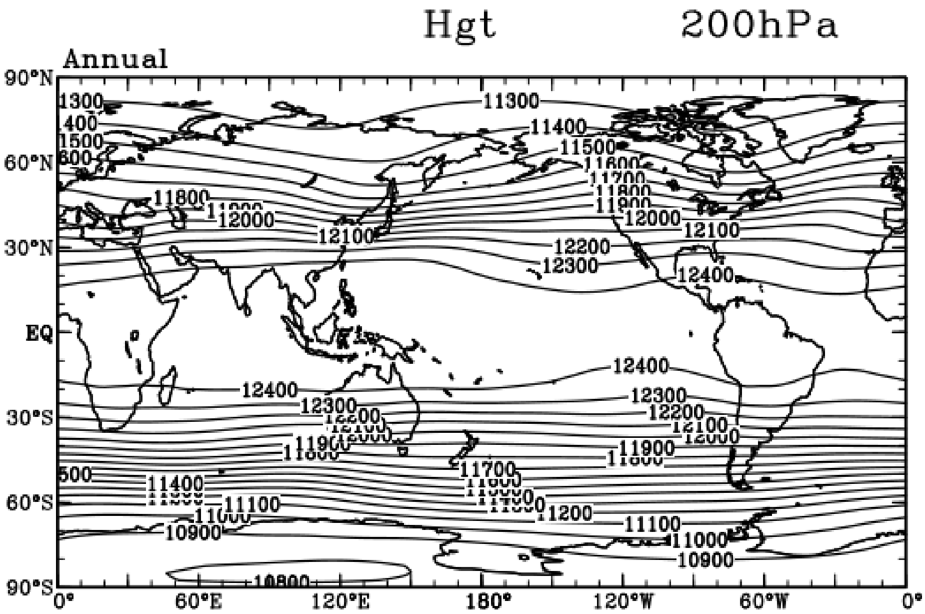
- NH: the eddy field Z^* shows a prominent **wave-2 to wave-3 pattern** at 500 hPa, with troughs over eastern North America and East Asia, and ridges over the eastern ocean basins.
- SH: eddy amplitudes are much weaker, consistent with the zonally symmetric nature of the SH circulation and the relative absence of topographic forcing.



Annual-mean $[\bar{v}]$ ($\mathrm{m\,s^{-1}}$). Note the much smaller contour interval compared to $[\bar{u}]$.

- The eddy amplitude generally increases with height (larger at 200 hPa than at 850 hPa), consistent with the equivalent-barotropic structure of stationary Rossby waves.

eddy(A^*)는 동서 평균에서 벗어난 파동 구조를 의미한다. NH에서는 대륙–해양 대비와 대규모 지형(히말라야, 로키) 때문에 eddy 진폭이 크고, SH에서는 작다. 이것이 NH 기후의 “비대칭성”의 핵심이다.



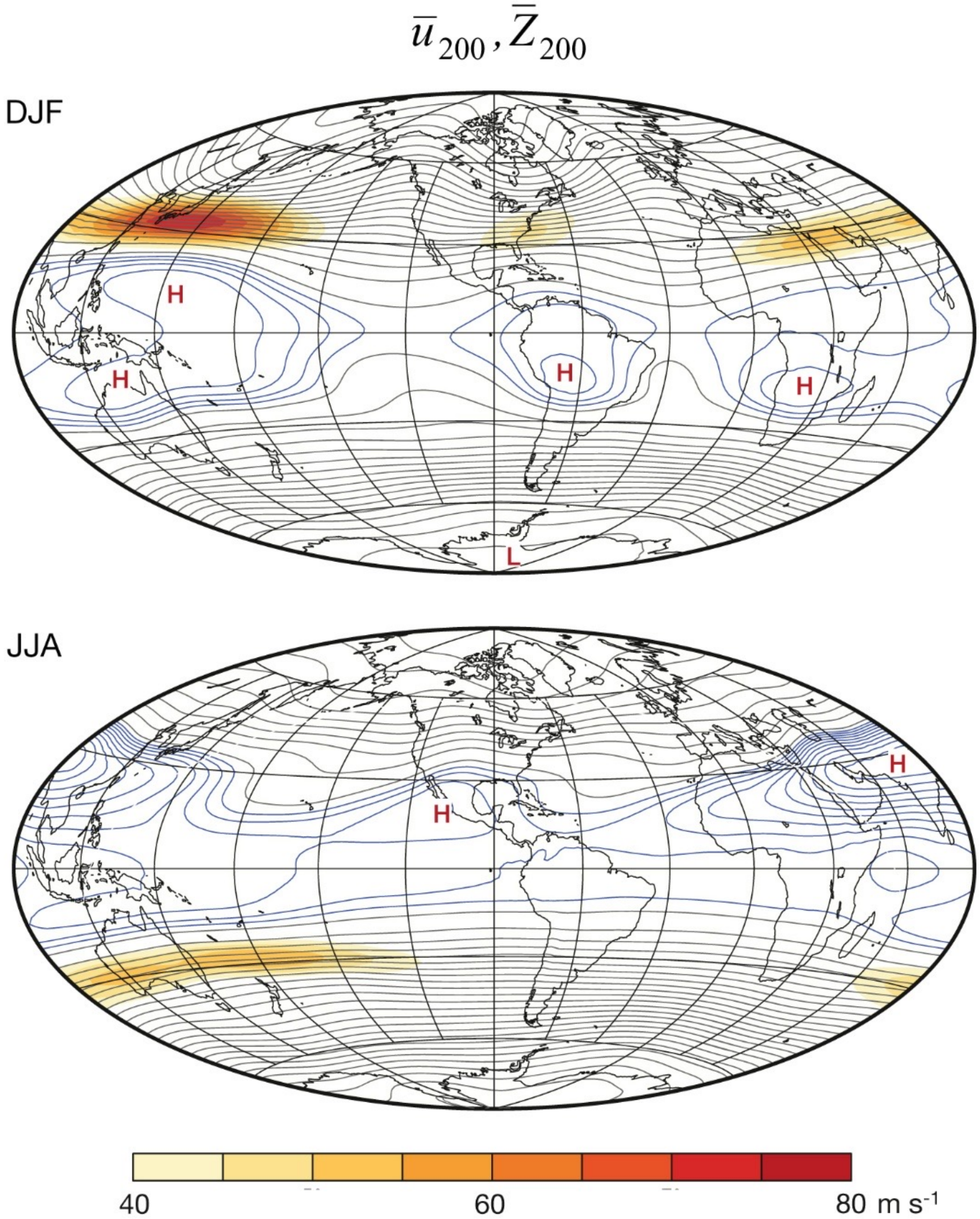
Annual-mean geopotential height at 500 hPa: (a) full field, (b) eddy field Z^*.

5.4. Thermal Wind and Jet Streams

The vertical shear of the geostrophic wind is related to the horizontal temperature gradient by the **thermal wind relation**:

$$\frac{\partial u_g}{\partial \ln p} = \frac{R}{f}\frac{\partial T}{\partial y}, \tag{5.2}$$

where R is the gas constant for dry air and f the Coriolis parameter. Because the meridional temperature gradient $\partial T/\partial y < 0$ in the NH (temperature decreases poleward), the westerly wind *increases with height,* producing the subtropical jet at $\sim$200 hPa.



Annual-mean (a) $[\bar{T}]$ (K) and (b) $[\bar{u}]$ ($\mathrm{m\,s^{-1}}$), illustrating the thermal wind relationship.

Example 1 | Thermal wind estimate of the subtropical jet

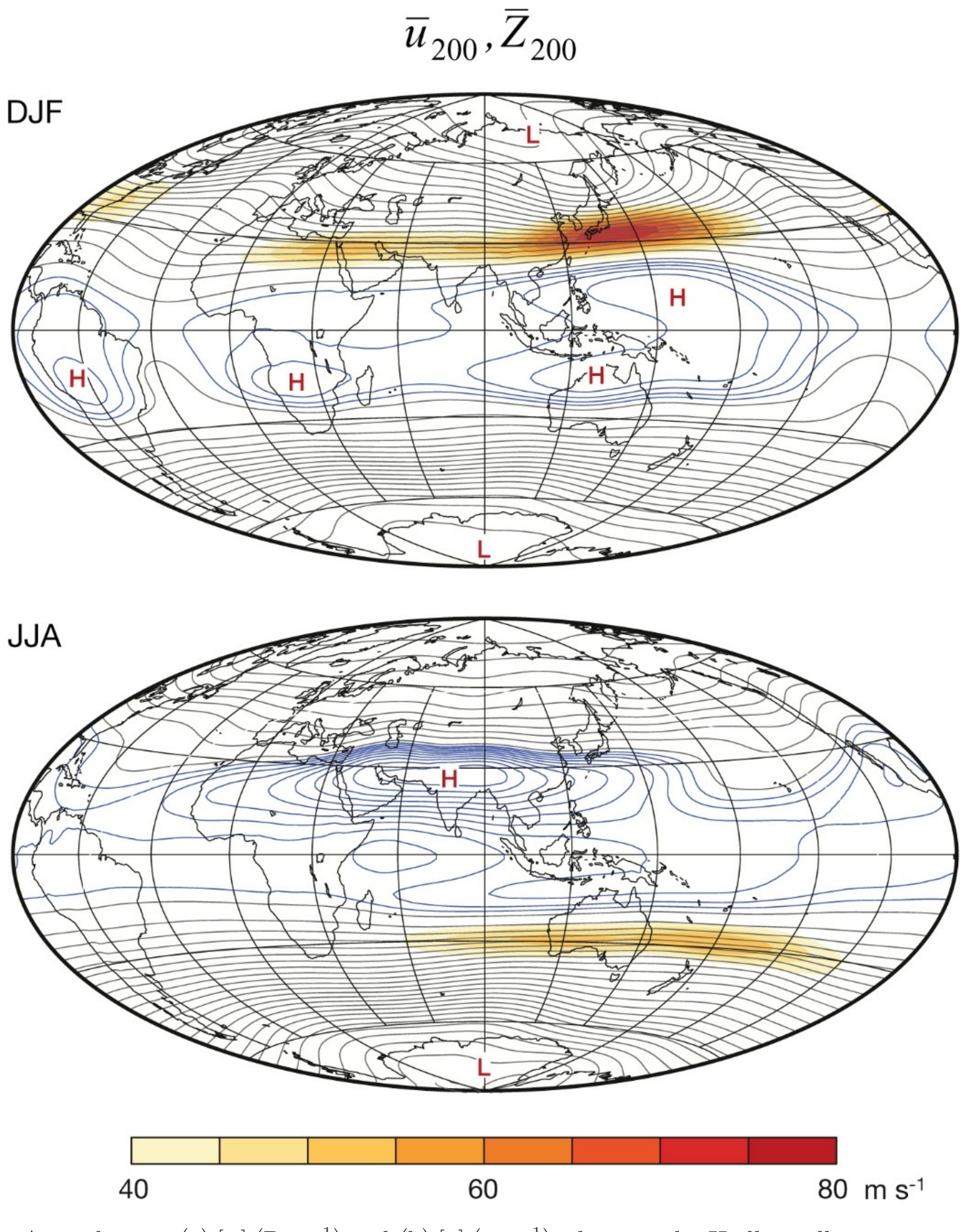
Take a meridional temperature gradient $\Delta T/\Delta y \approx -15\,\mathrm{K}$ over 20° latitude (~2200 km) between 15°N and 35°N. Then the thermal wind shear from 850 hPa to 200 hPa ($\Delta \ln p \approx 1.45$) gives:

$$\Delta u \approx \frac{R}{f}\frac{\Delta T}{\Delta y}\Delta \ln p \approx \frac{287}{7.3\times10^{-5}}\,\frac{15}{2.2\times10^{6}}\,1.45 \approx 39\ \mathrm{m\,s^{-1}}.$$

This is consistent with the observed subtropical jet speed of $\sim 40\,\mathrm{m\,s^{-1}}$.

thermal wind: 남북 온도 경도가 클수록 연직 바람 시어가 강해지고, 상층 제트가 강해진다. 이것이 아열대 제트의 핵심 메커니즘이다.

5.5. Meridional Overturning Circulation



Annual-mean (a) $[\bar{\omega}]$ ($\mathrm{Pa\,s^{-1}}$) and (b) $[\bar{v}]$ ($\mathrm{m\,s^{-1}}$), showing the Hadley cell structure.

Definition 5.7 | Hadley Cell — Annual Mean

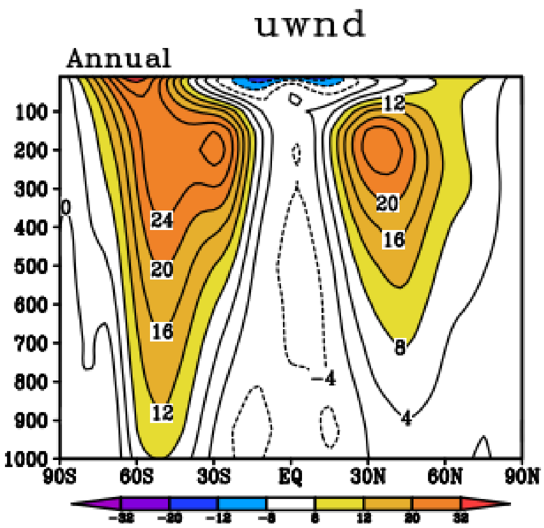
The annual-mean Hadley cell is characterised by:

- **Ascending motion** at the equator ($[\bar{\omega}] < 0$): deep convection in the ITCZ releases latent heat and drives upward mass flux.
- **Poleward flow** in the upper troposphere ($\sim$200 hPa): $[\bar{v}] > 0$ in the NH, $[\bar{v}] < 0$ in the SH.
- **Descending motion** in the subtropics ($\sim 30^\circ$ N/S, $[\bar{\omega}] > 0$): adiabatic warming maintains the subtropical highs and suppresses precipitation.
- **Equatorward return flow** at the surface: the trade winds.

적도에서 상승, 아열대에서 하강 — 이것이 해들리 셀의 기본 구조이다. 연평균에서는 NH 셀과 SH 셀이 거의 대칭이지만, 계절적으로는 겨울 반구 셀이 훨씬 강해지고 적도를 넘어 확장한다 (Section 6).

6. SEASONAL VARIATION

6.1. Sea-Level Pressure



Sea-level pressure (hPa) for (a) DJF and (b) JJA.

Remark 6.1 | Seasonal SLP contrast

DJF (boreal winter):
- Deep Aleutian Low
- Deep Icelandic Low
- Strong Siberian High (∼1040 hPa)
- Intense pressure gradient ⇒ strong midlatitude westerlies

JJA (boreal summer):
- Continental thermal lows (Asian, N. American)
- Strong oceanic subtropical highs
- Weakened Aleutian/Icelandic Lows
- Monsoon circulations dominate South/East Asia

시베리아 고기압은 DJF에만 존재하는 열적 고기압이다. 여름에는 대륙이 가열되어 열적 저기압으로 바뀐다.

6.2. Wind Seasonal Cycle

850 hPa – Monsoon Circulation

Remark 6.2 | Somali Jet and monsoon onset

In JJA, a powerful **cross-equatorial low-level jet** — the Somali jet — funnels moisture from the southern Indian Ocean into the Indian subcontinent. This jet is among the strongest low-level wind features on Earth, reaching 15–20 $\mathrm{m\,s^{-1}}$ near the coast of Somalia. It is a key element of the South Asian monsoon onset.

200 hPa – Subtropical Jet Intensification

DJF 동아시아 200 hPa 제트가 50 m/s를 초과하는 이유: 겨울에는 적도–극 온도 경도가 최대이고, 티벳 고원의 기계적/열적 강제가 합쳐지기 때문이다.

6.3. Zonal-Mean Zonal Wind – Seasonal

Remark 6.3 | Seasonal jet asymmetry

In DJF the NH subtropical jet intensifies dramatically to ∼28 $\mathrm{m\,s^{-1}}$ in the zonal mean (and much more locally), while the SH jet weakens. In JJA the pattern reverses. This is a direct consequence of the thermal wind relation (11.10): the **winter hemisphere** always has the steepest meridional temperature gradient and therefore the strongest jet.

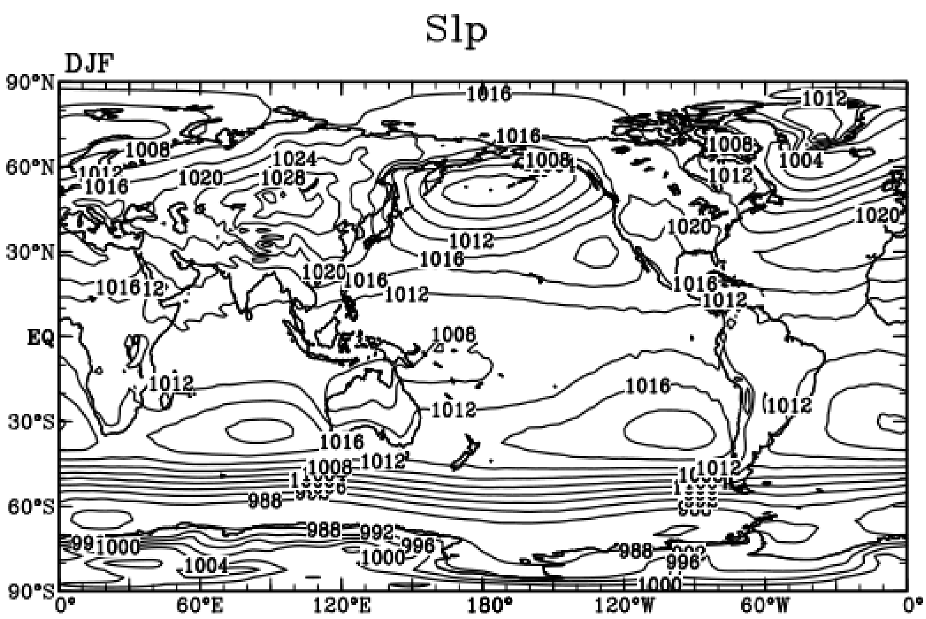
6.4. Zonal-Mean Meridional Wind – Seasonal

계절에 따라 해들리 셀의 상승 지역이 여름 반구 쪽으로 이동한다. DJF에는 SH에서 상승하고 NH에서 하강하는 "겨울 셀"이 우세하고, JJA에는 반대이다. 이 비대칭이 몬순과 ITCZ의 계절 이동을 구동한다.

6.5. Zonal-Mean Temperature

Definition 6.4 | Tropopause structure

The tropopause — the boundary between the troposphere and stratosphere — has a fundamentally different character in the tropics and extratropics:



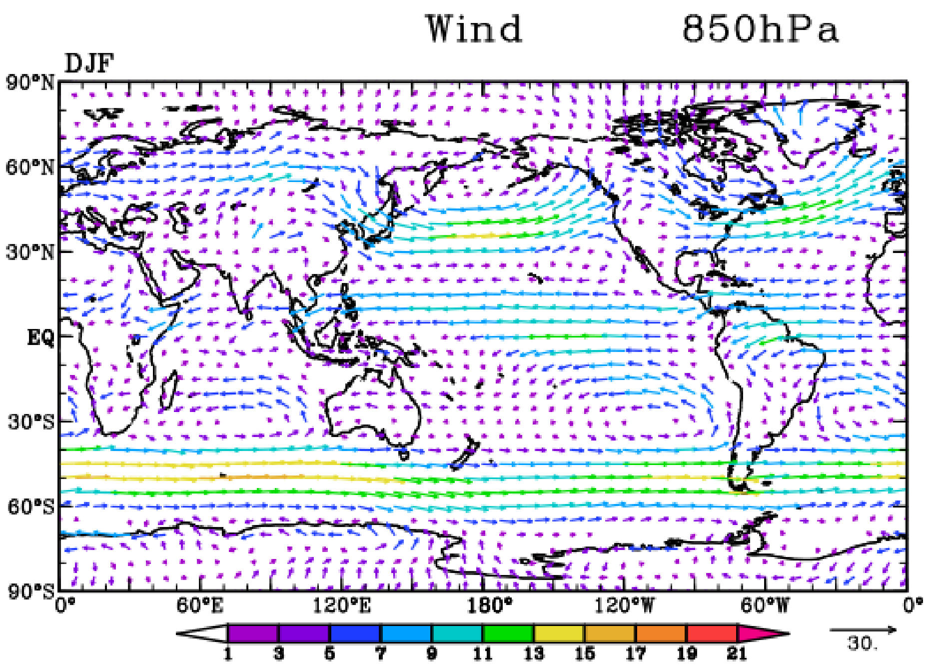
850 hPa wind ($m\,s^{-1}$) for (a) DJF and (b) JJA. Note the Somali jet in JJA.

Tropical tropopause:
- Height: ~100 hPa (~16–17 km)
- Temperature: ~190 K (−83°C)
- Set by deep convective overshooting and radiative balance

Extratropical tropopause:
- Height: ~300 hPa (~9–12 km)
- Temperature: ~220 K (−53°C)
- Set by baroclinic eddy mixing and radiative processes

The global-mean tropospheric lapse rate is approximately $\Gamma \approx 6.5\,K\,km^{-1}$.

6.6. Vertical Temperature Profiles



200 hPa wind ($m\,s^{-1}$) for (a) DJF and (b) JJA.

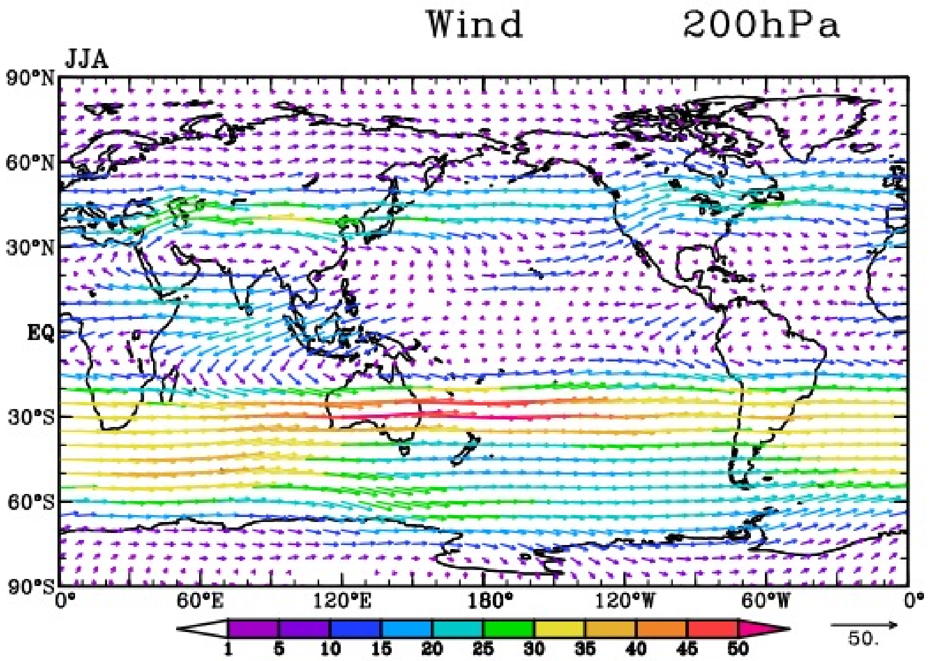
열대 대류권계면이 더 높고 더 차가운 것은, 깊은 대류(deep convection)가 공기를 높이까지 밀어올리기 때문이다. 반면 중위도 대류권계면은 경압 불안정에 의한 에디 혼합이 결정한다.

6.7. Quasi-Biennial Oscillation (QBO)

Definition 6.5 | Quasi-Biennial Oscillation (QBO)

The QBO is a quasi-periodic oscillation of the equatorial stratospheric zonal wind between easterly and westerly regimes, with:

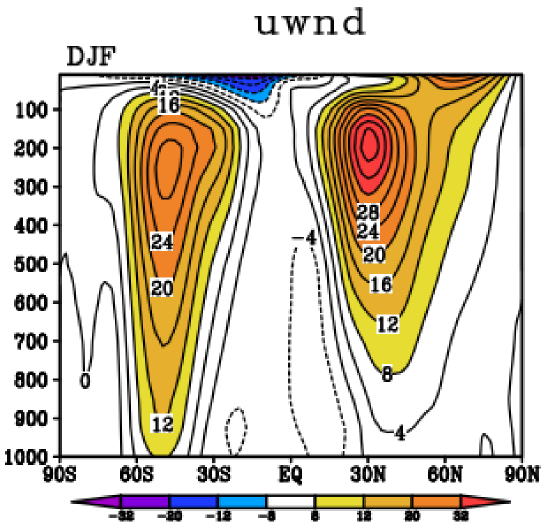
- **Period**: ~26–28 months (hence "quasi-biennial")



Zonal-mean zonal wind $[\bar{u}]$ ($\mathrm{m\,s^{-1}}$) for (a) DJF and (b) JJA.

- **Amplitude**: $\sim$20–30 $\mathrm{m\,s^{-1}}$ in zonal wind; $\sim$3 K in temperature
- **Vertical propagation**: phases descend from $\sim$10 hPa to $\sim$70 hPa
- **Mechanism**: driven by the interaction of vertically propagating equatorial waves (Kelvin and mixed Rossby–gravity waves) with the mean flow

QBO = 준 2년 진동. 적도 성층권에서 동풍과 서풍이 약 2년 주기로 교대한다. 이것은 대류권에서 발생한 적도 파동(Kelvin wave, MRG wave)이 성층권으로 전파되어 평균류를 가속/감속시키는 wave–mean flow interaction의 결과이다. QBO는 몬순, 극소용돌이 등 대류권 기후에도 원격 영향을 미친다.



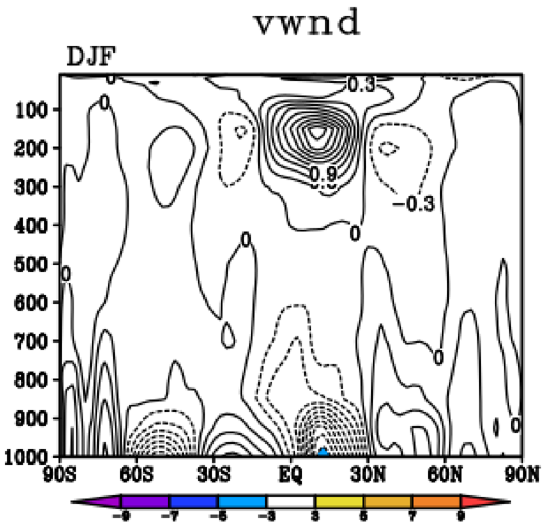
Zonal-mean meridional wind $[\bar{v}]$ ($\mathrm{m\,s^{-1}}$) for (a) DJF and (b) JJA, showing the cross-equatorial Hadley cell.

6.8. ITCZ Seasonal Migration

Remark 6.6 | ITCZ migration

The ITCZ (Intertropical Convergence Zone) follows the seasonal cycle of maximum insolation with a lag of ~1–2 months:

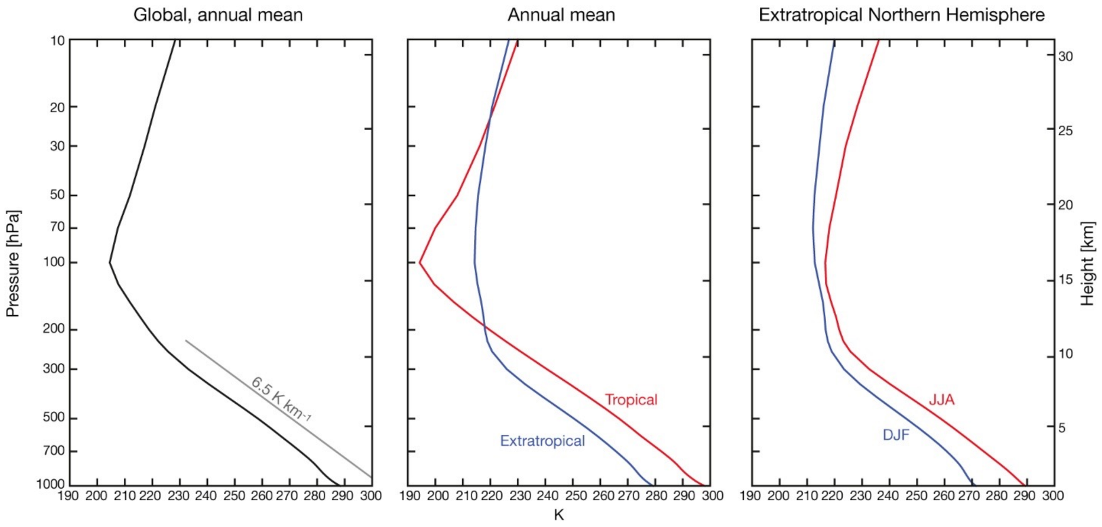
- DJF: ITCZ near ~5°S (over continents) to ~5°N (over oceans)
- JJA: ITCZ shifts to ~10°N over the continents (up to 25°N over South Asia — the monsoon)
- The ITCZ migrates farther from the equator over continents than over oceans, because land surfaces have lower thermal inertia.



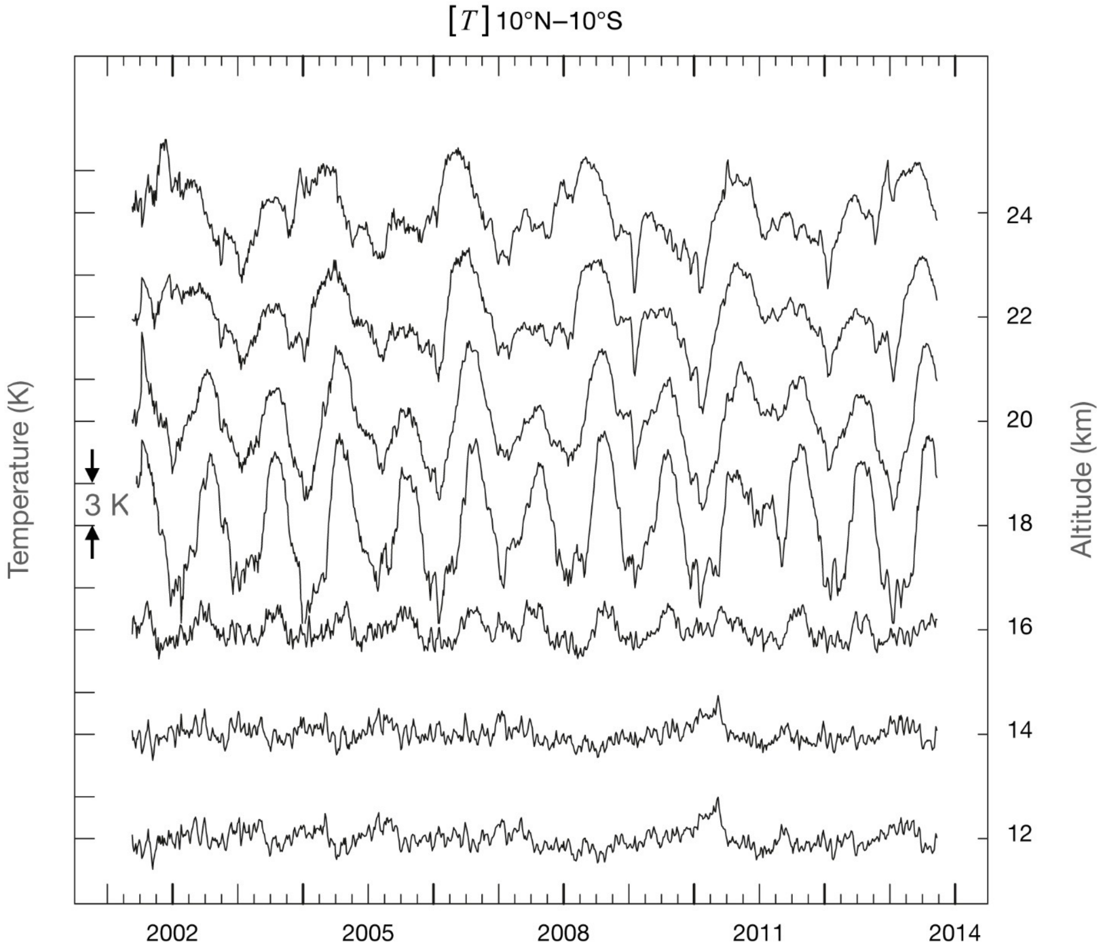
Zonal-mean temperature $[\bar{T}]$ (K) for (a) DJF and (b) JJA.

ITCZ의 계절 이동은 강수와 대규모 순환의 계절 변동을 이해하는 열쇠이다. ITCZ가 5°N에서 25°N까지 이동하는 것이 바로 아시아 몬순의 본질이다.

7. PRECIPITATION

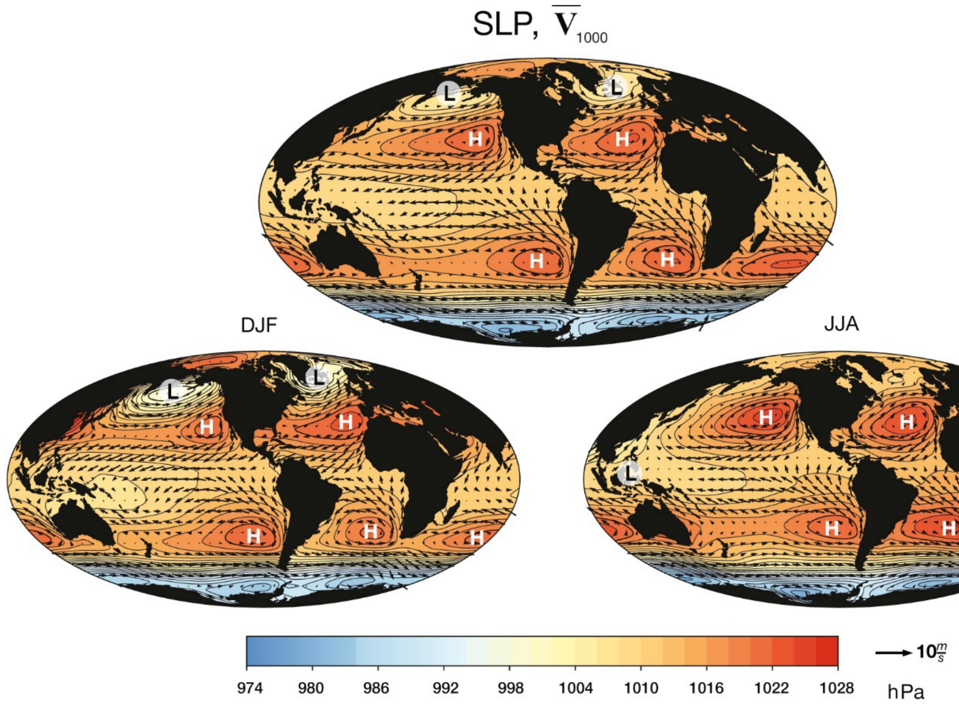


Vertical temperature profiles at representative latitudes.

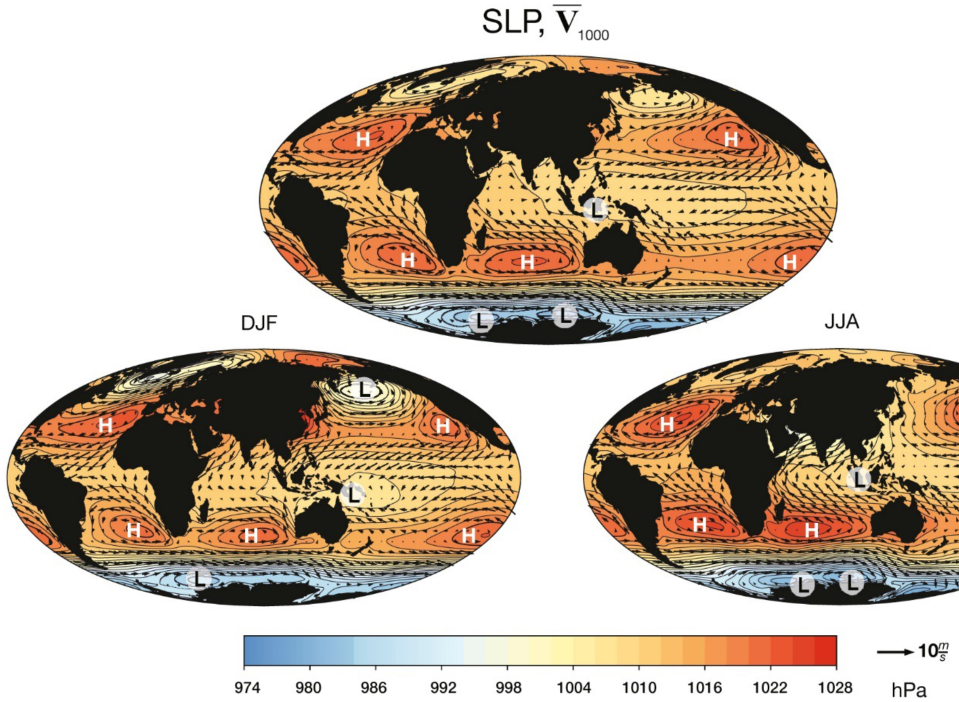


Quasi-biennial oscillation (QBO) in equatorial stratospheric temperature anomalies.

7.1. Annual-Mean Precipitation



Latitude–month cross-section of $[\bar{\omega}]$ at 500 hPa, illustrating the seasonal migration of the ITCZ.



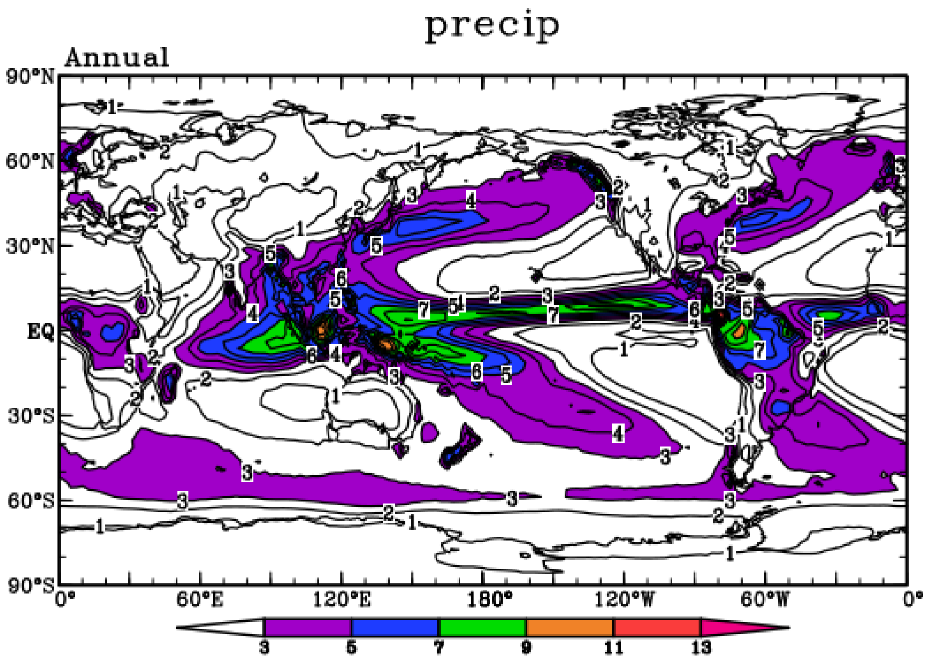
Annual-mean precipitation rate (mm day^{-1}).

Definition 7.1 | Major Precipitation Features

- **ITCZ** (Intertropical Convergence Zone): a narrow band near 5–10°N with precipitation rates of 7–13 mm day^{-1}, driven by low-level convergence and deep convection.
- **Warm pool / Maritime Continent**: the region of highest annual-mean precipitation (>9 mm day^{-1}), centred over the western Pacific and Indonesia.
- **SPCZ** (South Pacific Convergence Zone): extends south-eastward from the warm pool into the subtropical South Pacific.

- **Midlatitude storm tracks**: moderate precipitation (3–5 mm day^{-1}) associated with transient baroclinic eddies, especially over the North Pacific and North Atlantic.
- **Arid subtropics**: the descending branches of the Hadley cell suppress precipitation in the subtropical belt (∼15–30° N/S).

7.2. Precipitation and Vertical Motion



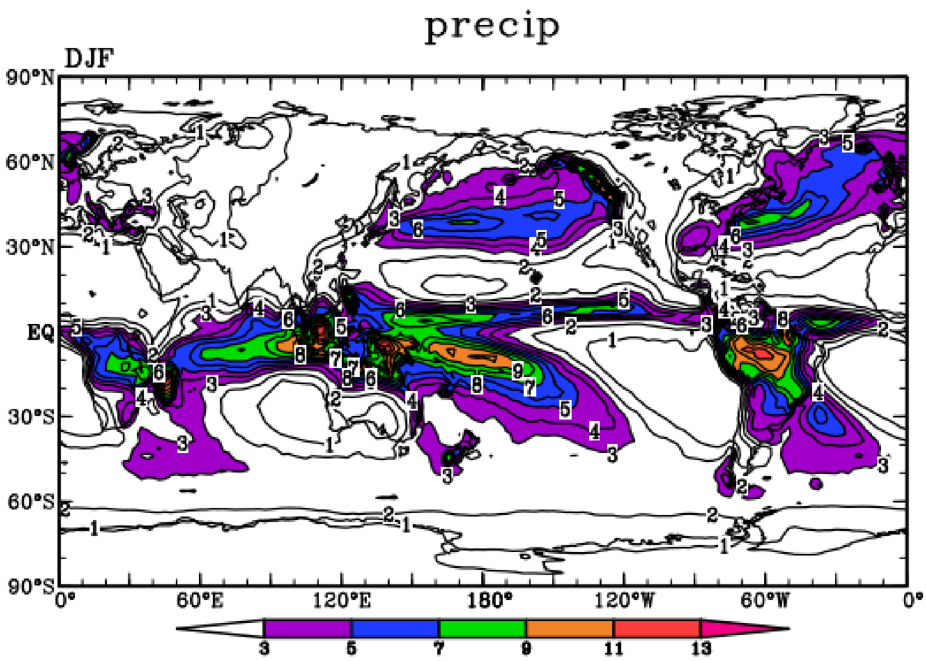
Spatial relationship between precipitation and 500 hPa ω.

Remark 7.2 | Precipitation–vertical motion linkage

There is a strong spatial correlation between annual-mean precipitation and mid-tropospheric ascent ($\omega_{500} < 0$). This reflects the fundamental mechanism: large-scale **upward motion** produces adiabatic cooling, saturation, condensation, and precipitation. Conversely, regions of large-scale descent are dry.

강수 = 상승 운동과 밀접하게 연결. 특히 ITCZ와 warm pool에서 $\omega < 0$이 강하고 강수가 많다. 아열대 고기압 지역에서는 $\omega > 0$ (하강)이고 강수가 적다.

7.3. Seasonal Precipitation



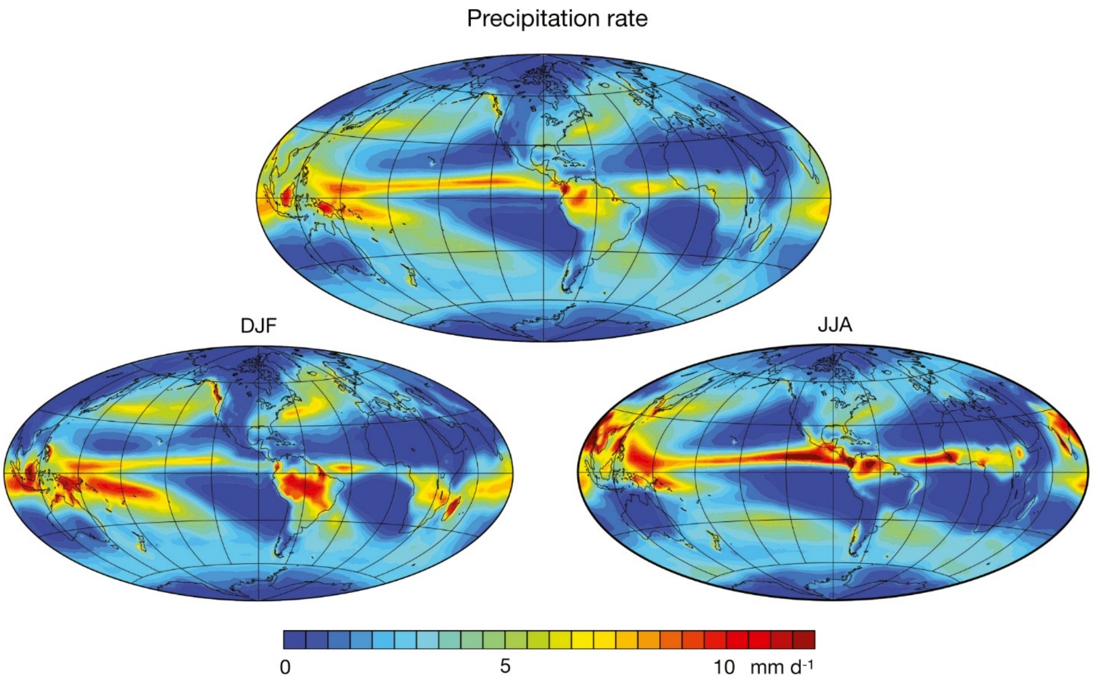
Precipitation rate ($\mathrm{mm\,day^{-1}}$) for (a) DJF and (b) JJA.

Remark 7.3 | Seasonal precipitation patterns

JJA:
- ITCZ shifts far northward
- Asian monsoon: intense rainfall over India, Bangladesh, Southeast Asia
- West African monsoon active
- NH storm tracks weaken

DJF:
- ITCZ near or south of equator
- Maritime Continent rainfall broadens
- Australian and South American monsoons
- NH storm tracks strengthen ⇒ enhanced mid-latitude precipitation

7.4. Diabatic Heating Rate



Diabatic heating rate Q averaged over the 850–500 hPa layer ($\mathrm{K\,day^{-1}}$).

Remark 7.4 | Diabatic heating: latent heat vs. radiative cooling

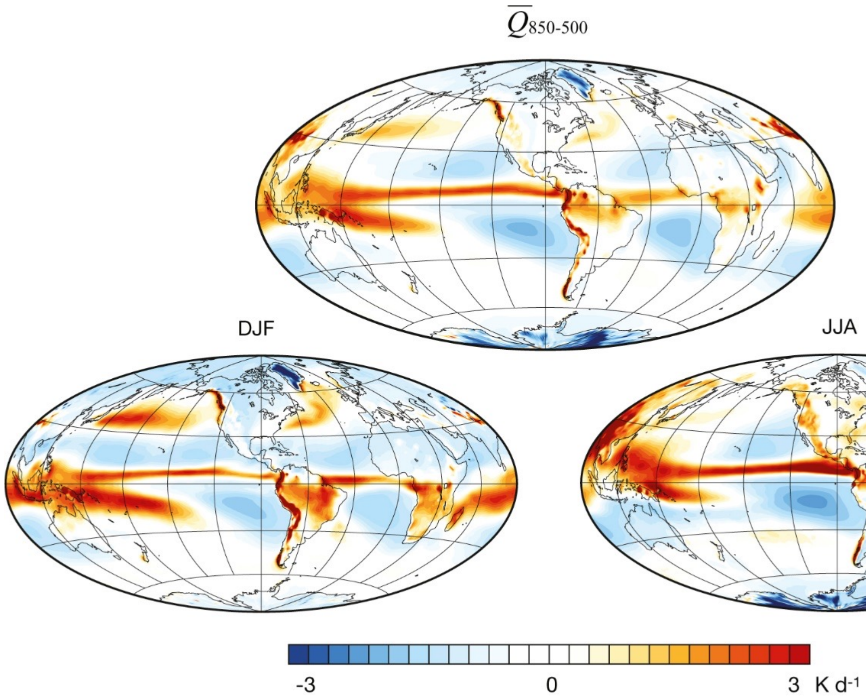
The net diabatic heating rate in the lower-to-mid troposphere is the residual between **latent heat release** (positive, associated with precipitation) and **radiative cooling** (negative, especially longwave emission to space). In the ITCZ and warm pool, latent heating strongly dominates ($Q > 0$). In the clear-sky subtropics, radiative cooling dominates ($Q < 0$).

8. SEA SURFACE TEMPERATURE (SST)

8.1. Annual-Mean SST Distribution

Definition 8.1 | Key SST Features

- **Warm pool** (western Pacific, >28°C): the warmest open-ocean waters on Earth, driven by the accumulation of warm water by the trade winds and weak upwelling.
- **Cold tongue** (eastern equatorial Pacific, ∼22–24°C): maintained by equatorial upwelling driven by the easterly trade winds (Ekman divergence).
- **Western boundary currents**: the Gulf Stream and Kuroshio create sharp SST fronts (∼5–10°C across ∼100 km).
- **Eastern boundary upwelling**: cold SST off the coasts of California, Peru/Chile, and Northwest Africa, driven by equatorward winds and offshore Ekman transport.



Annual-mean sea surface temperature (°C).

SST 분포의 동서 비대칭: 서태평양 warm pool vs. 동태평양 cold tongue. 이 비대칭은 무역풍(trade winds)에 의해 유지되며, ENSO 현상에서 이 비대칭이 약화/강화되면서 기후 변동이 발생한다.

8.2. SST and Precipitation

Remark 8.2 | SST–precipitation relationship

While precipitation is concentrated over the warmest SSTs, **high SST alone is not sufficient** for heavy rainfall. Two conditions must be met:

1. SST must exceed a threshold of ∼26–27°C to support deep convection.
2. **Large-scale low-level convergence** must be present to provide the moisture flux convergence and trigger convective instability.

The eastern equatorial Pacific often has SST > 26°C but relatively low precipitation because the Walker circulation drives low-level divergence there.

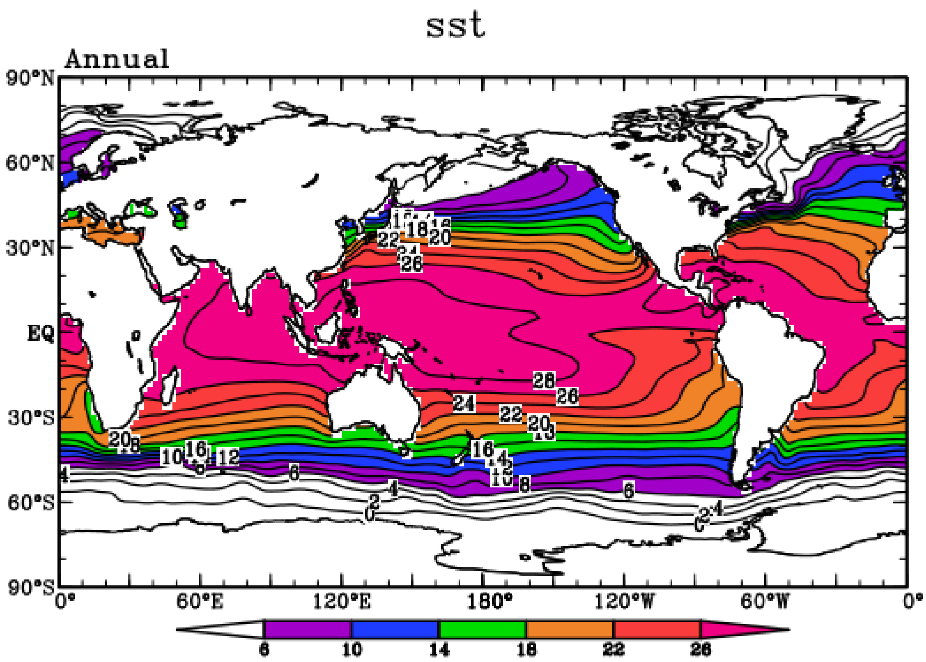
SST만 높다고 강수가 오는 건 아니다 — 대규모 수렴도 반드시 필요하다. 서태평양 warm pool에서는 SST가 높고 수렴도 있어서 강수가 강하지만, 동태평양에서는 SST가 비교적 따뜻해도 하강 기류(Walker circulation의 하강 지역) 때문에 강수가 적다. 이것은 열대 기후를 이해하는 데 핵심적인 개념이다.

8.3. SST and Surface Winds

Remark 8.3 | SST–wind coupling

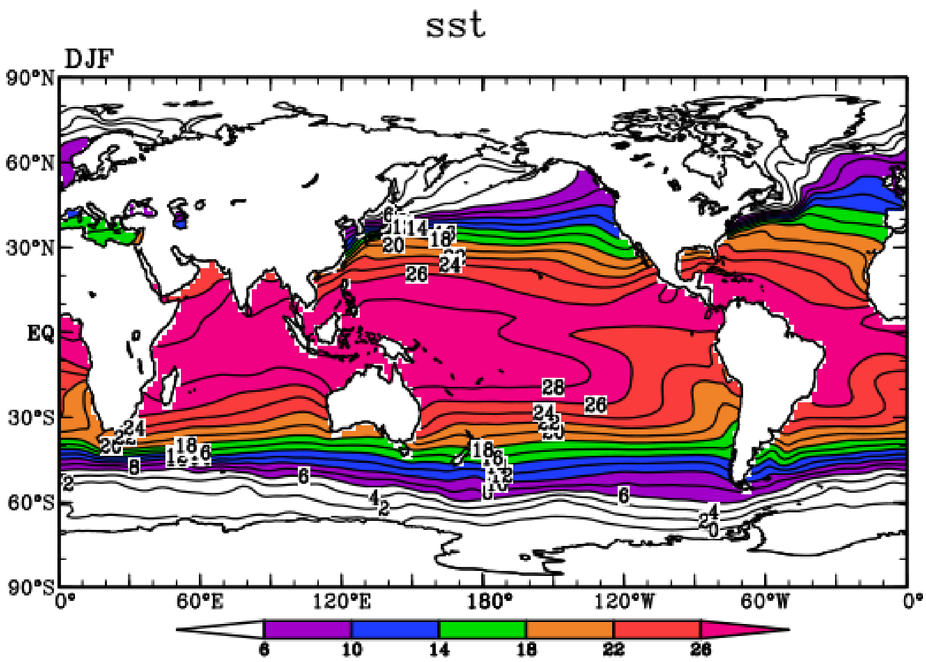
The relationship between SST and surface winds is fundamentally **two-way**:

- **Wind → SST**: trade winds drive equatorial Ekman divergence and upwelling, maintaining the cold tongue. Wind stress curl drives subtropical gyre circulation and western boundary currents.
- **SST → wind**: warm SSTs reduce atmospheric static stability in the boundary layer, enhancing surface turbulence and mixing momentum downward from the free troposphere, locally strengthening surface winds over warm SST fronts.



Relationship between SST and precipitation. Heavy rainfall occurs over the warmest SSTs, but convergence is also required.

> 무역풍이 적도 용승(equatorial upwelling)을 일으켜 cold tongue를 유지하고, cold tongue가 다시 동서 기압 경도를 강화하여 무역풍을 유지하는 양의 피드백 루프가 존재한다 (Bjerknes feedback). 이 피드백이 약해지면 El Niño가 발생한다.



SST (°C) with surface wind vectors, illustrating the wind–SST coupling.

Chapter Summary. The observed atmospheric general circulation is characterised by: (1) thermally driven overturning cells (Hadley, Ferrel, polar); (2) subtropical jet streams maintained by the thermal wind relationship; (3) strong land–sea contrasts that break zonal symmetry, especially in the NH; (4) dramatic seasonal variations driven by the migration of the solar heating maximum; and (5) tight coupling between SST, precipitation, and the large-scale wind field. These observations motivate the dynamical theories developed in subsequent chapters.

MAINTENANCE OF THE ZONALLY SYMMETRIC CI CULATION

9. MAINTENANCE OF THE ZONALLY SYMMETRIC CIRCULATION

This chapter addresses the central question of *how* the observed zonally averaged distributions of temperature, wind, and potential vorticity are maintained against dissipative processes. The key players are the **eddy fluxes**—the statistical covariances between departures from the zonal mean—together with diabatic heating and surface friction.

이 장의 핵심 질문: 관측된 동서 평균 상태는 어떻게 유지되는가? 에디 플럭스, 복사 가열, 마찰의 균형!

9.1. Maintenance of Zonally Averaged Temperature

Throughout this chapter we use the following notation conventions.

Definition 9.1 | Zonal-Mean and Eddy Decomposition

For any atmospheric variable $a(\lambda, \phi, p, t)$,

$$a = a_Z + a_E, \qquad a_Z \equiv \frac{1}{2\pi}\int_0^{2\pi} a\, d\lambda, \qquad (a_E)_Z = 0.$$

Here $(\cdot)_Z$ denotes the **zonal average** and $(\cdot)_E$ the **eddy (departure)** component. When a time mean is additionally taken we write $(\cdot)_{ZM}$.

Zonal mean: 경도 방향으로 한 바퀴 평균. Eddy: 그 평균으로부터의 편차. 시간 평균까지 하면 ZM.

Thermodynamic Energy Equation

The first law of thermodynamics applied to an ideal-gas atmosphere in pressure coordinates gives

$$\frac{\partial T}{\partial t} + \mathbf{V}\cdot\nabla T + \omega\frac{\partial T}{\partial p} - \frac{1}{c_p}\alpha\omega = \frac{1}{c_p}H, \tag{9.1}$$

where $\mathbf{V} = (u, v)$ is the horizontal wind, $\omega = Dp/Dt$ is the pressure-coordinate vertical velocity, $\alpha = 1/\rho$ is the specific volume, and H collects all diabatic heating rates (radiation, latent heat, sensible heat flux).

좌변 각 항의 의미:

- $\partial T/\partial t$: 국지적 온도 변화율
- $\mathbf{V}\cdot\nabla T$: 수평 이류에 의한 온도 변화
- $\omega\,\partial T/\partial p$: 연직 이류에 의한 온도 변화
- $-(1/c_p)\alpha\omega$: 단열 팽창/압축에 의한 온도 변화 ($\omega < 0$이면 상승, 팽창 냉각)

Using the equation of state $\alpha = RT/p$ we rewrite the adiabatic compression term:

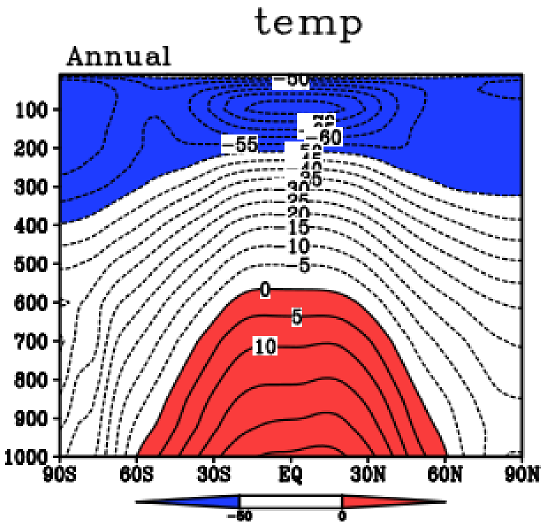
$$\frac{1}{c_p}\alpha\omega = \frac{R\omega T}{c_p p}.$$

Flux Form of the Thermodynamic Equation

The continuity equation in pressure coordinates states

$$\nabla\cdot\mathbf{V} + \frac{\partial\omega}{\partial p} = 0. \tag{9.2}$$

Multiplying (9.2) by T and adding to (9.1) converts the advection terms to **flux form**:



Zonal-mean temperature cross-section (annual mean) as a function of latitude and pressure. Warm temperatures (red) in the tropical lower troposphere and cold temperatures (blue) in the upper troposphere and polar regions illustrate the equator-to-pole temperature gradient that the general circulation must maintain. 연평균 동서 평균 온도 단면도. 적도 하층 대류권의 고온과 극지방/상층 대류권의 저온이 일반 순환의 원동력이 되는 적도–극 온도 경도를 보여준다.

$$\boxed{\frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{V}T) + \frac{\partial}{\partial p}(\omega T) - \frac{R}{c_p\, p}\,\omega T = \frac{1}{c_p}H.} \tag{9.3}$$

Remark 9.2

The flux form is essential because zonal averaging and vertical integration commute cleanly with the divergence operators.

Zonal Averaging

Proposition 9.3 | Vanishing of Zonal Derivatives

For any periodic function $a(\lambda)$ with period 2π,

$$\left(\frac{\partial a}{\partial \lambda}\right)_Z = 0. \tag{9.4}$$

주기 함수의 한 주기 적분은 항상 0. 이것이 zonal averaging의 핵심 성질!

Proof

$$\left(\frac{\partial a}{\partial \lambda}\right)_Z = \frac{1}{2\pi}\int_0^{2\pi} \frac{\partial a}{\partial \lambda}\, d\lambda = \frac{1}{2\pi}\big[a(2\pi) - a(0)\big] = 0.$$

■

Expanding the horizontal divergence in spherical coordinates,

$$\nabla \cdot (\mathbf{V}T) = \frac{1}{a\cos\phi}\frac{\partial(uT)}{\partial \lambda} + \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}(vT\cos\phi),$$

the first term vanishes upon zonal averaging by (9.4). Thus the **zonally averaged thermodynamic equation** is

$$\frac{\partial T_Z}{\partial t} + \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\big((vT)_Z\cos\phi\big) + \frac{\partial}{\partial p}(\omega T)_Z - \frac{R}{c_p\, p}(\omega T)_Z = \frac{1}{c_p}H_Z. \tag{9.5}$$

Vertical Averaging and Boundary Conditions

Integrating (9.5) from $p = 0$ (top of atmosphere) to $p = p_0$ (surface) and applying the boundary conditions

$$\omega = 0 \quad \text{at } p = 0 \quad \text{and} \quad p = p_0, \tag{9.6}$$

the vertical flux divergence integrates out:

$$\int_0^{p_0} \frac{\partial}{\partial p}(\omega T)_Z\, dp = \big[(\omega T)_Z\big]_0^{p_0} = 0.$$

$\omega = 0$인 경계 조건:
- $p = 0$ (대기 상한): 대기 위에 물질이 없으므로 $\omega = 0$
- $p = p_0$ (지표): 지표면이 평평하고 움직이지 않으므로 $\omega \approx 0$

이 경계 조건을 적용하면 ωT의 연직 플럭스 발산 항이 적분 시 사라진다.

Similarly, define the vertical mean $\langle\cdot\rangle \equiv \frac{1}{p_0}\int_0^{p_0}(\cdot)\, dp$. The adiabatic compression term integrates to

$$\frac{R}{c_p\, p_0}\int_0^{p_0} \frac{(\omega T)_Z}{p}\, dp \approx 0,$$

because the boundary conditions force ωT to vanish at both limits and the integrand is small in between (to leading order). Therefore the **vertically and zonally averaged** thermodynamic equation becomes

$$\boxed{\frac{\partial\langle T_Z\rangle}{\partial t} + \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\big(\langle(vT)_Z\rangle\cos\phi\big) = \frac{1}{c_p}\langle H_Z\rangle.} \tag{9.7}$$

Decomposition of the Meridional Heat Flux

Using $v = v_Z + v_E$ and $T = T_Z + T_E$:

$$(vT)_Z = v_Z T_Z + (v_E T_E)_Z. \tag{9.8}$$

Remark 9.4 | Mean-Flow Contribution is Negligible

For quasi-geostrophic flow, the geostrophic meridional wind v_g satisfies

$$v_g = \frac{1}{f}\frac{1}{a}\frac{\partial \Phi}{\partial \lambda},$$

and upon zonal averaging,

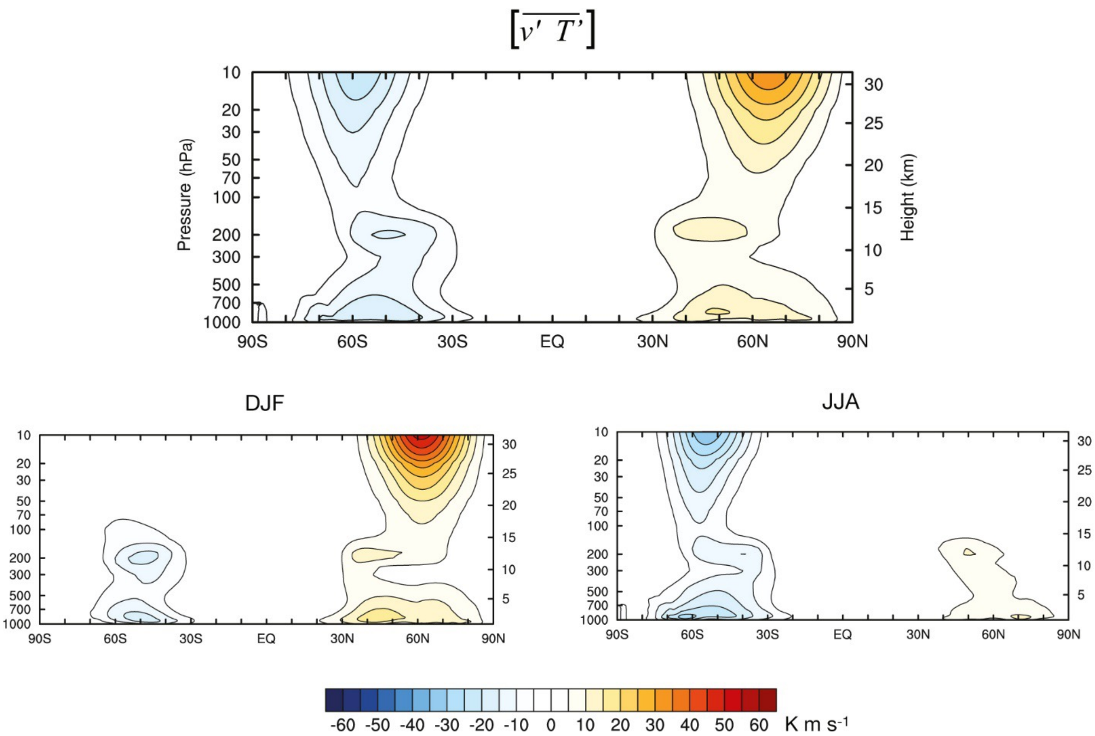
$$(v_g)_Z = \frac{1}{f}\frac{1}{a}\left(\frac{\partial \Phi}{\partial \lambda}\right)_Z = 0 \quad \text{by (9.4)}.$$

Hence $v_Z \approx 0$, and the mean-flow heat flux $v_Z T_Z \approx 0$.

지균 균형이면 v의 동서 평균은 거의 0. 따라서 열 수송은 거의 전부 에디에 의한 것!

Therefore

$$(vT)_Z \approx (v_E T_E)_Z. \tag{9.9}$$



Observed zonal-mean transient eddy heat flux $\overline{[v'T']}$ ($\mathrm{K\,m\,s^{-1}}$) cross-section: annual mean (top), DJF (bottom left), and JJA (bottom right). Poleward heat flux maxima appear in the midlatitude lower troposphere (∼40–60°, 700–850 hPa), strongest in the winter hemisphere, confirming that transient eddies dominate the meridional heat transport. 이동성 에디 열 플럭스 $[v'T']$ 관측 단면도. 중위도 하층 대류권에서 극 방향 열 수송이 최대이며, 겨울 반구에서 가장 강하다. 에디가 열 수송을 지배한다는 이론적 결과와 일치.

Steady-State Balance

Taking a **time mean** of (9.7) and assuming a **statistically steady state** ($\partial\langle T_Z\rangle/\partial t = 0$):

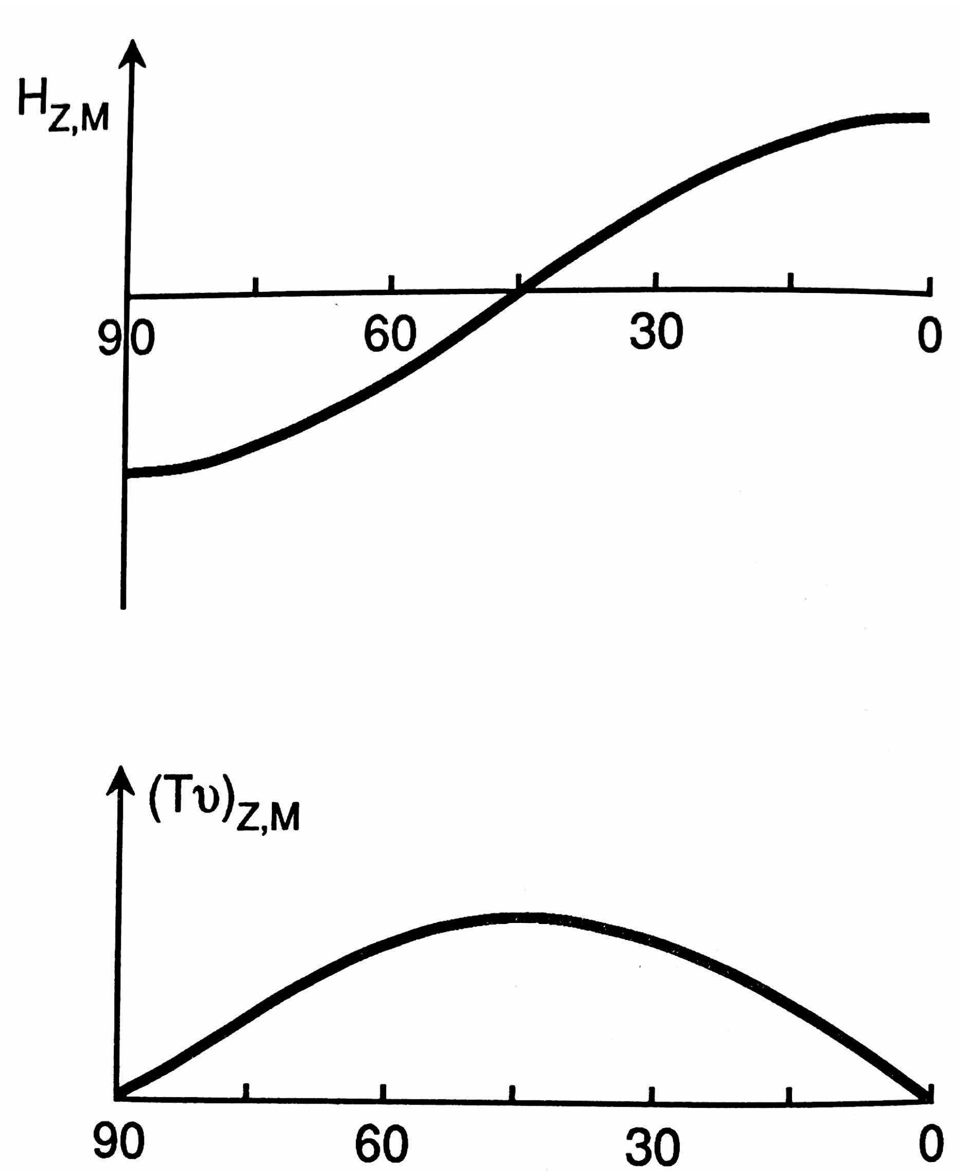
$$\boxed{\frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left(\overline{(vT)}_{ZM}\cos\phi\right) \approx \frac{1}{c_p}\overline{H}_{ZM}.} \tag{9.10}$$

Theorem 9.5 | Steady-State Heat Balance

In the time-mean, zonally and vertically averaged thermodynamic balance, the convergence of the meridional heat transport balances diabatic heating. Since the mean-flow contribution to $\overline{(vT)}_{ZM}$ is negligible, the balance is between **eddy heat transport** and diabatic heating:

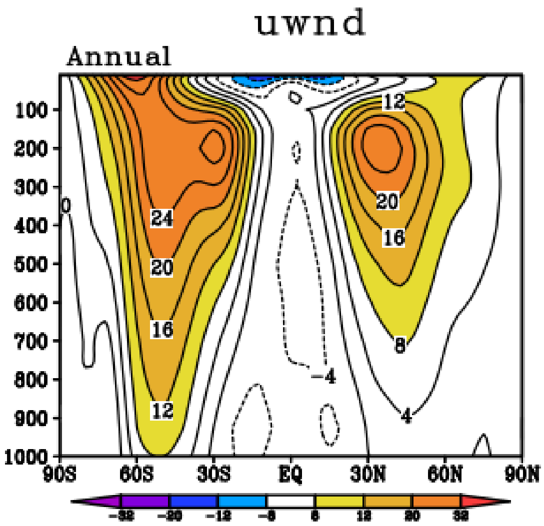
$$\frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left(\overline{(v_E T_E)}_{ZM}\cos\phi\right) \approx \frac{1}{c_p}\overline{H}_{ZM}.$$

열 수송 ≈ 에디에 의한 수송이 지배적. 평균 흐름의 기여는 무시할 수 있다. 정상 상태에서 에디 열 수송의 수렴 = 당뇨 가열 (복사 냉각/가열).



Idealized profiles of the zonally and vertically averaged diabatic heating H_{ZM} (top) and meridional heat transport $(Tv)_{ZM}$ (bottom) as functions of latitude. Diabatic heating is positive (warming) in the tropics and negative (radiative cooling) at high latitudes; the implied poleward heat transport peaks in the midlatitudes near ∼40°. 이상적인 당뇨 가열 H_{ZM} (위)과 자오면 열 수송 $(Tv)_{ZM}$ (아래)의 위도별 분포. 열대에서 가열, 고위도에서 냉각이며, 이를 균형시키는 극 방향 열 수송은 중위도에서 최대.

9.2. Maintenance of Zonally Averaged Wind



Zonal-mean zonal wind cross-section (annual mean, $\mathrm{m\,s^{-1}}$) as a function of latitude and pressure. The subtropical jet streams appear as westerly maxima ($>24\,\mathrm{m\,s^{-1}}$) near 200 hPa at $\sim 30°$ in both hemispheres, with easterlies in the tropical upper troposphere and weak surface westerlies in the midlatitudes. 연평균 동서 평균 동서풍 단면도. 아열대 제트류가 양 반구 $\sim 30°$, 200 hPa 부근에서 서풍 극대로 나타나며, 열대 상층에는 동풍이, 중위도 지표에는 약한 서풍이 존재한다.

Zonal Momentum Equation

The zonal momentum equation in spherical pressure coordinates is

$$\frac{\partial u}{\partial t} + \frac{u}{a\cos\phi}\frac{\partial u}{\partial \lambda} + \frac{v}{a}\frac{\partial u}{\partial \phi} + \omega\frac{\partial u}{\partial p} - \frac{uv\tan\phi}{a} = -\frac{1}{a\cos\phi}\frac{\partial \Phi}{\partial \lambda} + fv + F_\lambda, \tag{9.11}$$

where the $uv\tan\phi/a$ term is the metric (curvature) term arising from spherical geometry, and F_λ represents frictional and other sub-grid-scale forces.

좌변 각 항:
- $\partial u/\partial t$: 동서 바람의 국지적 변화율
- 이류 항 3개: 수평 + 연직 이류
- $-uv\tan\phi/a$: 구면 좌표계의 기하학적(metric) 항

우변:
- $-(a\cos\phi)^{-1}\partial\Phi/\partial\lambda$: 기압경도력
- fv: 코리올리 힘
- F_λ: 마찰력

Flux Form

Using the continuity equation (9.2) and the identity

$$v\frac{\partial u}{\partial\phi} - \frac{uv\tan\phi}{a} = \frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}(uv\cos^2\phi) - \frac{u}{a\cos\phi}\frac{\partial}{\partial\phi}(v\cos\phi),$$

the momentum equation is rewritten in **flux form**:

$$\boxed{\frac{\partial u}{\partial t} + \frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}(uv\cos^2\phi) + \frac{\partial}{\partial p}(u\omega) = -\frac{1}{a\cos\phi}\frac{\partial\Phi}{\partial\lambda} + fv + F_\lambda.} \tag{9.12}$$

Zonally and Vertically Averaged Equation

Applying zonal averaging to (9.12):

- The pressure-gradient term vanishes: $\left(\frac{1}{a\cos\phi}\frac{\partial\Phi}{\partial\lambda}\right)_Z = 0$ by (9.4).
- The Coriolis term: $fv_Z \approx 0$ (geostrophic balance $\Rightarrow v_Z \approx 0$).

Integrating vertically with boundary conditions (9.6):

$$\boxed{\frac{\partial\langle u_Z\rangle}{\partial t} = -\frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\big(\langle(uv)_Z\rangle\cos^2\phi\big) + \langle F_{\lambda Z}\rangle.} \tag{9.13}$$

동서 평균 바람의 시간 변화 = 운동량 플럭스의 수렴 + 마찰력. 기압경도력과 코리올리 힘은 zonal averaging 후 사라진다!

Decomposition of Momentum Flux

Analogously to the heat flux decomposition,

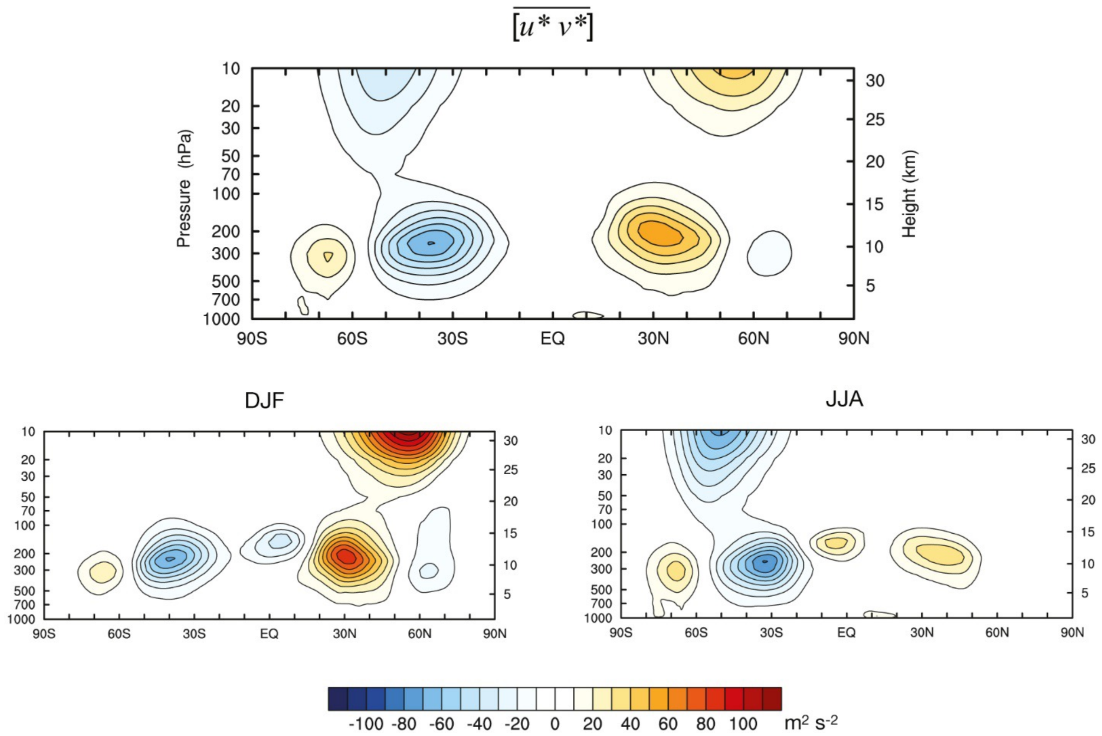
$$(uv)_Z = u_Z\,v_Z + (u_E\,v_E)_Z \approx (u_E\,v_E)_Z, \tag{9.14}$$

since $v_Z \approx 0$ by the geostrophic argument. The momentum budget is therefore controlled by **eddy momentum fluxes**.

Parameterization of Surface Friction

Friction enters through the stress divergence:

$$F_\lambda = g\frac{\partial\tau_\lambda}{\partial p}, \tag{9.15}$$



Observed zonal-mean eddy momentum flux $\overline{[u^*v^*]}$ ($\mathrm{m^2\,s^{-2}}$) cross-section (total eddies = stationary + transient): annual mean (top), DJF (bottom left), and JJA (bottom right). Positive values indicate poleward momentum transport, which converges into the midlatitude jet region, illustrating the upgradient nature of eddy momentum transport. 관측된 에디 운동량 플럭스 $\overline{[u^*v^*]}$ 단면도 (정상파 + 이동성 에디 합). 양의 값은 극 방향 운동량 수송을 나타내며, 중위도 제트류 영역으로 수렴하여 운동량의 upgradient 수송을 보여준다.

where τ_λ is the zonal component of the stress tensor. At the surface, the stress is parameterized by a **bulk aerodynamic formula**:

$$\tau_{\lambda Z}(p_0) \approx -C_d\,\rho\,V_0\,u_Z(p_0), \tag{9.16}$$

where C_d is the drag coefficient ($\sim 10^{-3}$), and V_0 is the near-surface wind speed. Integrating (9.15) vertically and linearizing:

$$\langle F_{\lambda Z}\rangle \approx -\varepsilon\,u_Z(p_0), \qquad \varepsilon \approx 3\times 10^{-6}\ \mathrm{s^{-1}}. \tag{9.17}$$

ε는 Rayleigh 마찰 계수. 지표 마찰을 선형 감쇠로 근사한 것. $1/\varepsilon \approx 3.9$일, 즉 약 4일의 감쇠 시간 규모.

Steady-State Momentum Balance

In a time-mean steady state ($\partial\langle u_Z\rangle/\partial t = 0$):

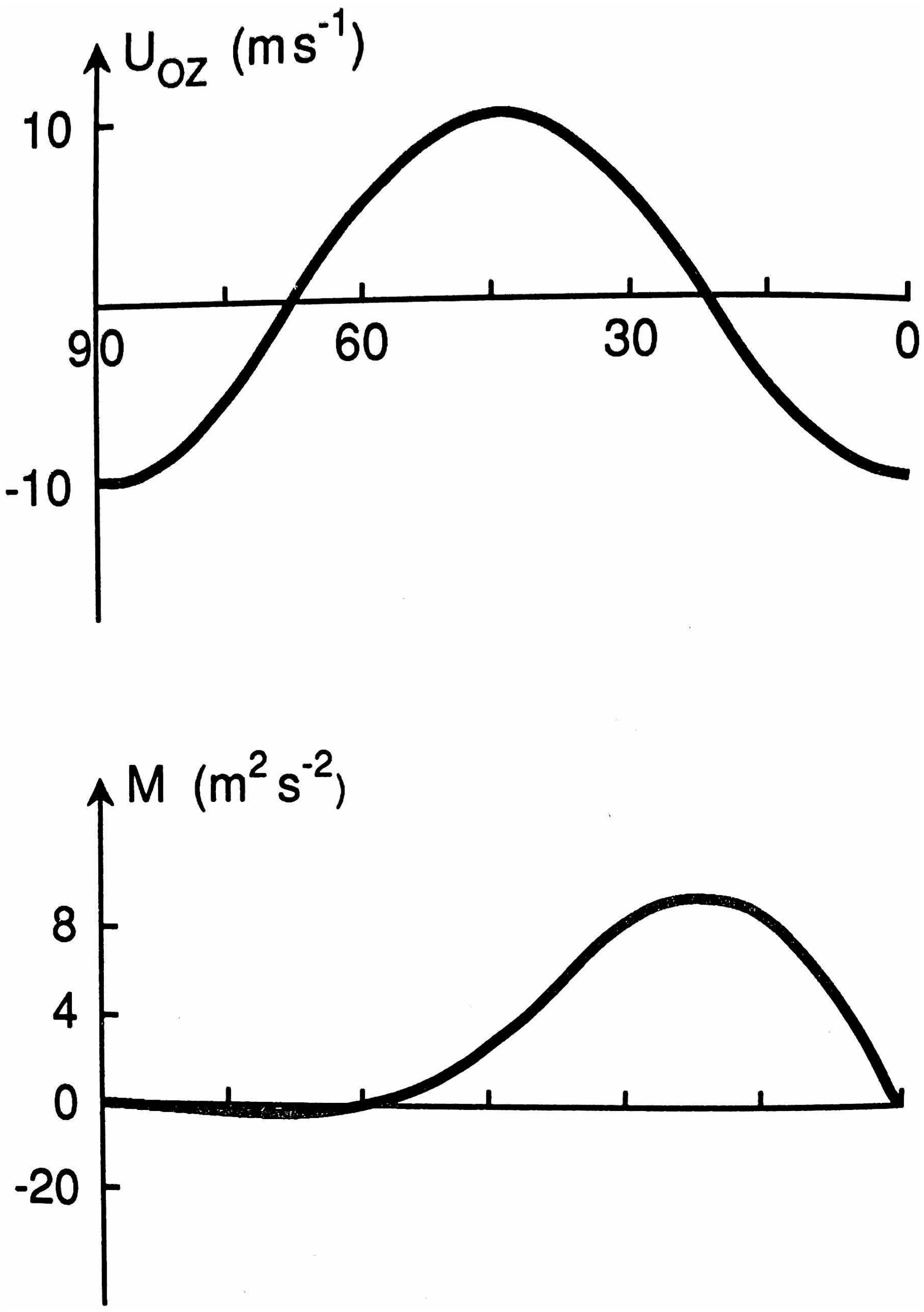
$$\boxed{\frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\left(\overline{(u_E v_E)}_{ZM}\cos^2\phi\right) \approx \varepsilon\,\overline{u}_Z(p_0).} \tag{9.18}$$

Theorem 9.6 | Steady-State Momentum Balance

The convergence of the eddy momentum flux balances surface friction. Where momentum flux

> converges (midlatitudes), westerlies are maintained; where it diverges (tropics and high latitudes), easterlies or weak westerlies result.

운동량 플럭스의 수렴 ≈ 소산 (마찰에 의한 소멸). 에디가 운동량을 중위도로 모아서 제트를 유지하고, 마찰이 이를 소산시킨다.



Idealized profiles of the zonal-mean surface zonal wind U_{OZ} (top, $\mathrm{m\,s^{-1}}$) and vertically averaged eddy momentum flux $M = \overline{(u_E v_E)}_{ZM}$ (bottom, $\mathrm{m^2\,s^{-2}}$) as functions of latitude. Surface westerlies peak near 45° where momentum flux converges; surface easterlies occur in the tropics and high latitudes where momentum flux diverges. 이상적인 지표 동서풍 U_{OZ} (위)과 에디 운동량 플럭스 M (아래)의 위도별 분포. 운동량 플럭스가 수렴하는 ~45° 부근에서 지표 서풍이 극대, 발산하는 열대와 고위도에서 동풍.

Heat Transport vs. Momentum Transport: A Fundamental Contrast

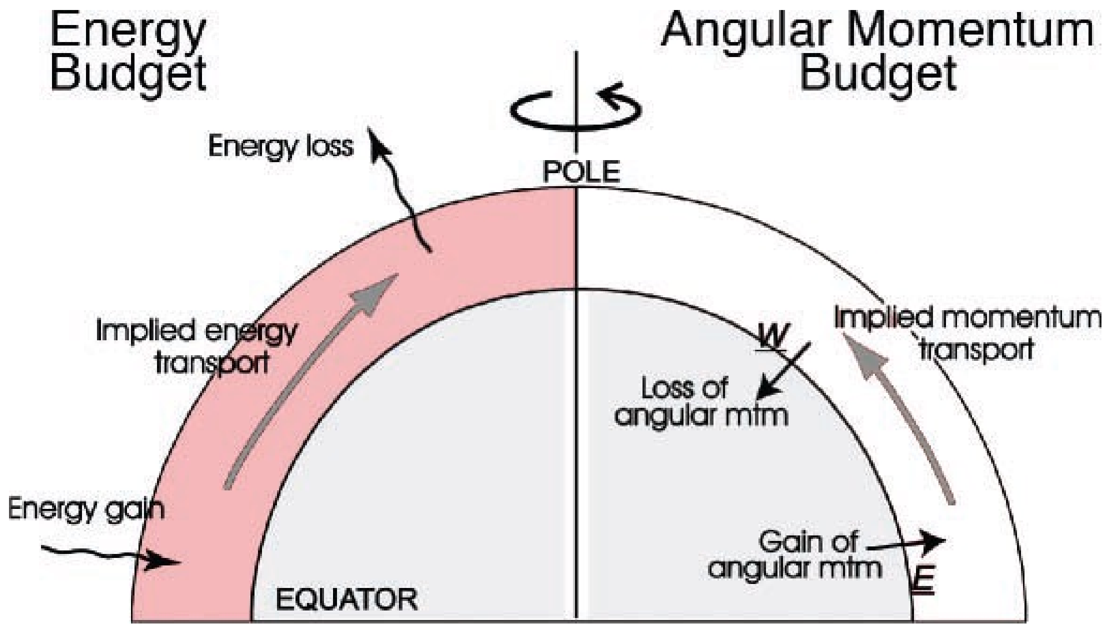
Heat Transport

- **Downgradient**: warm → cold (equator → pole)
- $\overline{(v_E T_E)}_{ZM} > 0$ in NH (northward)
- Can be parameterized as diffusion: $(v_E T_E)_Z \approx -K_T \frac{1}{a}\frac{\partial T_Z}{\partial \phi}$

Momentum Transport

- **Upgradient**: low-u → high-u regions
- From tropics & high latitudes → midlatitudes
- **Cannot** be parameterized as simple diffusion

열은 따뜻한 곳에서 차가운 곳으로 (downgradient), 운동량은 반대로 upgradient! 이게 핵심 차이. 열은 확산으로 모사 가능하지만, 운동량은 불가능. 이 차이를 해결하는 열쇠가 바로 PV 수송과 EP 플럭스 이론이다.



Schematic comparison of the energy budget (left) and the angular momentum budget (right). In the energy budget, there is a net radiative gain in the tropics and a net loss at the poles, requiring poleward energy transport. In the angular momentum budget, westerly surface winds at midlatitudes lose angular momentum to friction, while tropical easterlies gain it; eddy momentum transport from the tropics to midlatitudes maintains the balance. 에너지 수지 (좌)와 각운동량 수지 (우)의 비교 모식도. 에너지는 적도에서 극으로 수송 (복사 불균형 보상), 운동량은 열대와 고위도에서 중위도로 수송 (제트 유지와 마찰 보상).

Phase Relationships for Nonzero Fluxes

For the eddy fluxes to be nonzero in the zonal mean, *systematic phase relationships* must exist between the wave perturbations.

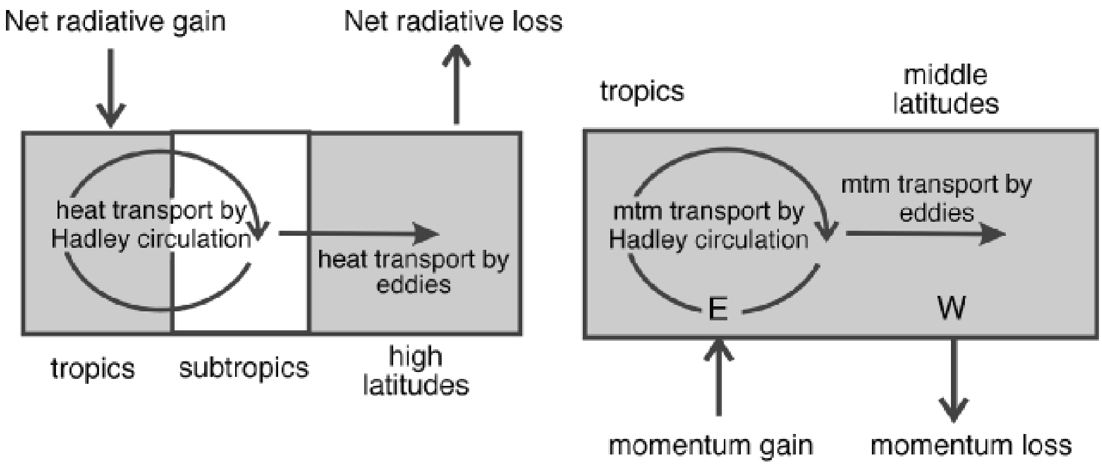
Proposition 9.7 | Phase Condition for Nonzero Heat Flux

Consider wave-form perturbations $v_E = \hat{v}\cos(k\lambda - \nu t)$ and $T_E = \hat{T}\cos(k\lambda - \nu t + \delta)$. Then

$$(v_E T_E)_Z = \tfrac{1}{2}\hat{v}\hat{T}\cos\delta.$$

For nonzero poleward heat flux, $\delta \neq \pm\pi/2$: *the temperature perturbation must lag the geopotential/velocity perturbation.*

$\delta = 0$이면 최대, $\delta = \pm\pi/2$이면 0. 관측에서는 온도가 기압골 뒤에 위치 ($\delta > 0$) ⇒ 북쪽으로 열 수송.



Box-diagram schematic of heat transport (left) and momentum transport (right) in the atmosphere. Heat is transported poleward by the Hadley circulation in the tropics and by eddies in the extratropics, balancing net radiative gain and loss. Momentum is transported poleward by the Hadley cell and equatorward by eddies, maintaining tropical easterlies (E) and midlatitude westerlies (W) against surface friction. 열 수송 (좌)과 운동량 수송 (우)의 박스 모식도. 해들리 순환은 열대에서, 에디는 중고위도에서 열 수송을 담당. 운동량은 해들리 셀이 극 방향으로, 에디가 적도에서 중위도로 수송하여 표면 바람 분포를 유지한다.

Proposition 9.8 | Phase Condition for Nonzero Momentum Flux

For nonzero $(u_E\, v_E)_Z$, the trough/ridge lines of the wave must be **tilted** in the horizontal plane. In the Northern Hemisphere, **NE–SW tilted** trough lines yield northward (poleward) momentum transport:

$$(u_E\, v_E)_Z > 0 \quad \Longleftrightarrow \quad \text{NE–SW tilt of trough axes.}$$

직관적 이해:
- 기압골 축이 NE–SW로 기울어져 있으면, 골 서쪽에서는 남서풍 ($u > 0, v < 0$이 아닌 $u > 0, v > 0$ 조합이 우세), 골 동쪽에서는 북서풍이지만 $u < 0$.
- 결과적으로 $uv > 0$인 영역이 $uv < 0$인 영역보다 넓어서 zonal mean이 양수.
- 이 기울기가 없으면 (남북 대칭이면) $(u_E v_E)_Z = 0$.

9.3. Potential Vorticity Transport

The upgradient nature of momentum transport poses a major challenge for parameterization. The resolution comes from considering the transport of **quasi-geostrophic potential vorticity (QGPV)**, which is *downgradient*—and whose transport encapsulates both heat and momentum fluxes simultaneously.

Quasi-Geostrophic Potential Vorticity

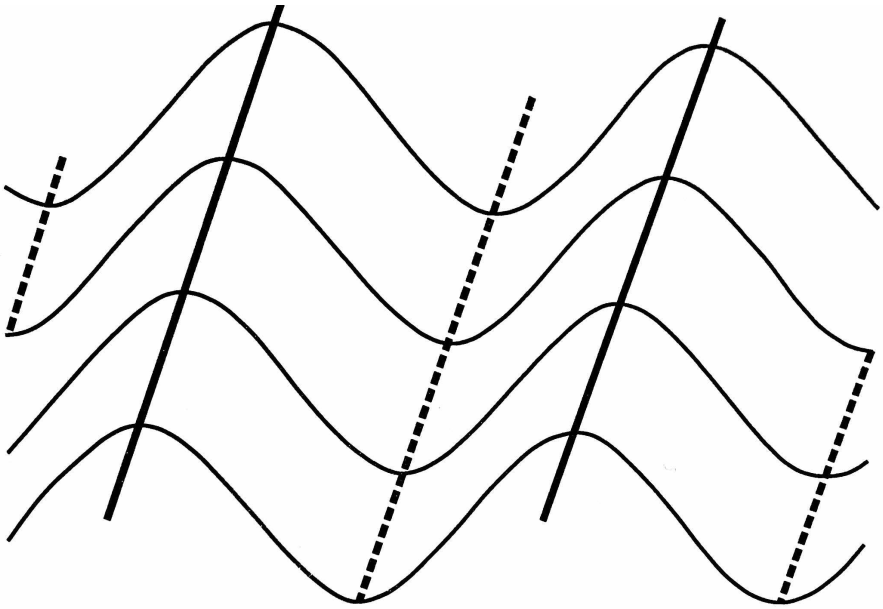
Definition 9.9 | QG Potential Vorticity

The quasi-geostrophic potential vorticity is defined as

$$Q = (\zeta_g + f) + \frac{\partial}{\partial p}\left(\frac{f^2}{\sigma}\frac{\partial \Psi}{\partial p}\right), \tag{9.19}$$

where ζ_g is the geostrophic relative vorticity, $\Psi = \Phi/f_0$ is the geostrophic streamfunction, and $\sigma = -(\alpha/\theta)(\partial\theta/\partial p)$ is the static stability parameter.

QG PV = 절대 소용돌이도 + 정적 안정도 항. 이것은 합으로 정의된다. Ertel PV는 곱으로 정의되는 것과 대조적: $P_E = -g\eta_a \cdot \nabla\theta$. QG PV는 Ertel PV의 선형화(quasi-geostrophic 근사)에 해당한다.



Schematic of NE–SW tilted trough and ridge axes in a midlatitude Rossby wave. Solid lines indicate troughs (low geopotential) and dashed lines indicate ridges. The systematic tilt yields a nonzero zonal-mean eddy momentum flux $(u_E v_E)_Z > 0$, resulting in poleward (northward in NH) momentum transport. 중위도 로스비 파의 기압골/기압능 축 기울기 모식도. NE–SW 방향으로 기울어진 기압골 축 (실선)은 $[u_E v_E] > 0$을 만들어 극 방향 운동량 수송을 일으킨다.

Meridional PV Flux Decomposition

The zonally averaged meridional PV flux can be decomposed into three contributions:

$$(vQ)_Z = \underbrace{f\, v_Z}_{\text{planetary}} + \underbrace{(v\zeta)_Z}_{\text{vorticity flux}} + \underbrace{\left[v \frac{\partial}{\partial p}\left(\frac{f^2}{\sigma}\frac{\partial \Psi}{\partial p} \right) \right]_Z}_{\text{thickness flux}}. \tag{9.20}$$

We now show that each dynamical term reduces to a divergence of a familiar flux.

FIRST TERM: PLANETARY VORTICITY.

Since f depends only on latitude, $f v_Z$ is simply the Coriolis acceleration of the mean meridional flow. In the vertically averaged, steady-state budget this term is negligible ($v_Z \approx 0$).

SECOND TERM: VORTICITY FLUX = CONVERGENCE OF M

Proposition 9.10 | Vorticity Flux Identity

$$(v\zeta)_Z = -\frac{1}{a\cos^2\phi}\frac{\partial}{\partial \phi}\left((uv\cos^2\phi)_Z\right). \tag{9.21}$$

Proof In the QG framework on the beta plane, $\zeta_g = -\partial u/\partial y + \partial v/\partial x$ (approximating $y = a\phi$,

$x = a\cos\phi\,\lambda$). Then

$$(v\zeta)_Z = \left(v\frac{\partial v}{\partial x}\right)_Z - \left(v\frac{\partial u}{\partial y}\right)_Z$$
$$= \frac{\partial}{\partial x}\left(\frac{v^2}{2}\right)_Z - \frac{\partial (uv)_Z}{\partial y} + \left(u\frac{\partial v}{\partial y}\right)_Z .$$

The first term vanishes under zonal averaging. Using the nondivergent geostrophic wind ($\partial u/\partial x + \partial v/\partial y = 0$ in the QG limit), the last term becomes $-(u\,\partial u/\partial x)_Z = -\partial(u^2/2)_Z/\partial x = 0$. Hence, in spherical coordinates with the metric factor:

$$(v\zeta)_Z = -\frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\left((uv)_Z\cos^2\phi\right).$$

■

$(v\zeta)_Z$는 운동량 플럭스 $(uv)_Z$의 수렴과 같다! 즉, 소용돌이도 수송을 알면 운동량 수송을 안다.

THIRD TERM: THICKNESS FLUX = VERTICAL DERIVATI

Proposition 9.11 | Thickness Flux Identity

$$\left[v\frac{\partial}{\partial p}\left(\frac{f^2}{\sigma}\frac{\partial\Psi}{\partial p}\right)\right]_Z = -\frac{\partial}{\partial p}\left[\frac{Rf}{\sigma\,p}(vT)_Z\right]. \tag{9.22}$$

Proof Using the hydrostatic relation $\partial\Psi/\partial p = \partial(\Phi/f_0)/\partial p = -\alpha/f_0 = -RT/(f_0 p)$ and integrating by parts in λ (noting boundary terms vanish by periodicity), one obtains

$$\left[v\frac{\partial}{\partial p}\left(\frac{f^2}{\sigma}\frac{\partial\Psi}{\partial p}\right)\right]_Z = -\frac{\partial}{\partial p}\left(\frac{f^2}{\sigma}\frac{(v\,\partial\Psi/\partial p)_Z}{1}\right) = -\frac{\partial}{\partial p}\left[\frac{Rf}{\sigma\,p}(vT)_Z\right].$$

■

thickness flux는 열 플럭스 $(vT)_Z$의 연직 미분과 관련된다. PV 수송 = 운동량 플럭스 수렴 + 열 플럭스의 연직 미분. 한 공식으로 둘 다 포함!

Full PV Transport Formula

Combining (9.21) and (9.22) (and neglecting the planetary term for the eddy part):

$$\boxed{(v_E Q_E)_Z = -\frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\left((u_E v_E)_Z\cos^2\phi\right) - \frac{\partial}{\partial p}\left[\frac{Rf}{\sigma\,p}(v_E T_E)_Z\right].} \tag{9.23}$$

Remark 9.12

This is a key result: the eddy PV flux is *entirely determined* by the eddy momentum flux and eddy heat flux.

Downgradient PV Transport and Parameterization

Observations and theory show that **eddy PV transport is downgradient**:

$$(v_E Q_E)_Z \approx -K_Q\,\frac{1}{a}\frac{\partial Q_Z}{\partial\phi}, \qquad K_Q > 0. \tag{9.24}$$

Theorem 9.13 | Resolution of the Upgradient Momentum Puzzle

The meridional gradient of mean QG PV is dominated by the planetary vorticity gradient $\beta = \partial f/\partial y$:

$$\frac{1}{a}\frac{\partial Q_Z}{\partial \phi} \approx \beta + (\text{smaller terms}).$$

Since the total PV flux is downgradient and includes both the (downgradient) heat flux and the (upgradient) momentum flux, the upgradient momentum transport is a **necessary consequence** of downgradient PV transport combined with the dominance of β in the PV gradient.

왜 운동량이 upgradient로 수송되는지 이해하기:
1. PV 수송은 downgradient (관측 사실 & 이론).
2. PV gradient $\approx \beta > 0$ (행성 소용돌이도 기울기가 지배적).
3. 따라서 $(v_E Q_E)_Z < 0$ (남쪽으로, downgradient).
4. PV 수송 = 운동량 수렴 + 열 수송의 연직 미분.
5. 열 수송이 downgradient (양수)이므로, 운동량 수렴 항이 이를 보상해야 함.
6. 결론: 운동량은 upgradient! 에디가 PV를 확산시키면서 동시에 운동량을 모은다.

이것이 대기 대순환에서 제트 기류가 유지되는 근본 메커니즘이다!

9.4. Eliassen-Palm (EP) Flux

The EP flux framework provides the most elegant and complete description of eddy–mean flow interaction in the zonally averaged circulation.

Zonally Averaged Equation Set

The complete set of zonally averaged quasi-geostrophic equations on the β-plane is:

$$\frac{\partial u_Z}{\partial t} - f v_Z = \frac{1}{a\cos\phi}\frac{\partial}{\partial \phi}\big((u_E v_E)_Z \cos\phi\big) \cdot \frac{-\cos\phi}{\cos\phi} + F_{\lambda Z} \tag{9.25}$$

$$\frac{\partial \theta_Z}{\partial t} + \omega_Z \frac{\partial \bar{\theta}}{\partial p} = -\frac{1}{a\cos\phi}\frac{\partial}{\partial \phi}\big((v_E \theta_E)_Z \cos\phi\big) + \frac{1}{c_p} H_Z \tag{9.26}$$

$$\frac{1}{a\cos\phi}\frac{\partial}{\partial \phi}(v_Z \cos\phi) + \frac{\partial \omega_Z}{\partial p} = 0 \tag{9.27}$$

$$f\frac{\partial u_Z}{\partial p} = \frac{R}{p}\frac{1}{a}\frac{\partial T_Z}{\partial \phi} \tag{9.28}$$

(9.25): 동서 운동량 방정식 (에디 운동량 플럭스 강제 포함)
(9.26): 열역학 방정식 (에디 열 플럭스 강제 포함)
(9.27): 연속 방정식 (평균 자오면 순환 v_Z, ω_Z 결정)
(9.28): 온도풍 관계 (진단 관계식)

Rearranging (9.25) more explicitly:

$$\frac{\partial u_Z}{\partial t} = f v_Z - \frac{1}{a\cos^2\phi}\frac{\partial}{\partial \phi}\big((u_E v_E)_Z \cos^2\phi\big) + F_{\lambda Z}. \tag{9.29}$$

The Non-Acceleration Theorem

Theorem 9.14 | Non-Acceleration Theorem (Charney and Drazin, 1961)

Under steady, conservative (adiabatic, non-dissipative) conditions ($H = 0$, $F_\lambda = 0$, $\partial/\partial t = 0$):

1. The eddy fluxes force a mean meridional circulation (v_Z, ω_Z), but
2. they do **not** force the zonal-mean zonal wind u_Z or the zonal-mean temperature θ_Z.

That is, the momentum flux convergence is **exactly counterbalanced** by the Coriolis torque of the induced mean meridional circulation:

$$f v_Z = \frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\left((u_E v_E)_Z \cos^2\phi\right) \implies \frac{\partial u_Z}{\partial t} = 0. \tag{9.30}$$

비가속 정리의 의미:
- 단열, 비마찰 조건에서 에디는 평균류를 가속시키지 못한다!
- 에디가 운동량을 수렴시키면, 그에 의해 유도된 평균 자오면 순환의 코리올리 힘이 정확히 상쇄한다.
- 따라서 평균류가 변하려면 반드시 마찰이나 비단열 가열이 있어야 한다.
- 이것은 대기 역학에서 가장 중요한 정리 중 하나!

Proof [Sketch] From (9.25) with $F_{\lambda Z} = 0$ and $\partial u_Z/\partial t = 0$:

$$f v_Z = \frac{1}{a\cos^2\phi}\frac{\partial}{\partial\phi}\left((u_E v_E)_Z \cos^2\phi\right).$$

From (9.26) with $H_Z = 0$ and $\partial\theta_Z/\partial t = 0$:

$$\omega_Z \frac{\partial\bar\theta}{\partial p} = -\frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left((v_E\theta_E)_Z \cos\phi\right).$$

These determine v_Z and ω_Z in terms of the eddy fluxes, but u_Z and θ_Z remain unchanged. The eddies drive a mean meridional circulation that exactly compensates their direct forcing of the mean state. ■

Definition of the EP Flux Vector

Definition 9.15 | Eliassen–Palm Flux

The **EP flux vector** $\mathbf{F}$ is defined (in the ϕ–p plane) as

$$\boxed{\mathbf{F} = \Bigg(\underbrace{-a(u_E v_E)_Z\cos\phi}_{F^{(\phi)}},\ \underbrace{\frac{f a\cos\phi\,(v_E\theta_E)_Z}{\partial\bar\theta/\partial p}}_{F^{(p)}}\Bigg).} \tag{9.31}$$

The meridional component $F^{(\phi)}$ is related to the eddy momentum flux, and the vertical component $F^{(p)}$ is related to the eddy heat flux.

EP 플럭스 벡터의 두 성분:
$F^{(\phi)}$: $-(u_E v_E)_Z$ 에 비례 → 운동량 플럭스
$F^{(p)}$: $(v_E\theta_E)_Z$ 에 비례 → 열 플럭스
이 두 에디 플럭스를 하나의 벡터로 통합한 것이 EP 플럭스!

EP Flux Divergence and Wave Activity

The zonally averaged zonal momentum equation can be written compactly as

$$\frac{\partial u_Z}{\partial t} - f v_Z = \frac{1}{a\cos\phi}\nabla\cdot\mathbf{F} + F_{\lambda Z}, \tag{9.32}$$

where

$$\nabla\cdot\mathbf{F} \equiv \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left(F^{(\phi)}\cos\phi\right) + \frac{\partial F^{(p)}}{\partial p}. \tag{9.33}$$

Theorem 9.16 | Non-Acceleration in EP Flux Form

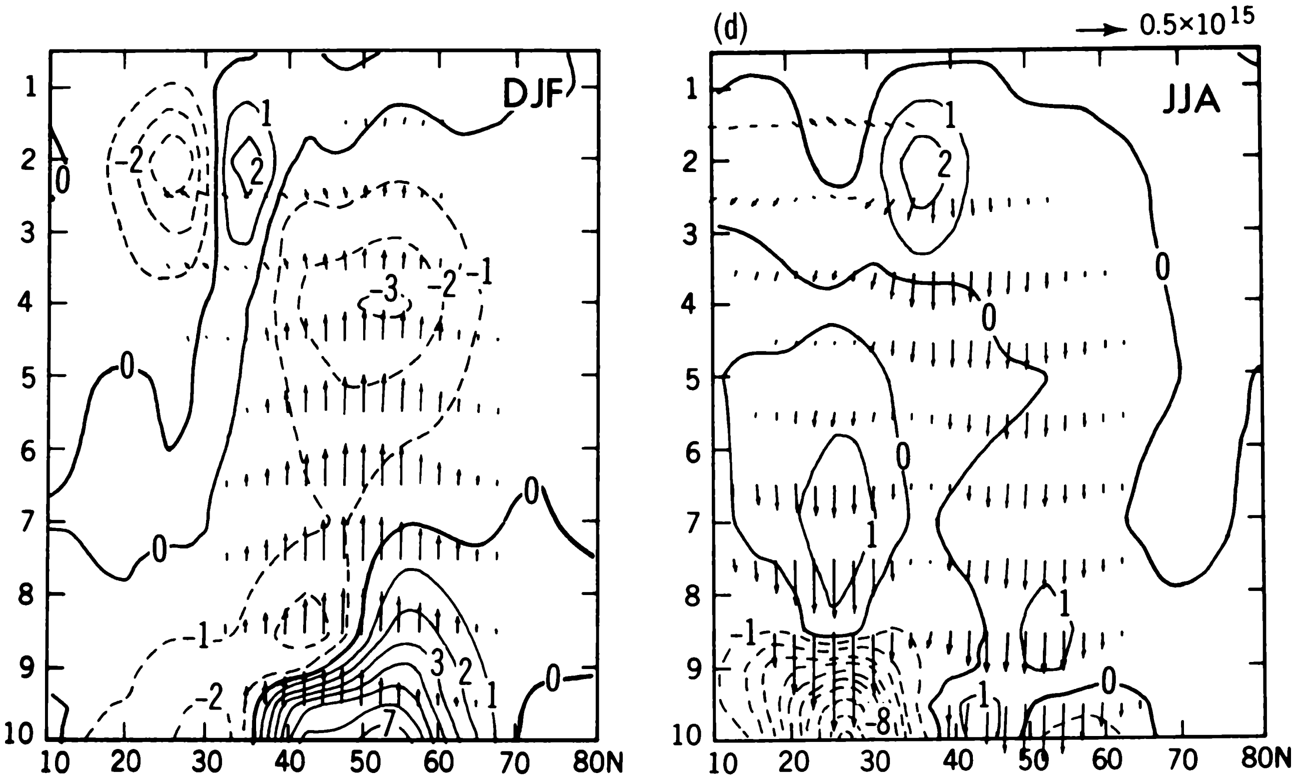
The non-acceleration condition is equivalent to

$$\nabla\cdot\mathbf{F} = 0. \tag{9.34}$$

The zonal-mean flow is accelerated (decelerated) where the EP flux is convergent (divergent):

$$\frac{\partial u_Z}{\partial t} \propto \nabla\cdot\mathbf{F}.$$

$\nabla\cdot\mathbf{F} > 0$: 서풍 가속 (EP flux가 발산하는 곳)
$\nabla\cdot\mathbf{F} < 0$: 동풍 가속 (EP flux가 수렴하는 곳, 파동이 소산/쇄파)
$\nabla\cdot\mathbf{F} = 0$: 비가속 조건 (보존적 파동의 전파)



EP flux cross-sections (arrows) and EP flux divergence (contours, $\mathrm{m\,s^{-1}\,day^{-1}}$) for DJF (left) and JJA (right) in the Northern Hemisphere. Arrows indicate wave-activity propagation: dominantly upward (poleward heat flux) and equatorward (poleward momentum flux). Convergence ($\nabla\cdot\mathbf{F} < 0$, dashed contours) indicates wave dissipation and easterly acceleration of the mean flow; divergence ($\nabla\cdot\mathbf{F} > 0$, solid contours) indicates westerly acceleration. EP 플럭스 단면도 (화살표)와 EP 플럭스 발산 (등치선): DJF (좌)와 JJA (우). 화살표는 파동 활동도의 전파 방향을 나타내며, 주로 위쪽 (열 플럭스)과 적도 쪽 (운동량 플럭스)을 향한다. 수렴 영역에서 동풍 가속, 발산 영역에서 서풍 가속이 발생한다.

Remark 9.17 | Interpretation of EP Flux Diagrams

In EP flux cross-section diagrams:

- Arrows show the direction of wave-activity propagation.
- **Upward** arrows indicate poleward heat flux ($(v_E\theta_E)_Z > 0$ in NH).
- **Equatorward** arrows indicate poleward momentum flux ($(u_E v_E)_Z > 0$ in NH $\Rightarrow F^{(\phi)} < 0$).
- Divergence/convergence of arrows indicates wave sources/sinks, hence acceleration/deceleration of the mean flow.

The General Case: Heating and Friction

In the general (non-conservative) case, the mean meridional circulation (v_Z, ω_Z) is forced by three agents:

1. **Mean diabatic heating** $\overline{H}_{ZM}$
2. **Mean friction** $F_{\lambda Z}$
3. **Eddy forcing** via $\nabla \cdot \mathbf{F}$

The mean meridional circulation satisfies

$$f v_Z = \frac{1}{a\cos\phi}\nabla\cdot\mathbf{F} + F_{\lambda Z} - \frac{\partial u_Z}{\partial t}, \tag{9.35}$$

$$\omega_Z \frac{\partial\bar{\theta}}{\partial p} = \frac{1}{c_p}H_Z - \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left((v_E\theta_E)_Z\cos\phi\right) - \frac{\partial\theta_Z}{\partial t}. \tag{9.36}$$

Remark 9.18 | The Ferrel Cell

The midlatitude **Ferrel cell** is an indirect (thermally indirect) circulation: warm air sinks and cold air rises. It is **not** driven by differential heating (like the Hadley cell) but is forced primarily by the eddies through the EP flux divergence. The eddy momentum flux convergence drives a poleward flow aloft ($f v_Z \approx \nabla\cdot\mathbf{F}/(a\cos\phi)$), which by continuity requires descent in the subtropics and ascent at high latitudes.

페렐 셀은 에디에 의해 구동되는 간접 순환! 열적으로는 따뜻한 곳에서 하강, 차가운 곳에서 상승하므로 ‘간접적’이다. 해들리 셀과 달리 직접적인 열적 구동이 아니라 에디 강제에 의한 응답이다.

9.5. Transformed Eulerian Mean (TEM) Formulation

The standard Eulerian-mean equations mask the net effect of eddies because the eddy-driven mean meridional circulation partially cancels the direct eddy forcing (non-acceleration theorem). The **TEM formulation** introduces a **residual circulation** that reveals the true eddy impact.

Residual Mean Meridional Circulation

Definition 9.19 | Residual Circulation

The residual meridional and vertical velocities are defined as

$$v^* \equiv v_Z - \frac{\partial}{\partial p}\left(\frac{(v_E\theta_E)_Z}{\partial\bar{\theta}/\partial p}\right), \tag{9.37}$$

$$\omega^* \equiv \omega_Z + \frac{1}{a\cos\phi}\frac{\partial}{\partial\phi}\left(\frac{(v_E\theta_E)_Z\cos\phi}{\partial\bar{\theta}/\partial p}\right). \tag{9.38}$$

잔차 순환 = Eulerian 평균 순환 − 에디에 의한 Stokes drift 보정. 이것은 실제 물질 수송에 더 가까운 순환이다.

The TEM momentum and thermodynamic equations become:

$$\frac{\partial u_Z}{\partial t} - fv^* = \frac{1}{a\cos\phi}\nabla\cdot\mathbf{F} + F_{\lambda Z}, \tag{9.39}$$

$$\frac{\partial \theta_Z}{\partial t} + \omega^*\frac{\partial\bar{\theta}}{\partial p} = \frac{1}{c_p}H_Z. \tag{9.40}$$

Remark 9.20

In the TEM formulation, (9.40) contains **no explicit eddy flux terms**—the eddy heat flux has been absorbed into the definition of ω^*. The entire eddy effect on the mean flow appears only through $\nabla\cdot\mathbf{F}$ in (9.39).

TEM의 장점:
- 열역학 방정식에서 에디 항이 사라짐 → 잔차 순환이 단열 가열에 의해 직접 결정.
- 운동량 방정식에서 에디 효과가 오직 $\nabla\cdot\mathbf{F}$ 하나로 표현.
- 비가속 정리가 자명해짐: $\nabla\cdot\mathbf{F}=0$이면 $v^*=0$, $\omega^*=0$, 따라서 $\partial u_Z/\partial t=0$.
- 잔차 순환은 Brewer–Dobson 순환과 밀접 (성층권에서 특히 유용).

9.6. Globally Averaged Energy Balance

The maintenance of the general circulation ultimately rests on the **differential radiative heating** between the tropics and poles. This section provides a quantitative overview.

Top-of-Atmosphere Radiation Balance

Definition 9.21 | Net Radiation at TOA

The net downward radiation at the top of the atmosphere is

$$R_{\text{net}}(\phi) = S_{\text{abs}}(\phi) - F_{\text{OLR}}(\phi), \tag{9.41}$$

where S_{abs} is the absorbed shortwave (solar) radiation and F_{OLR} is the outgoing longwave radiation.

The key observational facts:

- **Absorbed solar radiation**: ranges from > 300 W/m^2 in the tropics to < 100 W/m^2 at the poles—a factor of ~ 3 variation.
- **OLR**: much more uniform, ranging from ~ 250 W/m^2 in the tropics to ~ 175 W/m^2 at the poles—less than a factor of 1.5 variation.
- **Net radiation crossover**: $R_{\text{net}} = 0$ at $\sim 38°$ N/S latitude. Equatorward: net surplus; poleward: net deficit.

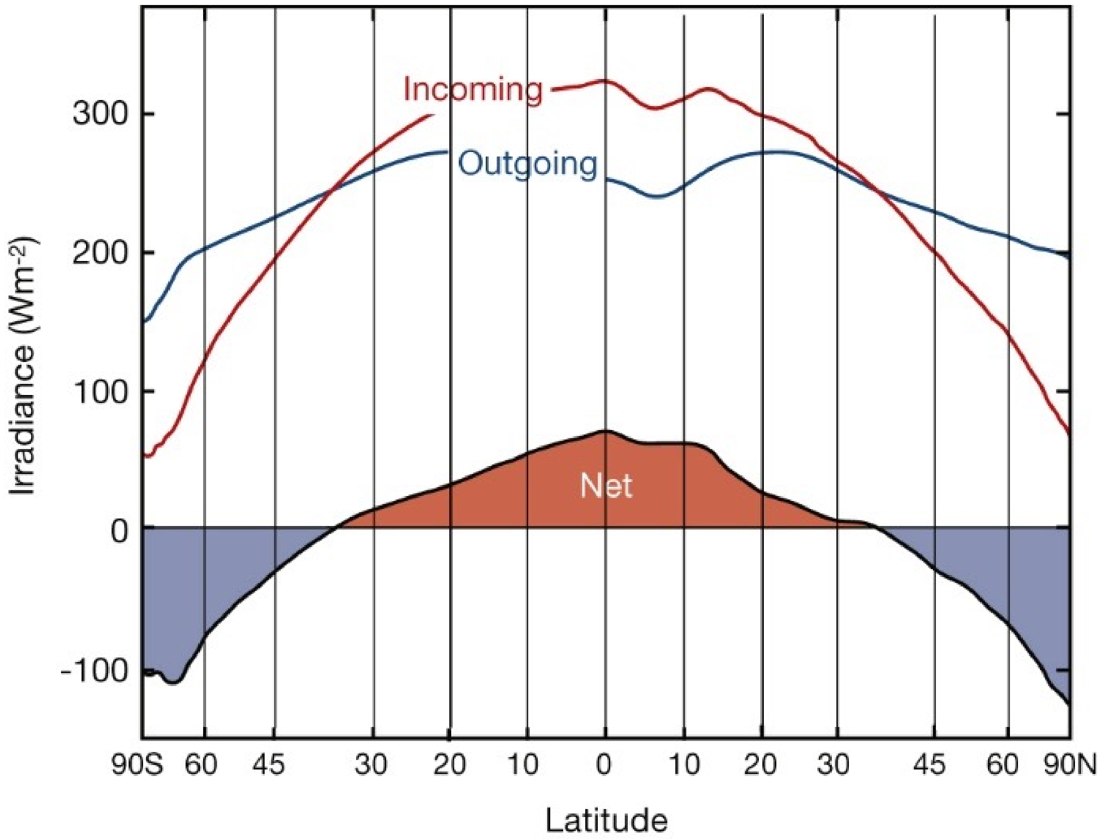
태양 복사는 적도–극 사이 3배 이상 차이가 나지만, OLR은 1.5배 차이에 불과. 이 불균형이 대기와 해양의 에너지 수송을 구동하는 원동력이다!

Poleward Energy Transport

From the requirement of global energy balance, the net poleward energy transport across latitude ϕ is

$$\mathcal{F}(\phi) = 2\pi a^2 \int_{\phi}^{\pi/2} R_{\text{net}}(\phi')\cos\phi'\,d\phi'. \tag{9.42}$$

Observational estimates yield:



Zonally averaged TOA radiation balance as a function of latitude: absorbed solar (incoming, red), outgoing longwave radiation (blue), and net radiation (shaded). The tropics have a net radiative surplus (red shading) and the extratropics a net deficit (blue shading), with the crossover near ∼38° latitude. This differential heating is the fundamental driver of the general circulation. 위도별 TOA 복사 수지: 흡수 태양 복사 (적색), 사출 장파 복사 (청색), 순 복사 (음영). 열대의 복사 잉여와 고위도의 복사 적자가 ∼38°에서 교차하며, 이 차등 가열이 대기 대순환의 근본 원동력이다.

- Total transport peaks at ~ 5–6 PW at $\sim 35°$ latitude.
- **Atmospheric transport** dominates in the **extratropics** ($> 30°$).
- **Oceanic transport** is significant mainly in the **tropics** ($< 20°$), contributing ~ 2 PW.

에너지 수송의 분담:
- 전구 총 수송 최대: ∼5–6 PW (1 PW = 10^{15} W), $\sim 35°$
- 대기: 중위도에서 지배적—주로 에디에 의한 현열 + 잠열 수송
- 해양: 열대에서 중요—서안 경계류 (걸프 스트림, 쿠로시오)
- 고위도에서는 거의 전부 대기가 담당

Decomposition of Atmospheric Energy Transport

The total atmospheric energy transport can be decomposed:

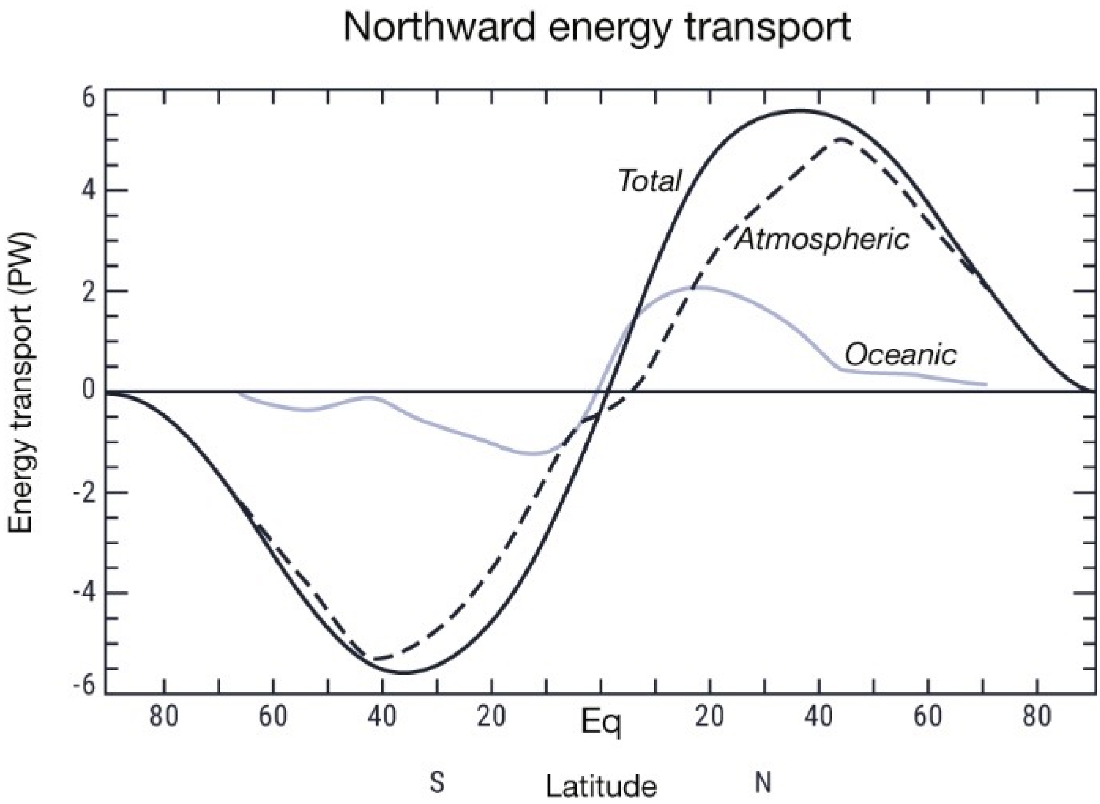
$$\mathcal{F}_{\text{atm}} = \underbrace{\mathcal{F}_{\text{MMC}}}_{\text{mean meridional circ.}} + \underbrace{\mathcal{F}_{\text{SE}}}_{\text{stationary eddies}} + \underbrace{\mathcal{F}_{\text{TE}}}_{\text{transient eddies}} . \tag{9.43}$$

Tropics ($|\phi| < 30°$):

- Hadley cell (MMC) dominates
- Transports dry static energy poleward
- Also large latent heat transport equatorward (lower branch)

Extratropics ($|\phi| > 30°$):

- Transient eddies dominate
- Sensible heat $c_p(v_E T_E)_Z$
- Latent heat $L_v(v_E q_E)_Z$
- Potential energy $(v_E \Phi_E)_Z$



Observed northward energy transport (PW) as a function of latitude: total (dashed), atmospheric (solid black), and oceanic (grey). The total poleward transport peaks at ~5–6 PW near 35° latitude; atmospheric transport dominates in the extratropics while oceanic transport contributes significantly in the tropics. 관측된 극 방향 에너지 수송 (PW): 총 수송 (점선), 대기 수송 (실선), 해양 수송 (회색). 총 수송은 ~35°에서 ~5–6 PW로 최대이며, 중위도에서는 대기, 열대에서는 해양이 지배적이다.

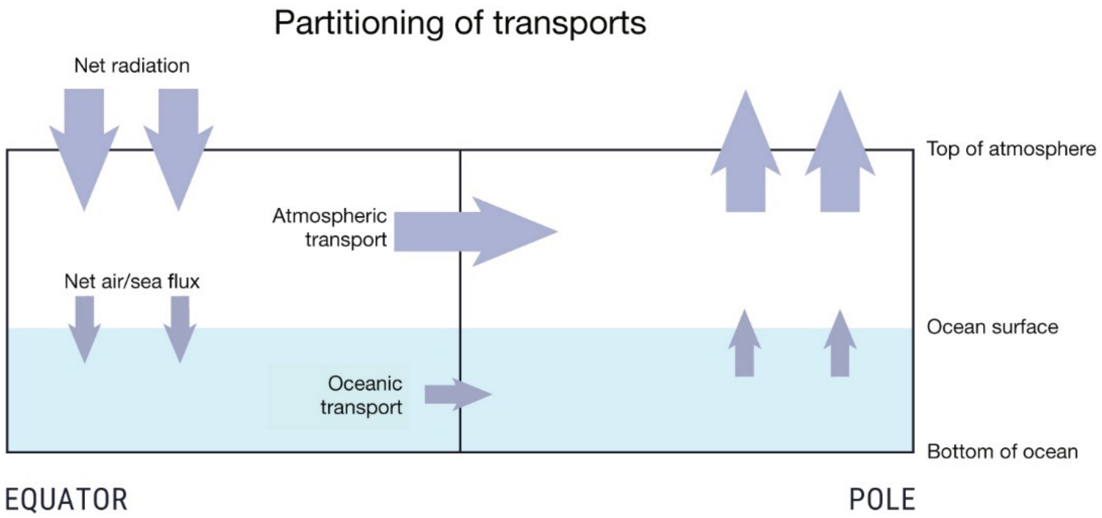
열대에서는 해들리 셀, 중위도에서는 에디(특히 이동성 저기압)가 에너지 수송의 주역.

Summary: The General Circulation as an Energy Engine

> The atmosphere acts as a heat engine that:
> 1. Absorbs energy preferentially in the tropics,
> 2. Transports it poleward via mean circulations and eddies,
> 3. Radiates it to space more uniformly across all latitudes.
>
> The eddies (baroclinic waves) are the primary agents of poleward transport in the extratropics, maintaining the observed temperature distribution against radiative forcing, and simultaneously maintaining the jet streams through upgradient momentum transport.

Chapter 4 핵심 정리:
1. 열 수송: downgradient, 에디가 지배, 확산으로 근사 가능.
2. 운동량 수송: upgradient, 에디가 지배, 확산으로 근사 불가.
3. PV 수송: downgradient, 운동량 + 열 수송을 통합. PV 확산으로 upgradient 운동량 수송 설명 가능.
4. EP 플럭스: 에디–평균류 상호작용의 통합적 기술. $\nabla \cdot \mathbf{F}$가 평균류 가속을 결정.
5. 비가속 정리: 보존적 파동은 평균류를 가속시키지 못함. 마찰/비단열이 있어야 변화.
6. TEM: 잔차 순환이 실제 수송에 더 가까움. 에디 효과가 $\nabla \cdot \mathbf{F}$로 깔끔하게 표현.
7. 에너지 균형: 적도–극 복사 불균형 → 대기/해양 수송 → 일반 순환 유지.

References: *Wallace, Battisti, Thompson, and Hartmann, "The Atmospheric General Circulation," Cambridge University Press*; *Holton and Hakim, "An Introduction to Dynamic Meteorology," Ch. 10.2–10.3*; *Vallis, "Atmospheric and Oceanic Fluid Dynamics," Ch. 12–13.*



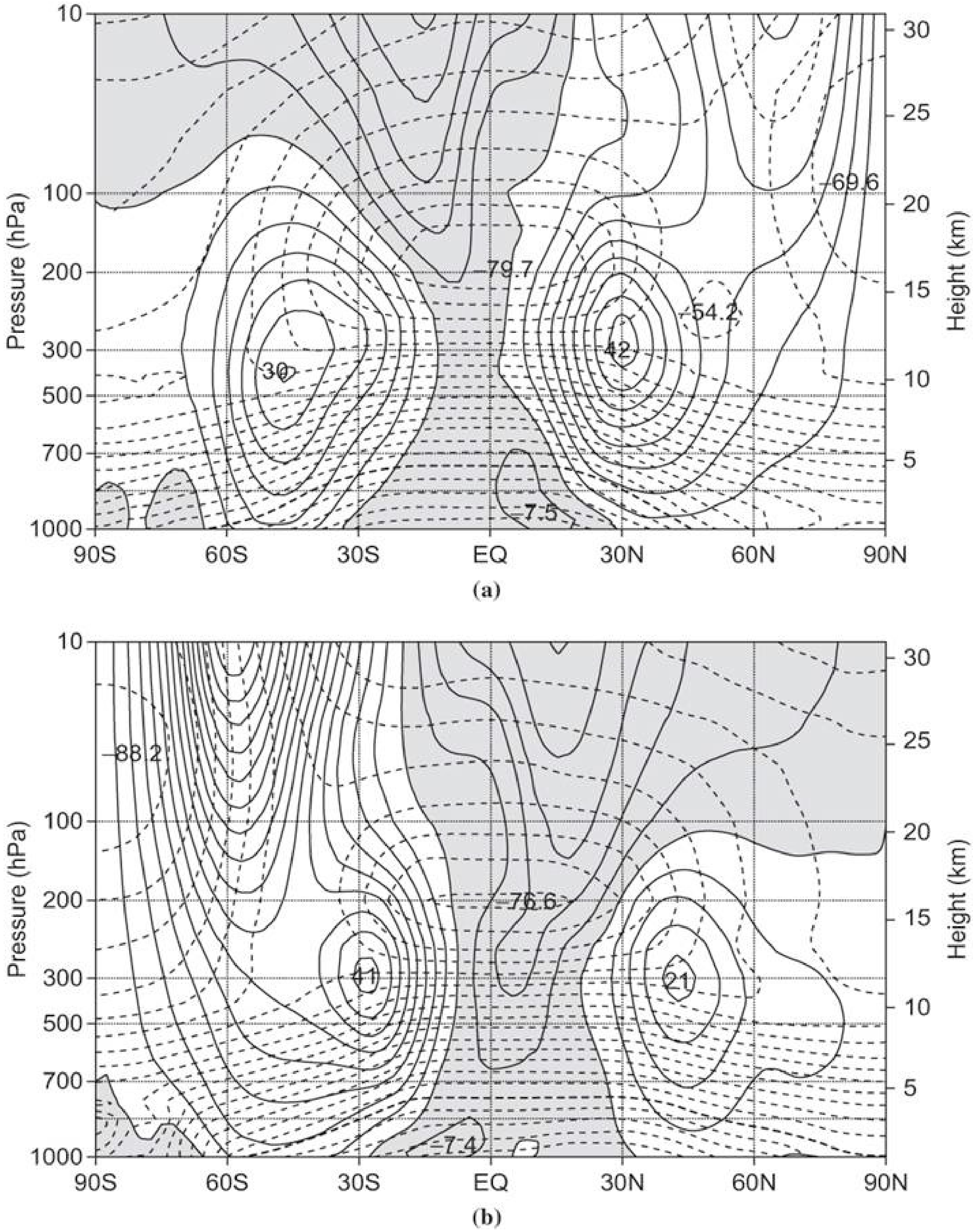
Schematic of the partitioning of poleward energy transports between the atmosphere and ocean. Net radiation is absorbed at the top of the atmosphere (mostly in the tropics) and radiated back to space (mostly at higher latitudes). The atmosphere and ocean each carry a portion of the required poleward transport, exchanging energy at the ocean surface through air–sea fluxes. 대기–해양 간 극 방향 에너지 수송 분배의 모식도. 적도에서 순 복사 에너지를 받아 대기 수송과 해양 수송으로 나뉘어 극 방향으로 운반된 후, 고위도에서 우주로 방출된다.

QUASI-GEOSTROPHIC ANALYSIS

10. OBSERVED STRUCTURE OF EXTRATROPICAL CIRCULATION

Before developing the mathematical apparatus of quasi-geostrophic (QG) theory, we briefly review the observational features that motivate it. The extratropical atmosphere is characterised by strong **asymmetries** in both space and time: synoptic disturbances are not uniformly distributed but instead are concentrated in **storm tracks** that coincide with the regions of strongest time-mean zonal winds.

경압 불안정 = 중위도 종관 규모 시스템의 주요 불안정 메커니즘



Observed zonally averaged cross-sections of (a) zonal wind $\bar{u}$ and (b) temperature $\bar{T}$ (DJF climatology). Solid contours denote westerlies / higher temperatures; dashed contours denote easterlies / lower temperatures. Note the subtropical jet maxima near 200 hPa and the strong meridional temperature gradient in the midlatitudes. 관측된 동서 평균 바람과 온도의 남북–연직 단면. 중위도의 강한 자오 온도 경도와 아열대 제트가 경압 불안정의 배경이 된다.

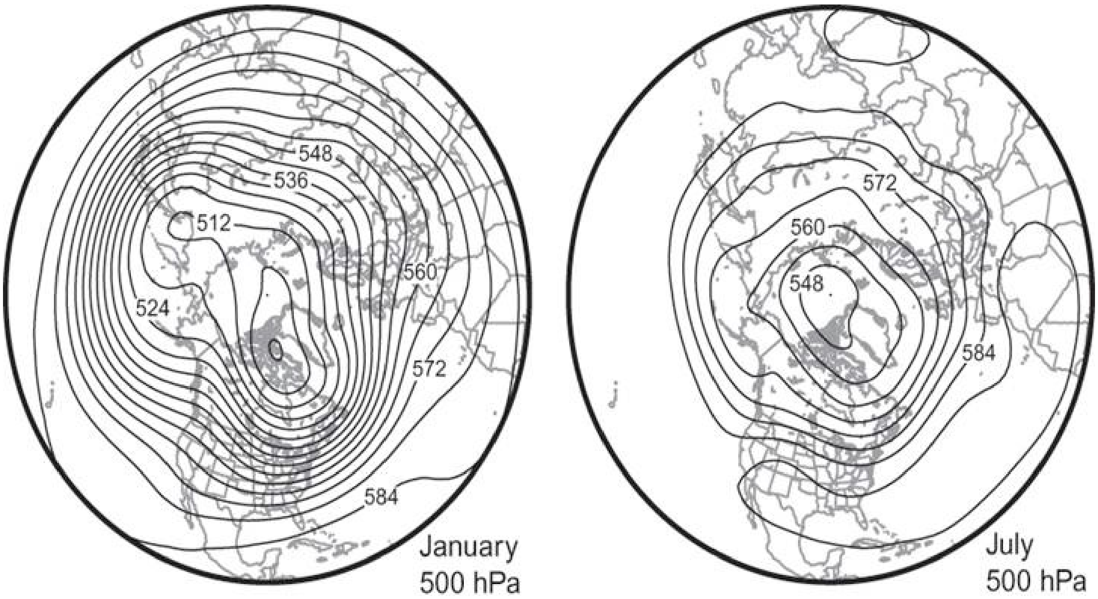
10.1. Storm Tracks and Jet Streams

Synoptic-scale disturbances—the familiar travelling cyclones and anticyclones of the midlatitudes—develop preferentially in regions of **maximum time-mean zonal wind**: the western Pacific jet and the North Atlantic jet. These regions of enhanced transient eddy activity are called **storm tracks**.

Remark 10.1 | Properties of Storm Tracks

- Storm tracks lie downstream of the strongest **baroclinicity** (meridional temperature gradient / vertical wind shear).
- They are maintained by a combination of local baroclinic energy conversion and downstream energy propagation.
- Strongest in winter, when meridional temperature gradients are largest.

Storm track는 시간 평균 제트 부근에서 발생한다. 서태평양과 북대서양의 제트가 가장 강하므로, 여기에서 종관 규모 요란이 가장 활발하다. 겨울에 자오 온도 경도가 크므로 storm track도 겨울에 가장 강하다.



Time-mean 500 hPa geopotential height for January (left) and July (right) over the Northern Hemisphere. The deep troughs over eastern North America and eastern Asia in January mark the entrance regions of the major storm tracks. 1월과 7월의 500 hPa 시간 평균 지위고도. 겨울에 동아시아와 북미 동해안의 깊은 기압골이 storm track 입구를 표시한다.

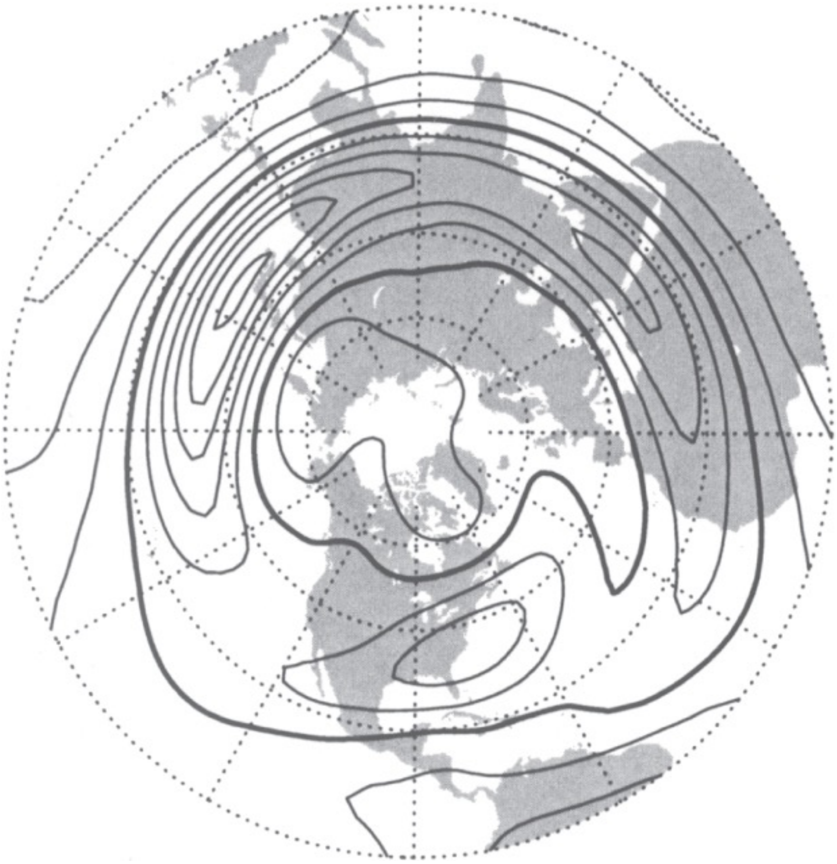
10.2. Baroclinic Instability

The **primary instability mechanism** for synoptic-scale systems in the midlatitudes is **baroclinic instability**. It extracts available potential energy (APE) from the mean flow and converts it into eddy kinetic energy. The necessary conditions for baroclinic instability are:

1. A nonzero meridional temperature gradient (equivalently, vertical wind shear via the thermal wind relation).
2. The disturbance must **tilt westward with height** so that warm air rises and cold air sinks, converting mean-flow PE to wave KE.

Baroclinic instability가 작동하려면:
- 자오 온도 경도 $\partial T/\partial y \neq 0$ (또는 연직 바람 시어 $\partial u/\partial z \neq 0$)
- 교란이 고도에 따라 서쪽으로 기울어져야 함 (westward tilt with height)

서쪽 기울기가 있어야 따뜻한 공기가 상승하고, 차가운 공기가 하강하여 평균류의 위치에너지를 파동의 운동에너지로 전환할 수 있다.



Northern Hemisphere mean 500 hPa geopotential height showing the planetary-scale wave pattern. Contours illustrate the quasi-stationary Rossby wave troughs and ridges that steer synoptic disturbances. 북반구 평균 500 hPa 지위고도: 행성 규모 정상 로스비파의 기압골/기압능 패턴.

10.3. Baroclinic Wave Lifecycle

The lifecycle of a typical baroclinic wave proceeds through three stages:

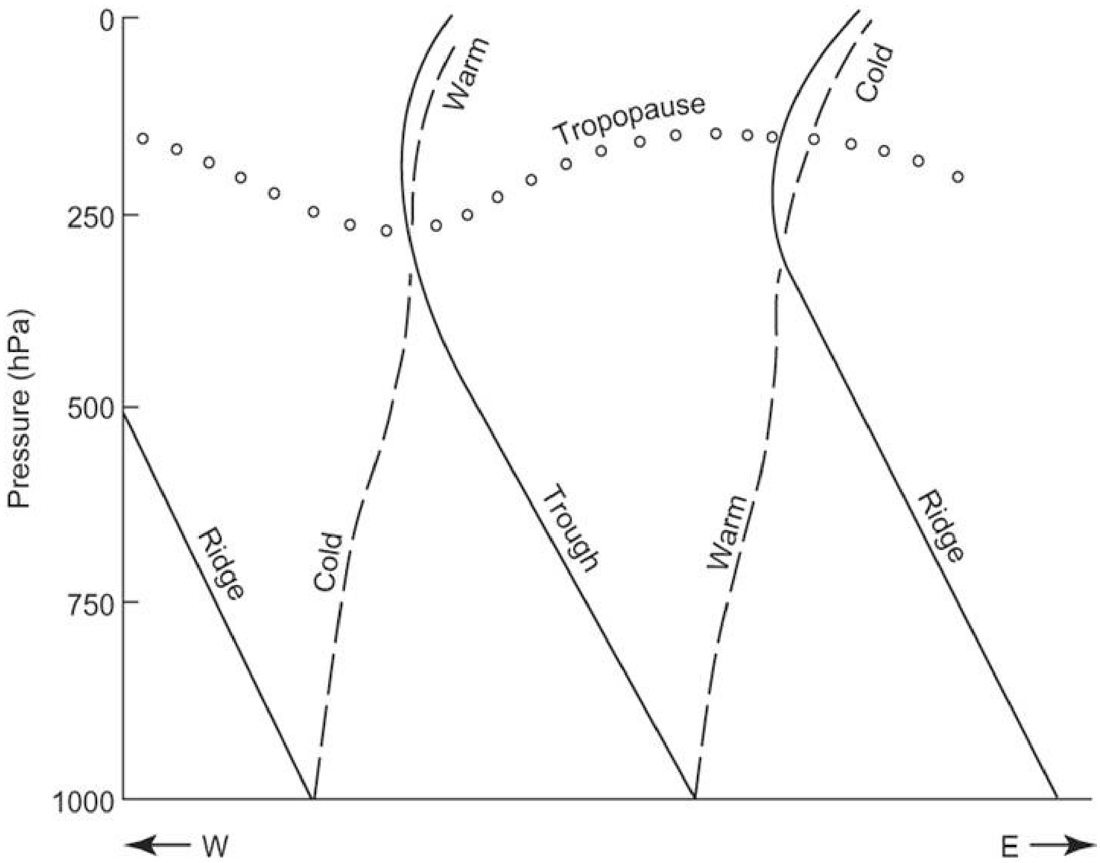
1. **Incipient developing stage**: A small-amplitude perturbation with a westward tilt with height begins to grow by extracting APE from the mean meridional temperature gradient.
2. **Rapid development stage**: The disturbance amplifies rapidly. Upper-level PV anomalies and surface temperature anomalies interact cooperatively, reinforcing each other (mutual amplification).
3. **Occlusion stage**: The wave matures, the surface fronts wrap around the low centre, the warm sector narrows, and the system begins to decay as the westward tilt is lost and energy conversion ceases.

경압파 생애주기: 발달 초기 → 급속 발달 → 폐색. 서쪽 기울기가 사라지면 에너지 변환이 멈추고 파동이 소멸.

11. DERIVATION OF QG EQUATIONS (z-COORDINATE)

The quasi-geostrophic (QG) equations constitute the single most important dynamical framework for understanding midlatitude weather systems. They are derived by systematically exploiting the **smallness of the Rossby number**, $\mathrm{Ro} = U/(fL) \sim 0.1$, to filter out fast-moving inertia-gravity waves while retaining the slow, balanced motions that dominate the extratropics.

QG 방정식 = Rossby 수가 작을 때 $(\mathrm{Ro} \sim 0.1)$, 느린 균형 운동만을 기술하는 근사 체계



Vertical cross-section of a developing baroclinic wave. Solid lines denote geopotential height contours (ridges and troughs); dashed lines denote isotherms. Note the characteristic **westward tilt with height** of the trough and ridge axes: warm air rises ahead of the trough while cold air sinks behind it, converting mean-flow APE to eddy KE. 경압파의 연직 단면. 기압골/기압능 축이 고도에 따라 서쪽으로 기울어져 있어 따뜻한 공기는 상승, 차가운 공기는 하강 — APE에서 KE로의 에너지 변환이 일어남.

11.1. Boussinesq Approximation

Definition 11.1 | Boussinesq Approximation

In the Boussinesq approximation, density variations are neglected everywhere except in the **buoyancy term** of the vertical momentum equation. A constant reference density ρ_0 replaces ρ in all other terms. The governing equations become:

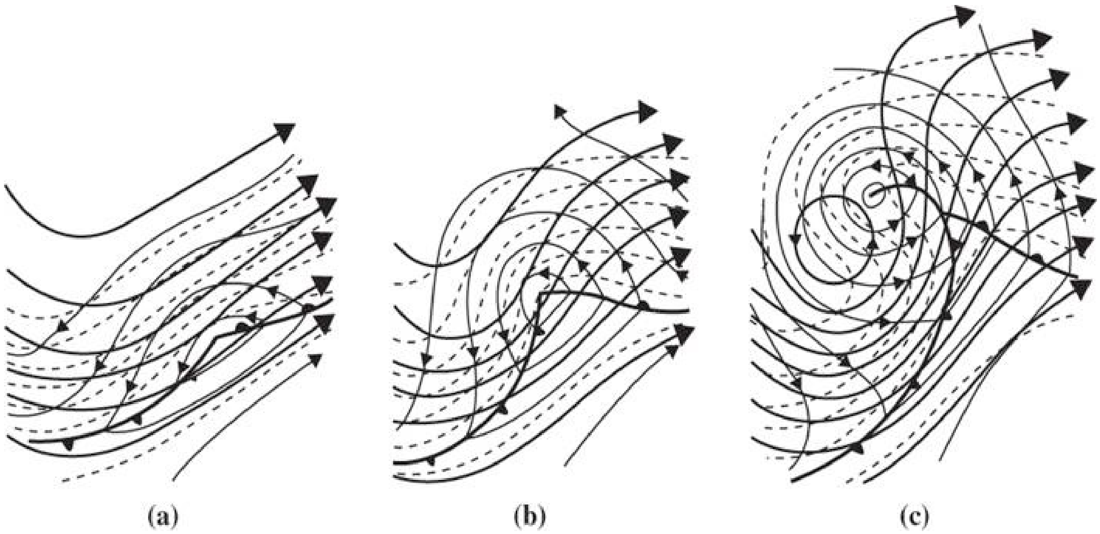
$$\frac{Du}{Dt} = -\frac{1}{\rho_0}\frac{\partial p}{\partial x} + fv + F_x, \tag{11.1}$$

$$\frac{Dv}{Dt} = -\frac{1}{\rho_0}\frac{\partial p}{\partial y} - fu + F_y, \tag{11.2}$$

$$\frac{Dw}{Dt} = -\frac{1}{\rho_0}\frac{\partial p}{\partial z} - g + g\frac{\theta}{\theta_0} + F_z, \tag{11.3}$$

where $\theta_{\text{tot}}(x, y, z, t) = \theta_0(z) + \theta(x, y, z, t)$ is the total potential temperature decomposed into a horizontally uniform reference profile $\theta_0(z)$ and a perturbation θ. 밀도 변화가 작으므로 부력항을 제외하고는 상수 ρ_0로 근사

Boussinesq 근사의 핵심 아이디어:
- 대기에서 밀도 변동 $\rho'/\rho_0 \sim 1\%$이므로, 관성항에서의 밀도 차이는 무시 가능
- 그러나 중력과 결합하면 $g\rho'/\rho_0$는 작지 않으므로 부력항에서는 반드시 유지



Schematic of baroclinic wave lifecycle at the surface: (a) incipient developing stage with small-amplitude perturbations, (b) rapid development stage with amplifying warm and cold fronts, and (c) occlusion stage where the warm sector narrows and the system begins to decay. Solid lines: isobars; dashed lines: fronts; arrows: surface wind. 경압파 생애주기 모식도: (a) 발달 초기, (b) 급속 발달기 (전선 강화), (c) 폐색기 (온난역 축소, 시스템 소멸).

- 연속 방정식이 $\nabla \cdot \mathbf{V} + \partial w/\partial z = 0$ (비압축성)으로 단순화

Under the Boussinesq approximation the **mass continuity equation** simplifies to the incompressible form:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{11.4}$$

11.2. Geostrophic Relations

At leading order in the Rossby number, the horizontal momentum equations (11.1)–(11.2) reduce to **geostrophic balance**:

$$u_g = -\frac{1}{\rho_0 f}\frac{\partial p}{\partial y}, \qquad v_g = +\frac{1}{\rho_0 f}\frac{\partial p}{\partial x}. \tag{11.5}$$

In vector form:

$$\mathbf{V}_g = \frac{1}{\rho_0 f}\hat{k} \times \nabla p. \tag{11.6}$$

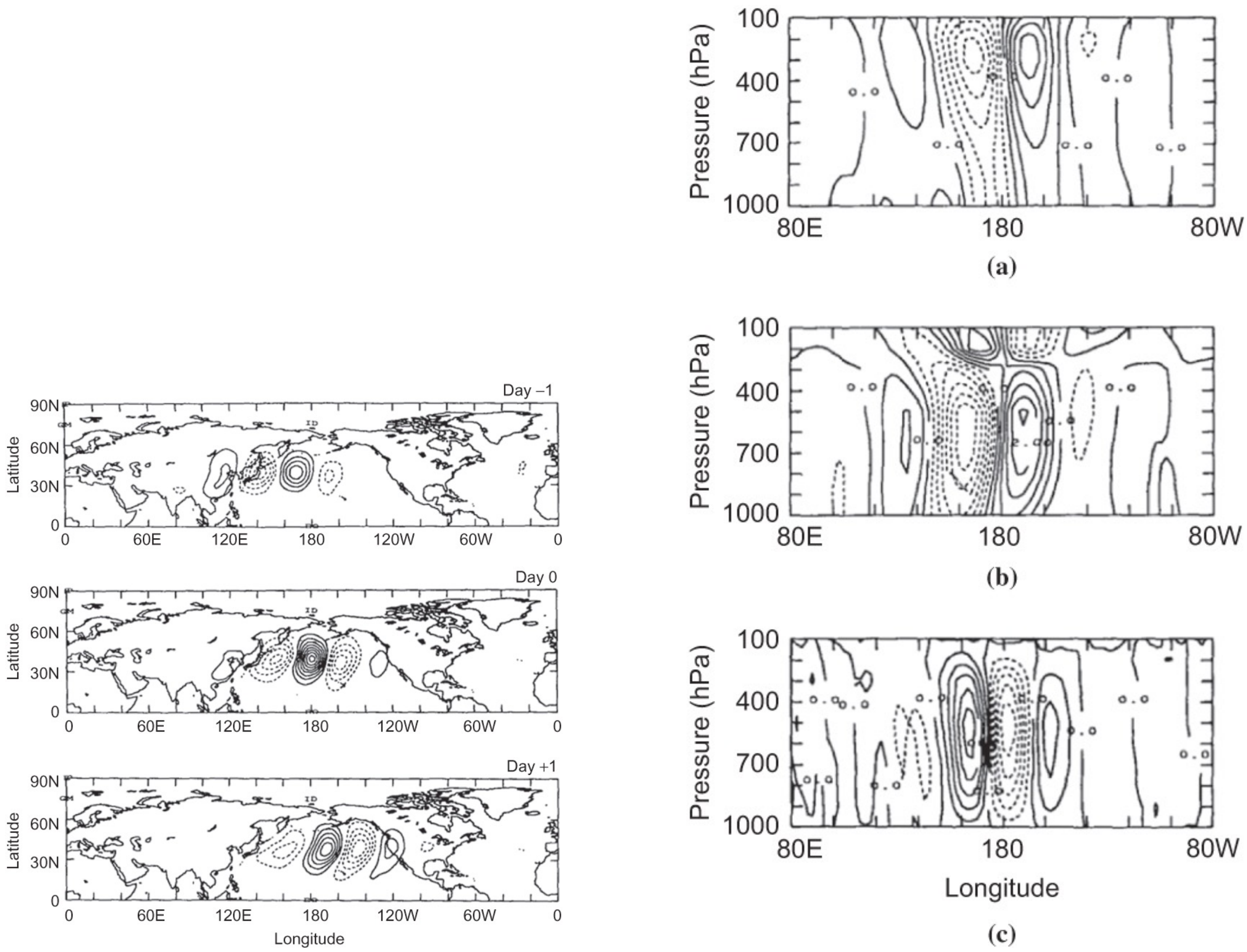
Remark 11.2 | Properties of Geostrophic Wind

- $\mathbf{V}_g$ is **non-divergent**: $\nabla \cdot \mathbf{V}_g = 0$ (on an f-plane).
- $\mathbf{V}_g$ is parallel to isobars, with low pressure to the left in the Northern Hemisphere.
- The geostrophic wind completely determines the horizontal pressure field (and vice versa) at any given level.

Geostrophic Vorticity

The vertical component of the **geostrophic vorticity** is obtained from $\mathbf{V}_g$:

$$\zeta_g = \frac{\partial v_g}{\partial x} - \frac{\partial u_g}{\partial y} = \frac{1}{\rho_0 f}\left(\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2}\right) = \frac{1}{\rho_0 f}\nabla^2 p. \tag{11.7}$$



Observed storm-track disturbance composite. Left: 500 hPa geopotential height anomaly at Day −1, Day 0, and Day +1, showing eastward propagation of the upper-level wave. Right: longitude–pressure cross-sections at the corresponding times, illustrating the westward tilt with height during the developing stage and the progressive vertical alignment toward occlusion. 관측 storm track 교란 합성장. 왼쪽: 500 hPa 고도 이상 (동진하는 상층파). 오른쪽: 경도–기압 단면 (발달기의 서쪽 기울기와 폐색기의 연직 정렬).

지균 소용돌이도는 압력장의 수평 라플라시안에 비례 — 압력장만으로 완전히 결정됨

QG Material Derivative

In the QG framework, the material derivative is evaluated using *only the geostrophic wind* for horizontal advection:

$$\frac{d}{dt} \equiv \frac{\partial}{\partial t} + u_g \frac{\partial}{\partial x} + v_g \frac{\partial}{\partial y}. \tag{11.8}$$

QG material derivative에서 주의할 점:
- 수평 이류는 지균풍만 사용 (비지균풍에 의한 이류는 $O(\mathrm{Ro})$ 작음)
- 연직 이류 $w\,\partial/\partial z$는 포함되지 않음 (열역학 방정식에서만 연직 운동이 나타남)
- 이것이 QG 방정식이 수학적으로 다루기 쉬운 핵심 이유

11.3. Hydrostatic Balance and Thermal Wind

The vertical momentum equation (11.3), under the hydrostatic and Boussinesq approximations, yields the **perturbation hydrostatic relation**:

$$\frac{\partial p}{\partial z} = \frac{\rho_0 \, g}{\theta_0} \, \theta. \tag{11.9}$$

Differentiating the geostrophic relations (11.5) with respect to z and substituting (11.9) gives the **thermal wind relation**:

$$\frac{\partial \mathbf{V}_g}{\partial z} = \frac{g}{f \, \theta_0} \, \hat{k} \times \nabla \theta. \tag{11.10}$$

In component form:

$$\frac{\partial u_g}{\partial z} = -\frac{g}{f \, \theta_0} \frac{\partial \theta}{\partial y}, \qquad \frac{\partial v_g}{\partial z} = +\frac{g}{f \, \theta_0} \frac{\partial \theta}{\partial x}. \tag{11.11}$$

열역풍 관계의 물리적 의미:
- 연직 바람 시어는 수평 온도 경도에 비례
- 남쪽이 따뜻하면 $(\partial\theta/\partial y < 0) \to \partial u_g/\partial z > 0 \to$ 상층으로 갈수록 서풍 증가
- 이것이 중위도 제트가 존재하는 이유: 적도-극 온도차가 상층 서풍 시어를 생성

11.4. Thermodynamic Energy Equation

In the QG framework, the thermodynamic equation for the perturbation potential temperature θ reads:

$$\frac{d\theta}{dt} = -w \, \frac{d\theta_0}{dz}, \tag{11.12}$$

where the left-hand side uses the QG material derivative (11.8) and the right-hand side represents adiabatic cooling/heating due to vertical motion in a stably stratified background.

Definition 11.3 | Brunt–Väisälä Frequency

The *Brunt–Väisälä frequency* N measures the static stability of the background stratification:

$$N = \left[\frac{g}{\theta_0} \frac{d\theta_0}{dz} \right]^{1/2}. \tag{11.13}$$

Typical tropospheric values: $N \approx 0.01\,\mathrm{s}^{-1}$. For a stably stratified atmosphere, $d\theta_0/dz > 0$ so $N^2 > 0$. $N^2 > 0$이면 안정, $N^2 < 0$이면 정적 불안정 (대류 발생)

Using (11.13), the thermodynamic equation (11.12) can be written as:

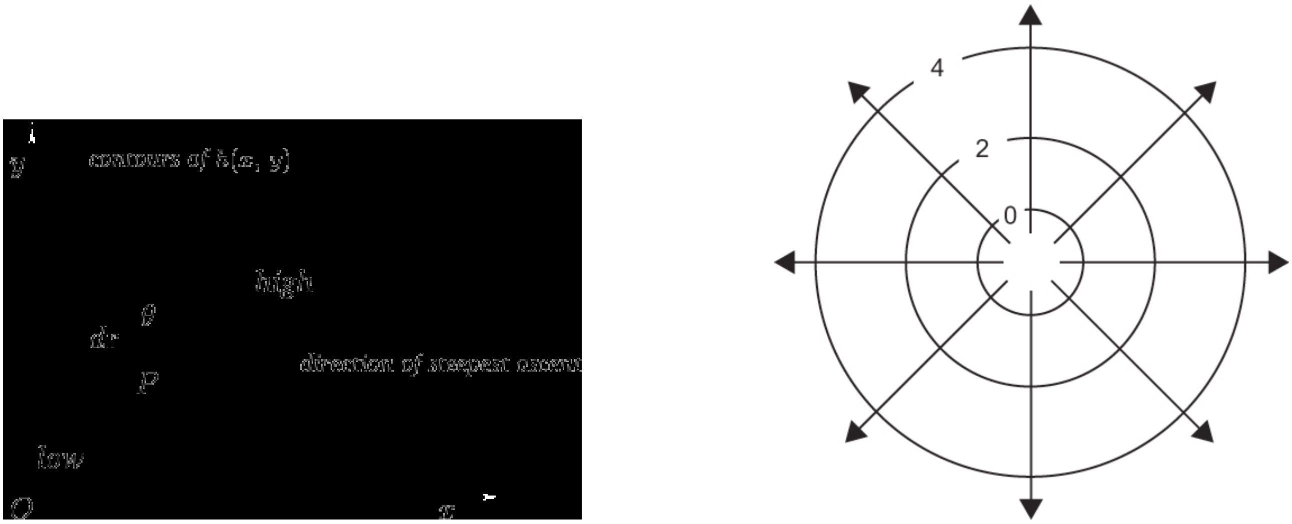
$$\frac{d\theta}{dt} = -\frac{\theta_0 N^2}{g} \, w. \tag{11.14}$$

11.5. Gradient and Laplacian – Geometric Intuition

Two differential operators appear repeatedly in QG theory and deserve careful geometric interpretation.

The Gradient

The gradient $\nabla\Phi$ points in the direction of **steepest ascent** of the scalar field Φ and has magnitude equal to the maximum rate of change. Contours of Φ are perpendicular to $\nabla\Phi$.



Left: Contours of a scalar field $h(x, y)$ with the gradient vector ∇h pointing in the direction of steepest ascent, perpendicular to isolines. Right: Illustration of gradient vectors radiating outward from a local minimum, showing the relationship between contour spacing and gradient magnitude. 왼쪽: 등고선에 수직인 경도(gradient) 벡터 — 최대 상승 방향을 가리킨다. 오른쪽: 극소점 주변의 경도 벡터 분포.

The Laplacian

The Laplacian $\nabla^2\Phi$ measures how the value of Φ at a point deviates from its average over a small neighbourhood:

Remark 11.4 | Sign Convention of the Laplacian

- $\nabla^2\Phi > 0$ at a **local minimum** of Φ: the value at the point is *less than* the neighbourhood average.
- $\nabla^2\Phi < 0$ at a **local maximum** of Φ: the value at the point is *greater than* the neighbourhood average.

라플라시안이 양수 = 극소점 (주변보다 값이 작음), 음수 = 극대점 (주변보다 값이 큼)

QG 이론에서의 응용:
- $\nabla^2 p > 0 \rightarrow$ 기압 극소 $\rightarrow \zeta_g > 0 \rightarrow$ 저기압성 소용돌이도
- $\nabla^2 p < 0 \rightarrow$ 기압 극대 $\rightarrow \zeta_g < 0 \rightarrow$ 고기압성 소용돌이도

따라서 저기압 중심에서는 항상 양의 (저기압성) 소용돌이도를 갖는다.

12. QG POTENTIAL VORTICITY

The concept of **potential vorticity** (PV) is the single most powerful idea in geophysical fluid dynamics. In the QG framework it takes a particularly elegant and useful form.

12.1. From Ertel PV to QG PV

Ertel PV under the Boussinesq Approximation

The exact (Ertel) potential vorticity for a Boussinesq fluid is:

$$\Pi = \frac{\boldsymbol{\omega}_a \cdot \nabla\theta_{\text{tot}}}{\rho_0}, \tag{12.1}$$

where $\boldsymbol{\omega}_a = \nabla \times \mathbf{V} + f\hat{k}$ is the absolute vorticity vector. For adiabatic, frictionless flow, Ertel PV is materially conserved:

$$\frac{D\Pi}{Dt} = 0. \tag{12.2}$$

QG Scaling of Ertel PV

Expanding $\theta_{\text{tot}} = \theta_0(z) + \theta(x, y, z, t)$ and $\boldsymbol{\omega}_a = (\xi, \eta, \zeta + f)$ where (ξ, η) are horizontal vorticity components:

$$\Pi = \frac{1}{\rho_0}\left[\xi\frac{\partial\theta}{\partial x} + \eta\frac{\partial\theta}{\partial y} + (\zeta + f)\left(\frac{d\theta_0}{dz} + \frac{\partial\theta}{\partial z}\right)\right]. \tag{12.3}$$

Under QG scaling ($\text{Ro} \ll 1$):

1. The horizontal vorticity terms $\xi\,\partial\theta/\partial x$ and $\eta\,\partial\theta/\partial y$ are $O(\text{Ro}^2)$ smaller than the vertical terms, so we neglect them.
2. In the vertical term, $\zeta\,\partial\theta/\partial z$ is $O(\text{Ro})$ compared to $f\,d\theta_0/dz$, so we keep only the leading-order contributions.
3. We retain ζ_g (not the full ζ) and replace ζ by ζ_g.

After careful ordering, the QG-relevant part of Ertel PV reduces to:

$$\Pi \approx \frac{1}{\rho_0}\left[(\zeta_g + f)\frac{d\theta_0}{dz} + f\frac{\partial\theta}{\partial z}\right]. \tag{12.4}$$

Dividing by the reference stability $(1/\rho_0)\,d\theta_0/dz$ and rearranging, we define the **QG potential vorticity**:

Definition 12.1 | QG Potential Vorticity, z-coordinate

$$\boxed{Q = (\zeta_g + f) + f\,\frac{\partial}{\partial z}\left(\frac{\theta}{\partial\theta_0/\partial z}\right).} \tag{12.5}$$

The three terms represent:
1. ζ_g: relative (geostrophic) vorticity
2. f: planetary vorticity
3. $f\,\dfrac{\partial}{\partial z}\left(\dfrac{\theta}{\partial\theta_0/\partial z}\right)$: **stretching vorticity** (vertical redistribution of stability)

QG PV의 세 가지 구성 요소:
1. 상대 소용돌이도 ζ_g: 유체 자체의 회전
2. 행성 소용돌이도 f: 지구 자전에 의한 기여
3. 신축 소용돌이도: 연직 안정도의 변형에 의한 소용돌이도
Ertel PV는 소용돌이도와 안정도의 곱이지만, QG PV는 합으로 정의됨. 소용돌이도가 증가하면 안정도가 감소해야 PV가 보존된다.

12.2. QG PV Conservation

The fundamental result of QG theory is that Q is materially conserved following the *geostrophic flow*:

Theorem 12.2 | Conservation of QG PV

For adiabatic, frictionless, quasi-geostrophic flow:

$$\boxed{\frac{dQ}{dt} = \left(\frac{\partial}{\partial t} + u_g\frac{\partial}{\partial x} + v_g\frac{\partial}{\partial y}\right) Q = 0.} \tag{12.6}$$

QG PV는 지균류를 따라 보존된다 — QG 이론의 핵심 결과! 이 하나의 방정식이 QG 역학 전체를 지배한다.

Proof We derive this from the QG vorticity and thermodynamic equations.

Step 1. The QG vorticity equation (derived in Sec. 12.3) gives:

$$\frac{d}{dt}(\zeta_g + f) = f\frac{\partial w}{\partial z}.$$

Step 2. The QG thermodynamic equation (11.12) gives:

$$\frac{d\theta}{dt} = -w\frac{d\theta_0}{dz}.$$

Dividing by $d\theta_0/dz$ and taking $\partial/\partial z$:

$$\frac{\partial}{\partial z}\left(\frac{1}{d\theta_0/dz}\frac{d\theta}{dt}\right) = -\frac{\partial w}{\partial z}.$$

Since $d\theta_0/dz$ depends only on z (not on x, y, t), we can move the material derivative outside:

$$\frac{d}{dt}\frac{\partial}{\partial z}\left(\frac{\theta}{d\theta_0/dz}\right) = -\frac{\partial w}{\partial z}.$$

Multiplying by f:

$$f\frac{d}{dt}\frac{\partial}{\partial z}\left(\frac{\theta}{d\theta_0/dz}\right) = -f\frac{\partial w}{\partial z}.$$

Step 3. Adding the vorticity equation (Step 1) and the rearranged thermodynamic equation (Step 2):

$$\frac{d}{dt}\left[(\zeta_g + f) + f\frac{\partial}{\partial z}\left(\frac{\theta}{d\theta_0/dz}\right)\right] = 0,$$

which is precisely $dQ/dt = 0$. ■

12.3. QG Vorticity Equation

The QG vorticity equation is obtained from the horizontal momentum equations by taking $\partial v_g/\partial x - \partial u_g/\partial y$ and using QG scaling. The result is:

$$\boxed{\frac{d}{dt}(\zeta_g + f) = f\frac{\partial w}{\partial z}.} \tag{12.7}$$

Remark 12.3 | Physical Interpretation

- Left-hand side: rate of change of **absolute geostrophic vorticity** following the geostrophic flow.

- Right-hand side: vorticity production by **stretching/compression** of planetary vorticity tubes.
- If $\partial w/\partial z > 0$ (stretching): columns become taller and thinner → spin-up (cyclonic vorticity increase).
- If $\partial w/\partial z < 0$ (compression): columns become shorter and wider → spin-down (anticyclonic vorticity increase).

절대 소용돌이도의 변화 = 행성 소용돌이 관(vortex tube)의 신축(stretching)에 의한 것

유도 과정의 핵심:
1. 수평 운동 방정식에서 $\partial/\partial x$(v-eq) $-$ $\partial/\partial y$(u-eq) 취함
2. 지균풍이 비발산이므로 $\nabla \cdot \mathbf{V}_g = 0$
3. 비지균풍의 수평 발산 $\partial u_a/\partial x + \partial v_a/\partial y$가 연속 방정식으로부터 $-\partial w/\partial z$와 연결
4. 결과적으로 소용돌이도 변화가 연직 운동의 수렴/발산으로 결정됨

12.4. QG Mass Continuity

The total horizontal wind is decomposed as $\mathbf{V} = \mathbf{V}_g + \mathbf{V}_a$ where $\mathbf{V}_a$ is the **ageostrophic wind**. Since $\nabla \cdot \mathbf{V}_g = 0$ (on an f-plane), the Boussinesq continuity equation (11.4) becomes:

$$\frac{\partial u_a}{\partial x} + \frac{\partial v_a}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{12.8}$$

지균풍은 비발산이므로, 수평 발산은 전적으로 비지균풍에 의한 것이다. 이 비지균 발산이 연직 운동을 구동한다.

12.5. QG PV in Terms of Pressure Alone

A remarkable feature of the QG system is that all dynamical quantities can be expressed in terms of a **single variable**: the pressure perturbation p. Substituting $\zeta_g = (1/\rho_0 f)\nabla^2 p$ and $\theta = (\theta_0/\rho_0 g)\,\partial p/\partial z$ (from the hydrostatic relation (11.9)) into (12.5):

$$\boxed{f\,(Q - f) = \frac{1}{\rho_0}\nabla^2 p + \frac{1}{\rho_0}\frac{\partial}{\partial z}\left(\frac{f^2}{N^2}\frac{\partial p}{\partial z}\right) \equiv Lp.} \tag{12.9}$$

The operator L is a three-dimensional elliptic operator (an "invertibility operator") that maps the pressure field to the PV distribution.

QG PV는 압력장 하나로 완전히 결정 — 매우 강력한 결과!

이것이 강력한 이유: PV 분포 $Q(x, y, z)$를 알면, 적절한 경계 조건과 함께 $Lp = f(Q - f)$를 풀어서 p를 구하고, p로부터 바람, 온도, 소용돌이도를 모두 진단(diagnose)할 수 있다. 이것이 **PV 역전(inversion)** 원리이다:

$$Q \xrightarrow{\text{inversion}} p \xrightarrow{\text{geostrophic}} \mathbf{V}_g,\ \theta,\ \zeta_g.$$

13. QG HEIGHT TENDENCY EQUATION

The QG height tendency equation is a **prognostic equation** that predicts how the pressure field evolves in time, given the current distribution of wind and temperature. It is the theoretical foundation of synoptic weather forecasting.

13.1. Derivation

We start from QG PV conservation (12.6) and the invertibility relation (12.9).

Step 1. From $dQ/dt = 0$:

$$\frac{\partial Q}{\partial t} = -\mathbf{V}_g \cdot \nabla Q. \tag{13.1}$$

Step 2. Noting that f is time-independent on a β-plane (i.e. $\partial f/\partial t = 0$), we have $\partial Q/\partial t = (1/f)\,\partial(Lp)/\partial t$, and ∇Q includes $\nabla f = \beta\hat{j}$ plus eddy PV gradients. Combining:

$$L\frac{\partial p}{\partial t} = -f\,\mathbf{V}_g \cdot \nabla(fQ). \tag{13.2}$$

Step 3. Expanding $fQ = f(\zeta_g + f) + f^2\,\partial_z(\theta/(d\theta_0/dz))$ and using the identity $\mathbf{V}_g \cdot \nabla f = \beta v_g$:

$$\boxed{L\frac{\partial p}{\partial t} = -\mathbf{V}_g \cdot \nabla(f\zeta_g) + f^2\frac{\partial}{\partial z}\left[\frac{\mathbf{V}_g \cdot \nabla\theta}{\partial\theta_0/\partial z}\right].} \tag{13.3}$$

13.2. Physical Interpretation

The two forcing terms on the right-hand side of (13.3) have clear physical interpretations:

Remark 13.1 | Height Tendency Forcing Terms

1. **Vorticity advection** ($-\mathbf{V}_g \cdot \nabla(f\zeta_g)$): Advection of absolute vorticity by the geostrophic wind.
   - Cyclonic vorticity advection ($-\mathbf{V}_g \cdot \nabla\zeta_g > 0$) $\rightarrow$ $L(\partial p/\partial t) > 0$ $\rightarrow$ since L reverses the sign (like ∇^2), this implies **pressure falls** $\rightarrow$ low deepens.
   - Anticyclonic vorticity advection $\rightarrow$ pressure rises $\rightarrow$ high builds.
2. **Differential temperature advection**: Upward increase of warm advection ($\partial/\partial z[-\mathbf{V}_g \cdot \nabla(-\theta)] > 0$) $\rightarrow$ pressure falls $\rightarrow$ low deepens.
   - Warm advection increasing with height destabilises the column, requiring convergence at low levels, which spins up a surface low.
   - Cold advection increasing with height stabilises the column, producing divergence at low levels and building a surface high.

저기압 발달 조건: 저기압성 소용돌이도 이류 + 온난 이류의 상향 증가. 이 두 효과가 같은 위치에서 협력하면 급속한 저기압 발달 (폭풍 발달)이 일어남!

실제 일기예보에서의 응용:
- 500 hPa 절대 소용돌이도 이류(AVA)가 양이면 $\rightarrow$ 지상 기압 하강 예상
- 하층에서 온난 이류 + 상층에서 더 강한 온난 이류 $\rightarrow$ 기압 하강 촉진
- 두 효과가 겹치는 곳에서 explosive cyclogenesis 가능

14. QG VERTICAL MOTION EQUATION

Vertical motion in the atmosphere cannot be measured directly; it must be **diagnosed** from other fields. The QG omega equation provides a powerful tool for this purpose.

14.1. Omega Equation (z-coordinate)

The omega equation is derived by eliminating $\partial p/\partial t$ between the QG vorticity equation (12.7) and the QG thermodynamic equation (11.12).

Step 1. Take $\partial/\partial z$ of the vorticity equation:

$$\frac{\partial}{\partial z}\left[\frac{d}{dt}(\zeta_g + f)\right] = f\frac{\partial^2 w}{\partial z^2}.$$

Step 2. Take ∇^2 of the thermodynamic equation (11.12):

$$\nabla^2\left[\frac{d\theta}{dt}\right] = -\frac{d\theta_0}{dz}\,\nabla^2 w.$$

Step 3. Rearrange and combine (using the thermal wind relation to convert $\partial\zeta_g/\partial z$ terms into $\nabla\theta$ terms):

$$\boxed{\underbrace{\left(\nabla^2 + \frac{f^2}{N^2}\frac{\partial^2}{\partial z^2}\right)}_{\tilde{L}} w = \frac{g}{\theta_0 N^2}\nabla^2(-\mathbf{V}_g\cdot\nabla\theta) - \frac{f}{N^2}\frac{\partial}{\partial z}(-\mathbf{V}_g\cdot\nabla\zeta_g).} \tag{14.1}$$

Remark 14.1 | Properties of the Omega Equation

- The operator $\tilde{L}$ is an **elliptic operator** (like a 3D Laplacian, with the vertical stretched by f/N). For an elliptic operator, $\tilde{L}w > 0$ implies $w < 0$ locally, and vice versa. Therefore the sign of w is **opposite** to the sign of the right-hand side.
- The equation is **diagnostic**: it gives w at any instant from the known geostrophic fields, without time integration.
- Two forcing terms:
  1. Laplacian of temperature advection: warm advection $(-\mathbf{V}_g\cdot\nabla\theta > 0) \to \nabla^2(\text{warm adv.}) > 0$ at the advection maximum $\to$ upward motion.
  2. Differential vorticity advection: cyclonic vorticity advection increasing with height $\to$ upward motion.

오메가 방정식 = 진단 방정식: 수직 운동을 지균풍의 이류 항들로부터 진단한다. 직접 측정 불가능한 w를 구하는 유일한 방법!

14.2. Application: Cyclogenesis

The QG framework provides a complete picture of how midlatitude cyclones develop through the cooperative interaction of upper-level and surface disturbances.

Remark 14.2 | QG Mechanism of Cyclogenesis

1. An **upper-level PV anomaly** (e.g. a trough approaching from the west) induces cyclonic circulation and ascent downstream and below.
2. This ascent produces **low-level convergence**, which spins up a surface cyclone (stretching of planetary vorticity tubes: $f\,\partial w/\partial z > 0$).
3. The surface cyclone, with its associated warm and cold fronts, generates a **surface temperature anomaly** (warm air drawn poleward ahead of the low).
4. This surface warm anomaly is equivalent to a positive PV anomaly at the surface, which induces **cyclonic circulation aloft**, reinforcing the upper-level trough.
5. The system amplifies as long as the disturbance maintains a **westward tilt with height**, ensuring continued conversion of mean-flow APE to eddy KE.

경압 저기압 발달의 핵심 메커니즘:
- 상층 PV 이상 + 지표 온도 이상 → 상호 강화 (cooperative interaction)
- 서쪽으로 기울어진 교란만이 평균류의 PE를 파동의 KE로 전환 가능
- 저기압 위의 발산 → 지표 저기압 강화 → 연직 신축이 소용돌이도 증가 유발
- 기울기가 사라지면 (연직 정렬) → 에너지 변환 중단 → 폐색 단계

15. QG EQUATIONS IN PRESSURE COORDINATES

For practical applications—especially weather forecasting and analysis of upper-air charts—it is convenient to formulate the QG equations in **pressure coordinates** (x, y, p, t), where pressure replaces height as the vertical coordinate.

15.1. Beta-Plane Approximation

Definition 15.1 | Beta-Plane Approximation

The Coriolis parameter is expanded to first order about a reference latitude ϕ_0:

$$f = f_0 + \beta y, \qquad f_0 = 2\Omega \sin\phi_0, \qquad \beta = \frac{2\Omega\cos\phi_0}{a}, \tag{15.1}$$

where a is the Earth's radius, Ω its rotation rate, and $y = a(\phi - \phi_0)$ is the northward distance from the reference latitude.

The scale analysis confirms that the β-effect is an $O(\mathrm{Ro})$ correction to f_0:

$$\frac{\beta L}{f_0} \sim \mathrm{Ro} \sim 0.1. \tag{15.2}$$

β-평면: f의 위도 변화를 선형으로 근사. $\beta L/f_0 \sim \mathrm{Ro}$이므로 β 효과는 1차 보정.

15.2. QG Momentum Equation (p-coordinate)

In pressure coordinates the geostrophic wind is:

$$u_g = -\frac{1}{f_0}\frac{\partial\Phi}{\partial y}, \qquad v_g = +\frac{1}{f_0}\frac{\partial\Phi}{\partial x}, \tag{15.3}$$

where $\Phi = gz$ is the geopotential. The QG momentum equation is:

$$\boxed{\frac{d\mathbf{V}_g}{dt} = -f_0\,\hat{k}\times\mathbf{V}_a - \beta y\,\hat{k}\times\mathbf{V}_g,} \tag{15.4}$$

where the QG material derivative is $d/dt = \partial/\partial t + \mathbf{V}_g\cdot\nabla$ (horizontal advection by the geostrophic wind only).

지균풍의 시간 변화 = 비지균풍에 의한 코리올리 힘 $(-f_0\hat{k}\times\mathbf{V}_a)$ + 베타 효과 $(-\beta y\hat{k}\times\mathbf{V}_g)$

물리적 의미:
- 첫째 항: 비지균풍(ageostrophic wind)에 대한 코리올리 힘이 지균풍을 변화시킴. 비지균풍이 없으면 지균풍은 변하지 않는다!
- 둘째 항: β 효과로 인한 보정. 위도에 따라 f가 변하므로 같은 기압 경도라도 다른 위도에서는 다른 지균풍을 줌.
- 이 방정식은 ageostrophic motion이 balanced flow의 진화를 구동함을 보여줌.

15.3. QG Thermodynamic Equation (p-coordinate)

The thermodynamic energy equation in pressure coordinates reads:

$$\boxed{\left(\frac{\partial}{\partial t} + \mathbf{V}_g \cdot \nabla\right)\left(-\frac{\partial \Phi}{\partial p}\right) - \sigma\omega = \frac{\kappa J}{p},} \tag{15.5}$$

where $-\partial\Phi/\partial p = RT/p = \alpha$ is proportional to temperature (via the hydrostatic relation), $\omega = Dp/Dt$ is the pressure-coordinate vertical velocity, $\sigma(p) = -(\alpha/\theta)\, d\theta/dp > 0$ is the **static stability parameter**, J is the diabatic heating rate, and $\kappa = R/c_p$.

p-좌표에서 열역학 방정식의 각 항:
- $(\partial/\partial t + \mathbf{V}_g \cdot \nabla)(-\partial\Phi/\partial p)$: 두께(온도)의 지균 이류를 포함한 시간 변화
- $-\sigma\omega$: 연직 운동에 의한 단열 냉각/가열 ($\omega < 0$이면 상승, 냉각)
- $\kappa J/p$: 비단열 가열 (복사, 잠열, 현열 등)

15.4. QG Vorticity Equation (p-coordinate)

Taking $(\partial/\partial x)$ of the v_g-equation minus $(\partial/\partial y)$ of the u_g-equation, and using the continuity equation $\nabla \cdot \mathbf{V}_a + \partial\omega/\partial p = 0$:

$$\boxed{\frac{d\zeta_g}{dt} = -\mathbf{V}_g \cdot \nabla(\zeta_g + f) + f_0 \frac{\partial \omega}{\partial p}.} \tag{15.6}$$

Remark 15.2 | Three Mechanisms for Vorticity Change

1. **Relative vorticity advection**: $-\mathbf{V}_g \cdot \nabla\zeta_g$
2. **Planetary vorticity advection**: $-\mathbf{V}_g \cdot \nabla f = -\beta v_g$
3. **Stretching term**: $f_0\, \partial\omega/\partial p$

각 항의 물리:
- 상대 소용돌이도 이류: 소용돌이가 바람에 의해 이동
- 행성 소용돌이도 이류: 공기가 북쪽으로 이동하면 ($v_g > 0$) f 증가 → 상대 소용돌이도 감소 (β 효과)
- 신축항: $\partial\omega/\partial p > 0$ (하강 약화 또는 상승 강화) → 수평 수렴 → 소용돌이도 증가

15.5. Wave Propagation: Short Waves vs. Long Waves

To understand how **relative vorticity advection** and **planetary vorticity advection** compete, consider a simple geopotential field:

$$\Phi = \Phi_0(p) - f_0\bar{u}\, y + f_0 A \sin(kx)\cos(ly), \tag{15.7}$$

where $\bar{u}$ is a constant zonal-mean zonal wind and A is the wave amplitude. From (15.3), the geostrophic wind perturbations are:

$$u_g' = -\frac{1}{f_0}\frac{\partial \Phi'}{\partial y} = Al\sin(kx)\sin(ly), \qquad v_g' = \frac{1}{f_0}\frac{\partial \Phi'}{\partial x} = Ak\cos(kx)\cos(ly).$$

The geostrophic relative vorticity is:

$$\zeta_g = \frac{1}{f_0}\nabla^2\Phi' = -\frac{(k^2+l^2)}{f_0}\, f_0 A\sin(kx)\cos(ly) = -(k^2+l^2)\, A\sin(kx)\cos(ly).$$

Relative Vorticity Advection

$$-\bar{u}\frac{\partial \zeta_g}{\partial x} = \bar{u}(k^2+l^2)\,Ak\cos(kx)\cos(ly).$$

This acts to propagate the wave **eastward** (in the direction of $\bar{u}$).

Planetary Vorticity Advection

$$-\beta v_g = -\beta Ak\cos(kx)\cos(ly).$$

This acts to propagate the wave **westward** (opposing $\bar{u}$).

Ratio of the Two Effects

The ratio of planetary to relative vorticity advection is:

$$\boxed{\frac{\text{Planetary vorticity advection}}{\text{Relative vorticity advection}} = \frac{\beta}{\bar{u}(k^2+l^2)}.} \tag{15.8}$$

Remark 15.3 | Short Waves vs. Long Waves

- **Short waves** (k^2+l^2 large): Relative vorticity advection dominates → net eastward propagation. Short waves are swept eastward by the mean flow.
- **Long waves** (k^2+l^2 small): Planetary vorticity advection dominates → net westward propagation (retrogression). Very long Rossby waves can propagate westward against the mean flow.
- **Stationary waves**: when the two effects exactly cancel, i.e. $\beta = \bar{u}(k^2+l^2)$, the wave is stationary. The stationary wavenumber is $K_s = \sqrt{\beta/\bar{u}}$.

단파: 상대 소용돌이도 이류 우세 → 동진. 장파: 행성 소용돌이도 이류 우세 → 서진 (후퇴). 이것이 “TRICKY” 한 부분!

정상파 조건: $k^2+l^2=\beta/\bar{u}$
이 조건을 만족하는 파수의 파동만이 지형이나 가열에 의해 강제되어 시간 평균적으로 제자리에 머물 수 있다. $\bar{u}$가 크면 정상파의 파수가 작아져서 (파장이 길어져서) 소수의 긴 파동만 정상일 수 있다. 이것이 중위도 대기의 planetary wave 패턴을 결정한다.

15.6. QG Geopotential Tendency Equation (p-coordinate)

Define the **geopotential tendency**:

$$\chi \equiv \frac{\partial \Phi}{\partial t}. \tag{15.9}$$

From QG PV conservation in pressure coordinates, one obtains:

$$\boxed{\underbrace{\left[\nabla^2 + \frac{\partial}{\partial p}\left(\frac{f_0^2}{\sigma}\frac{\partial}{\partial p}\right)\right]\chi}_{\text{Term A (tendency)}} = \underbrace{-f_0\,\mathbf{V}_g\cdot\nabla\left(\frac{1}{f_0}\nabla^2\Phi + f\right)}_{\text{Term B (vorticity advection)}} \underbrace{-\frac{\partial}{\partial p}\left[-\frac{f_0^2}{\sigma}\mathbf{V}_g\cdot\nabla\left(-\frac{\partial\Phi}{\partial p}\right)\right]}_{\text{Term C (thickness advection)}}.} \tag{15.10}$$

Remark 15.4 | Interpretation of Terms

- **Term A**: An elliptic operator acting on the tendency χ. Since $\nabla^2\chi \sim -\chi$ at the centre of an anomaly (as for any Laplacian), the sign of χ is *opposite* to the sign of the right-hand side.

- **Term B — Absolute vorticity advection**: Advection of absolute vorticity by the geostrophic wind. Positive (cyclonic) vorticity advection → $\chi < 0$ → geopotential **falls** → deepening low.
- **Term C — Differential thickness advection**: Vertical derivative of the geostrophic temperature advection weighted by f_0^2/σ. Warm advection increasing with height → $\chi < 0$ → geopotential falls.

일기예보에서의 활용:
- Term B: 500 hPa 절대 소용돌이도 이류 지도에서 양의 AVA 지역 → 기압골 접근, 지상 기압 하강
- Term C: 1000–500 hPa 두께 이류 → 온난 이류 증가 지역에서 지상 저기압 발달
- 두 항이 같은 위치에서 협력하면 → 급속한 저기압 발달 (explosive cyclogenesis)

15.7. QG PV in Pressure Coordinates

Collecting the vorticity and stretching contributions in pressure coordinates:

Definition 15.5 | QG Potential Vorticity, p-coordinate

$$\boxed{Q = \frac{1}{f_0}\nabla^2\Phi + f + \frac{\partial}{\partial p}\left(\frac{f_0}{\sigma}\frac{\partial\Phi}{\partial p}\right).} \tag{15.11}$$

The three terms are:
1. $\dfrac{1}{f_0}\nabla^2\Phi$: geostrophic relative vorticity
2. $f = f_0 + \beta y$: planetary vorticity (Coriolis parameter)
3. $\dfrac{\partial}{\partial p}\left(\dfrac{f_0}{\sigma}\dfrac{\partial\Phi}{\partial p}\right)$: stretching vorticity

p-좌표 QG PV도 지위고도(Φ) 하나로 완전히 결정됨. PV 역전(inversion)의 기초!

15.8. QG Omega Equation (p-coordinate)

The omega equation in pressure coordinates is obtained by eliminating the tendency χ between the vorticity equation (15.6) and the thermodynamic equation (15.5).

Step 1. Apply $(\partial/\partial p)(f_0^2/\sigma)\cdot(\cdot)$ to the vorticity equation to get the stretching contribution to the tendency.

Step 2. Apply ∇^2 to the thermodynamic equation.

Step 3. Subtract to eliminate $\partial\Phi/\partial t$ terms.

The result is the **QG omega equation**:

$$\boxed{\underbrace{\left(\nabla^2 + \frac{f_0^2}{\sigma}\frac{\partial^2}{\partial p^2}\right)\omega}_{\text{Term A}} = \underbrace{\frac{f_0}{\sigma}\frac{\partial}{\partial p}\left[\mathbf{V}_g\cdot\nabla\left(\frac{1}{f_0}\nabla^2\Phi + f\right)\right]}_{\text{Term B}} + \underbrace{\frac{1}{\sigma}\nabla^2\left[\mathbf{V}_g\cdot\nabla\left(-\frac{\partial\Phi}{\partial p}\right)\right]}_{\text{Term C}}.} \tag{15.12}$$

Remark 15.6 | Interpretation of the Omega Equation

- **Term A**: Elliptic operator on ω. Since $\nabla^2\omega \sim -\omega$ at the centre of a vertical velocity anomaly, the sign of ω is **opposite** to the sign of (B + C). Recall that in p-coordinates, $\omega < 0$ means **upward motion**.
- **Term B — Differential vorticity advection**: Vertical derivative of absolute vorticity ad-

vection.
  - Cyclonic vorticity advection increasing with height ($\partial/\partial p[\text{AVA}] < 0$ since p decreases upward) $\rightarrow \omega < 0 \rightarrow$ **ascent**.
  - Physically: upper-level PV advection produces divergence aloft, requiring compensating ascent through the column.
- **Term C — Laplacian of thickness advection**: Laplacian of temperature advection by the geostrophic wind.
  - Warm advection at a local maximum ($\nabla^2[\text{warm adv.}] < 0$) $\rightarrow \omega < 0 \rightarrow$ **ascent** in the warm front region.
  - Cold advection at a local maximum $\rightarrow \omega > 0 \rightarrow$ **descent** behind the cold front.

Term B: 상층에서 양의 소용돌이도 이류가 증가 → 상승. Term C: 온난전선 → 상승, 한랭전선 뒤 → 하강.

오메가 방정식의 실용적 중요성:
- ω는 직접 측정이 거의 불가능 → 반드시 진단 방정식으로 계산
- Term B는 주로 상층 일기도(500/300 hPa)의 트로프/릿지 패턴에서 추정
- Term C는 지표 전선 구조와 두께 이류 패턴에서 추정
- 상승 운동 → 구름, 강수 → 일기예보의 핵심!

15.9. Barotropic vs. Baroclinic Atmosphere

The distinction between barotropic and baroclinic atmospheres is central to understanding QG dynamics and the nature of instabilities.

Definition 15.7 | Barotropic Atmosphere

A fluid is **barotropic** if density depends only on pressure: $\rho = \rho(p)$. Equivalently, isobaric surfaces and isopycnal (constant density) surfaces are **parallel**. Consequences:
- No horizontal temperature gradients on pressure surfaces.
- No thermal wind: $\partial \mathbf{V}_g/\partial z = 0$, so the geostrophic wind is independent of height.
- No available potential energy (APE) to convert: the atmosphere is in its minimum-PE state.

순압 대기: 등압면과 등밀도면이 평행 → 바람이 고도에 따라 변하지 않음

Definition 15.8 | Baroclinic Atmosphere

A fluid is **baroclinic** if density depends on both pressure and temperature: $\rho = \rho(p, T)$. Equivalently, isobaric surfaces and isothermal/isopycnal surfaces **tilt relative to each other**. Consequences:
- Horizontal temperature gradients exist on pressure surfaces.
- The thermal wind relation gives nonzero vertical wind shear: $\partial \mathbf{V}_g/\partial z \neq 0$.
- Available potential energy (APE) exists and can be converted to kinetic energy through baroclinic instability.

경압 대기: 등압면과 등온면이 기울어져 있음 → 위치에너지를 운동에너지로 변환 가능!

순압 vs. 경압의 핵심 차이:

순압 (**Barotropic**)
- $\rho = \rho(p)$ only
- $\partial \mathbf{V}_g/\partial z = 0$
- APE $= 0$
- 경압 불안정 없음
- 순압 불안정만 가능

경압 (**Baroclinic**)
- $\rho = \rho(p, T)$
- $\partial \mathbf{V}_g/\partial z \neq 0$
- APE > 0
- 경압 불안정 가능
- 중위도 종관 시스템의 원동력

실제 대기는 항상 경압적이다 — 적도-극 온도 경도가 존재하기 때문. 순압 대기는 이론적 이상화에 불과하지만, 성층권이나 열대 대기의 일부 측면을 이해하는 데 유용하다.

16. SUMMARY OF QG EQUATIONS

We collect the complete set of QG equations in both coordinate systems for easy reference.

16.1. QG System in z-coordinates (Boussinesq)

1. **Geostrophic wind**: $\mathbf{V}_g = \frac{1}{\rho_0 f}\hat{k} \times \nabla p$
2. **Thermal wind**: $\frac{\partial \mathbf{V}_g}{\partial z} = \frac{g}{f\theta_0}\hat{k} \times \nabla\theta$
3. **QG vorticity equation**: $\frac{d}{dt}(\zeta_g + f) = f\frac{\partial w}{\partial z}$
4. **QG thermodynamic equation**: $\frac{d\theta}{dt} = -w\frac{d\theta_0}{dz}$
5. **QG PV conservation**: $\frac{dQ}{dt} = 0, \quad Q = (\zeta_g + f) + f\frac{\partial}{\partial z}\left(\frac{\theta}{d\theta_0/dz}\right)$
6. **Omega equation**: $\left(\nabla^2 + \frac{f^2}{N^2}\frac{\partial^2}{\partial z^2}\right) w = \frac{g}{\theta_0 N^2}\nabla^2(-\mathbf{V}_g \cdot \nabla\theta) - \frac{f}{N^2}\frac{\partial}{\partial z}(-\mathbf{V}_g \cdot \nabla\zeta_g)$

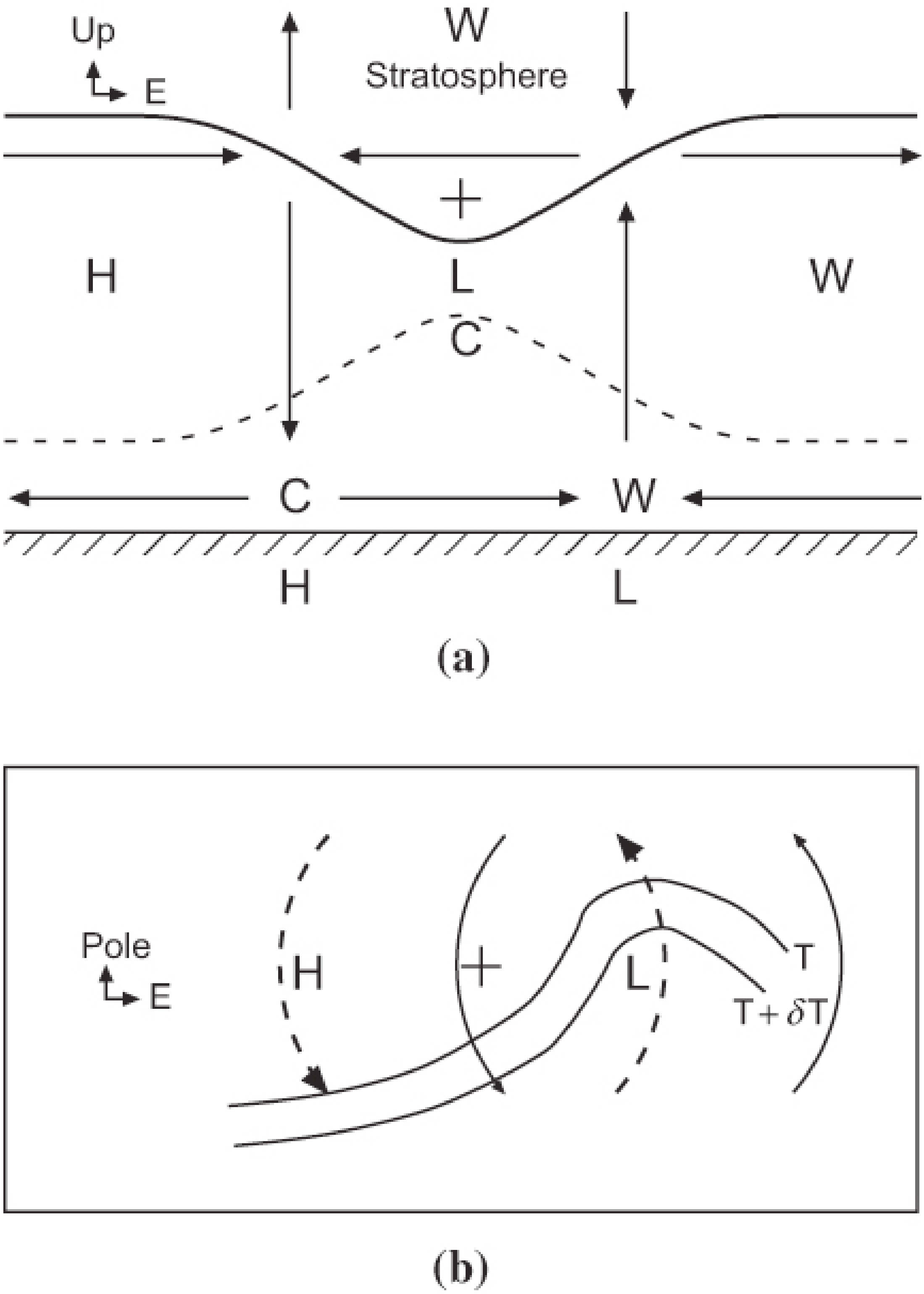
16.2. QG System in p-coordinates

1. **Geostrophic wind**: $u_g = -\frac{1}{f_0}\frac{\partial\Phi}{\partial y}, \quad v_g = \frac{1}{f_0}\frac{\partial\Phi}{\partial x}$
2. **QG momentum equation**: $\frac{d\mathbf{V}_g}{dt} = -f_0\hat{k} \times \mathbf{V}_a - \beta y\hat{k} \times \mathbf{V}_g$
3. **QG vorticity equation**: $\frac{d\zeta_g}{dt} = -\mathbf{V}_g \cdot \nabla(\zeta_g + f) + f_0\frac{\partial\omega}{\partial p}$
4. **QG thermodynamic equation**: $\left(\frac{\partial}{\partial t} + \mathbf{V}_g \cdot \nabla\right)\left(-\frac{\partial\Phi}{\partial p}\right) - \sigma\omega = \frac{\kappa J}{p}$
5. **QG PV conservation**: $\frac{dQ}{dt} = 0, \quad Q = \frac{1}{f_0}\nabla^2\Phi + f + \frac{\partial}{\partial p}\left(\frac{f_0}{\sigma}\frac{\partial\Phi}{\partial p}\right)$
6. **Geopotential tendency equation**: See Eq. (15.10)
7. **Omega equation**: See Eq. (15.12)

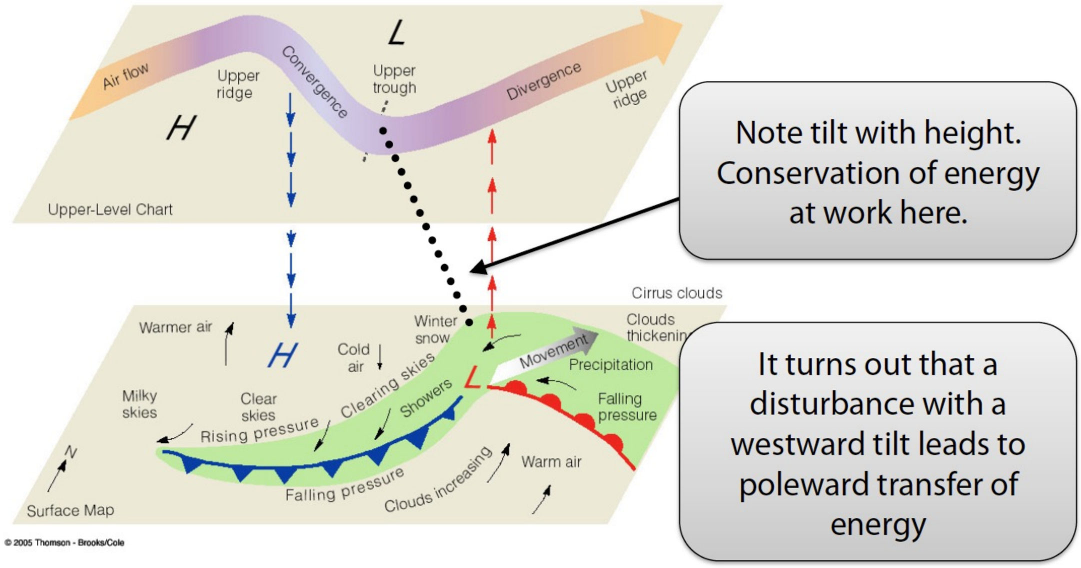
이 요약표를 시험 전에 반드시 외울 것! QG 체계의 모든 것이 여기에 있다.

QG 이론의 핵심 메시지 요약:
1. QG PV 하나가 전체 역학을 지배한다 ($dQ/dt = 0$).
2. 모든 동역학 변수는 단일 스칼라 장 (p 또는 Φ)으로 결정된다 (PV 역전).
3. 연직 운동과 지위 고도 변화는 소용돌이도 이류와 온도 이류로 진단된다.
4. 단파는 동진, 장파는 서진 — 정상파는 $K_s = \sqrt{\beta/\bar{u}}$.
5. 경압 불안정은 APE를 KE로 전환하며, 서쪽 기울기가 필수적이다.

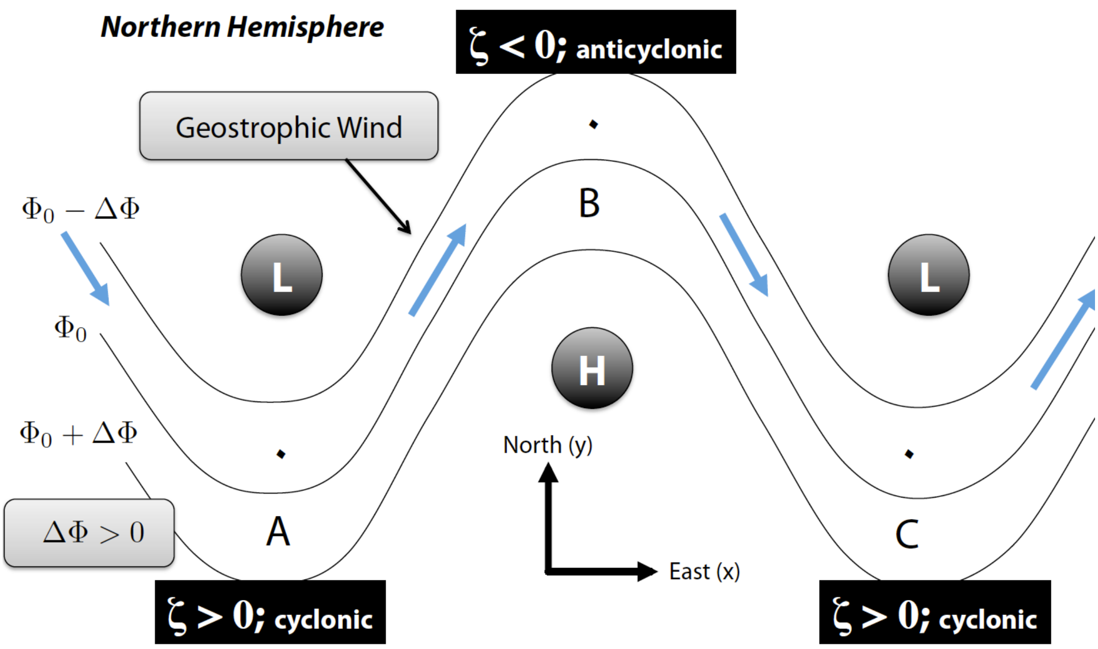
Ref: Holton and Hakim, Ch. 6; Vallis, *Atmospheric and Oceanic Fluid Dynamics*, Ch. 5.



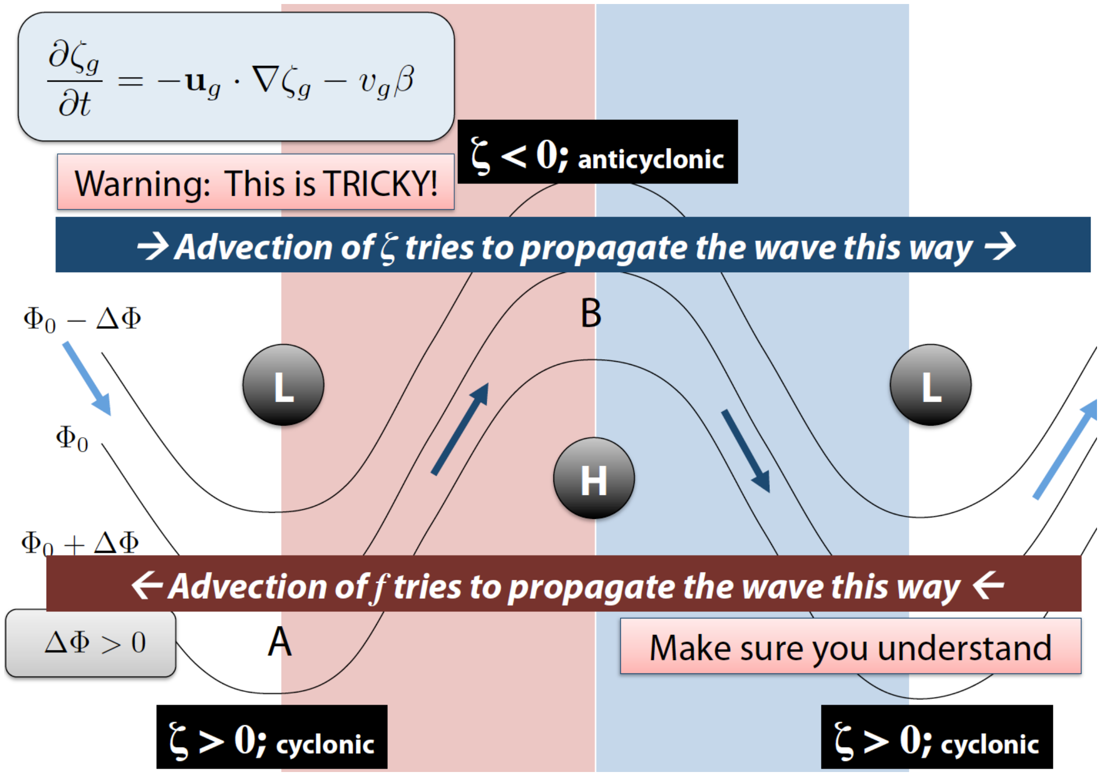
Schematic cross-section of midlatitude cyclogenesis showing the interaction between upper-level and surface disturbances. (a) Vertical cross-section: an upper-level trough (with positive PV anomaly) induces ascent and low-level convergence, spinning up a surface low. The surface warm anomaly (equivalent to a surface PV anomaly) reinforces the upper trough. Labels H and L denote high and low centres; W and C denote warm and cold air. (b) Plan view: the surface cyclone with warm and cold fronts, showing warm air advection ahead of the low and cold air intrusion behind. 저기압 발생 모식도. (a) 연직 단면: 상층 기압골이 하강류·수렴을 유도하여 지표 저기압 강화. (b) 평면도: 온난/한랭 전선과 지표 저기압의 배치.



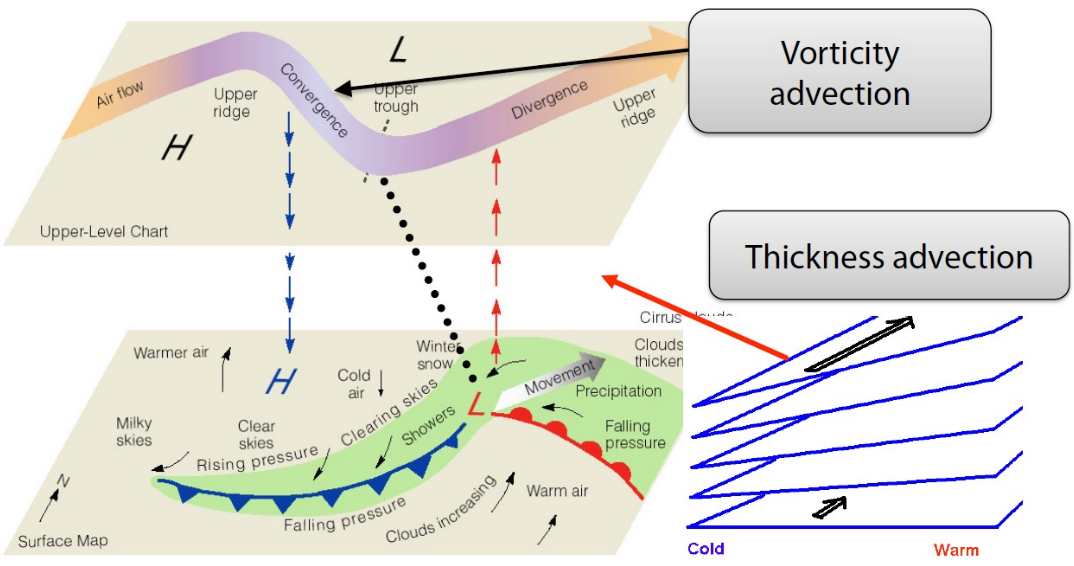
Three-dimensional schematic of the relationship between upper-level and surface charts during cyclogenesis. Upper panel: 500 hPa chart with trough (L) and ridge (H) axes; convergence upstream and divergence downstream of the upper trough are indicated. Lower panel: surface weather map showing the associated cyclone with warm and cold fronts. The westward tilt of the trough axis with height is essential for continued energy conversion. 상층 일기도와 지표 일기도의 관계. 상층 기압골의 하류에서 발산 → 지표 저기압 발달. 서쪽 기울기(tilt with height)가 에너지 변환의 핵심.



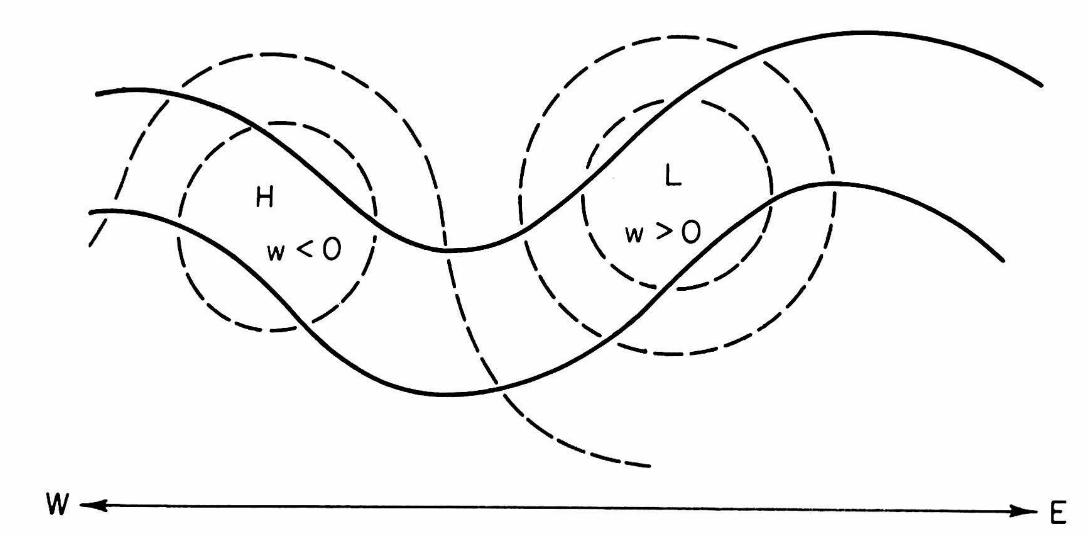
Upper-tropospheric wave pattern on a β-plane showing geopotential contours (thin lines) with troughs (L, cyclonic, $\zeta > 0$) and ridges (H, anticyclonic, $\zeta < 0$). The geostrophic wind blows along the contours; arrows indicate the wind direction. The β-effect and relative vorticity advection compete to determine the net wave propagation direction. β-평면 위의 상층 파동 패턴. 기압골(L, $\zeta > 0$)과 기압능(H, $\zeta < 0$)의 배치 및 지균풍 방향.



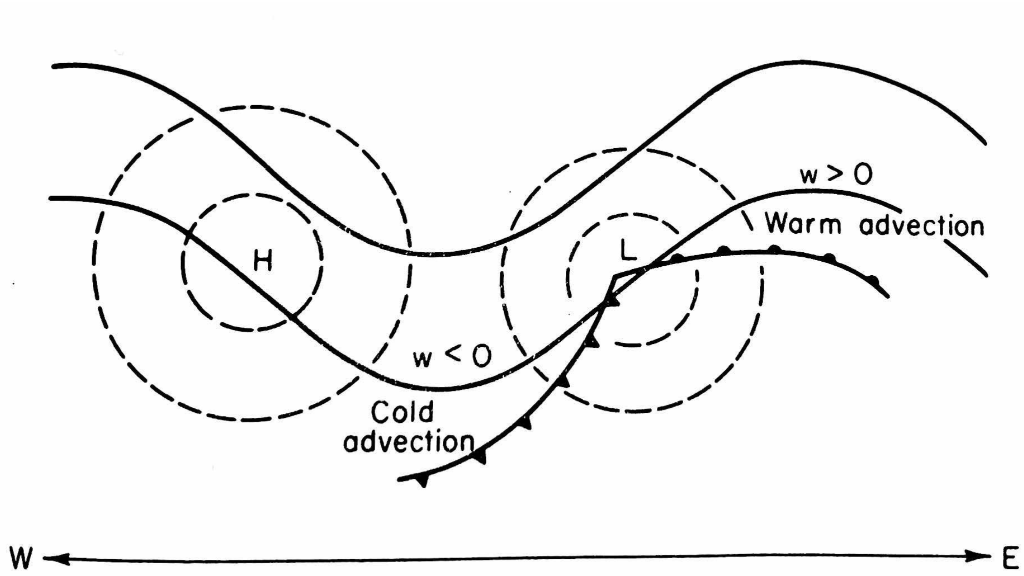
Competition between relative vorticity advection (eastward propagation) and planetary vorticity advection (westward propagation). For **short waves**, relative vorticity advection dominates and the wave propagates eastward; for **long waves**, planetary vorticity advection dominates and the wave retrogresses westward. 상대 소용돌이도 이류(동진)와 행성 소용돌이도 이류(서진)의 경쟁. 단파는 동진, 장파는 서진. 이 경쟁이 파동 전파 방향을 결정한다.



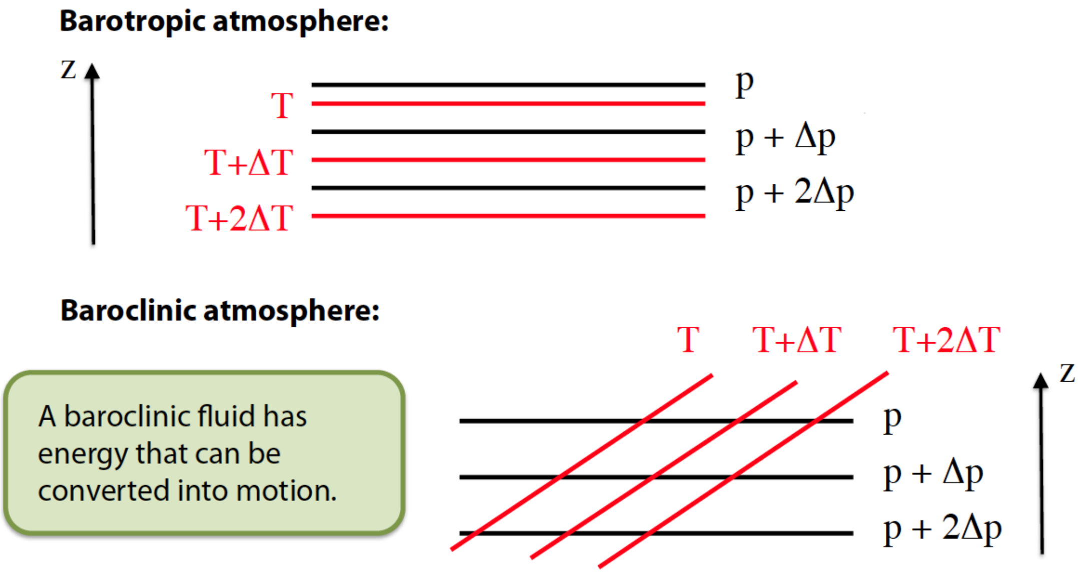
Relationship between the geopotential tendency equation forcing terms and synoptic weather patterns. Upper panel: 500 hPa chart showing vorticity advection (Term B) downstream of the upper trough. Lower panel: surface map with thickness advection (Term C) indicated by the thermal wind vectors. Where positive vorticity advection and warm advection increasing with height coincide, rapid surface development occurs. 지위 고도 경향 방정식의 강제항과 일기도의 관계. 상층 소용돌이도 이류(Term B)와 두께 이류(Term C)가 겹치는 곳에서 지표 저기압이 급속히 발달한다.



Schematic plan view of upper-level (solid) and lower-level (dashed) geopotential contours for a developing baroclinic wave. H marks the high (ridge) and L marks the low (trough). Descending motion ($w < 0$) occurs in the high-pressure region; ascending motion ($w > 0$) occurs in the low-pressure region. The westward tilt with height between the upper and lower wave patterns is evident. 발달 중인 경압파의 상층(실선)–하층(점선) 지위고도 등치선. 고기압 영역에서 하강($w < 0$), 저기압 영역에서 상승($w > 0$).



As in Fig. 59 but with frontal boundaries and temperature advection indicated. Warm advection ($w > 0$, ascent) occurs ahead of the surface low where the warm front lifts warm air over cold air; cold advection ($w < 0$, descent) occurs behind the low. This pattern is consistent with the omega-equation forcing by the Laplacian of temperature advection (Term C). 전선과 온도 이류가 표시된 경압파 평면도. 온난 전선 → 상승(온난 이류), 한랭 전선 뒤 → 하강(한랭 이류). 오메가 방정식 Term C의 강제에 해당.



Comparison of barotropic and baroclinic atmospheres. Top: in a **barotropic** atmosphere, isobaric surfaces (black) and isothermal surfaces (red) are parallel — no horizontal temperature gradient on pressure surfaces, no vertical wind shear, and no available potential energy (APE). Bottom: in a **baroclinic** atmosphere, isobaric and isothermal surfaces are tilted relative to each other — horizontal temperature gradients exist, the thermal wind produces vertical shear, and APE is available for conversion to kinetic energy via baroclinic instability. 순압 대기 vs. 경압 대기. 순압: 등압면과 등온면이 평행 (APE = 0). 경압: 등압면과 등온면이 기울어져 있어 에너지 변환이 가능.